

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z3,Z4,Z5

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z3 Z4 Z5: $\frac{Z_1 Z_3 Z_4 Z_5 g_m - Z_1 Z_3 Z_4}{2 Z_1 Z_3 Z_4 g_m + 2 Z_1 Z_3 Z_5 g_m + 2 Z_1 Z_3 + Z_1 Z_4 Z_5 g_m + Z_1 Z_4 + Z_3 Z_4 + 2 Z_3 Z_5 + Z_4 Z_5}$

$$H(z) = \frac{Z_1 Z_3 Z_4 Z_5 g_m - Z_1 Z_3 Z_4}{2 Z_1 Z_3 Z_4 g_m + 2 Z_1 Z_3 Z_5 g_m + 2 Z_1 Z_3 + Z_1 Z_4 Z_5 g_m + Z_1 Z_4 + Z_3 Z_4 + 2 Z_3 Z_5 + Z_4 Z_5}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3}{2 C_1 C_4 R_3 R_5 s^2 + 2 R_3 g_m + R_5 g_m + s (C_1 R_3 + C_1 R_5 + 2 C_4 R_3 R_5 g_m + 2 C_4 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_5} \sqrt{2 R_3 g_m + R_5 g_m + 1}}{C_1 R_3 + C_1 R_5 + 2 C_4 R_3 R_5 g_m + 2 C_4 R_3}$
 wo: $\frac{\sqrt{2} \sqrt{2 R_3 g_m + R_5 g_m + 1}}{2 \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_3 + C_1 R_5 + 2 C_4 R_3 R_5 g_m + 2 C_4 R_3}{2 C_1 C_4 R_3 R_5}$
 K-LP: $\frac{R_3 R_5 g_m - R_3}{2 R_3 g_m + R_5 g_m + 1}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.2 LP-2 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4}{2 C_1 C_4 R_3 R_4 R_5 s^2 + 2 R_3 R_4 g_m + 2 R_3 R_5 g_m + 2 R_3 + R_4 R_5 g_m + R_4 + s (C_1 R_3 R_4 + 2 C_1 R_3 R_5 + C_1 R_4 R_5 + 2 C_4 R_3 R_4 R_5 g_m + 2 C_4 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{2 R_3 R_4 g_m + 2 R_3 R_5 g_m + 2 R_3 + R_4 R_5 g_m + R_4}}{C_1 R_3 R_4 + 2 C_1 R_3 R_5 + C_1 R_4 R_5 + 2 C_4 R_3 R_4 R_5 g_m + 2 C_4 R_3 R_4}$
 wo: $\frac{\sqrt{2} \sqrt{2 R_3 R_4 g_m + 2 R_3 R_5 g_m + 2 R_3 + R_4 R_5 g_m + R_4}}{2 \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_3 R_4 + 2 C_1 R_3 R_5 + C_1 R_4 R_5 + 2 C_4 R_3 R_4 R_5 g_m + 2 C_4 R_3 R_4}{2 C_1 C_4 R_3 R_4 R_5}$
 K-LP: $\frac{R_3 R_4 R_5 g_m - R_3 R_4}{2 R_3 R_4 g_m + 2 R_3 R_5 g_m + 2 R_3 + R_4 R_5 g_m + R_4}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.3 LP-3 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{C_1 C_3 R_4 R_5 s^2 + 2R_4 g_m + 2R_5 g_m + s(C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{R_4}\sqrt{R_5}\sqrt{R_4 g_m + R_5 g_m + 1}}{C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4}$
 wo: $\frac{\sqrt{2R_4 g_m + 2R_5 g_m + 2}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_4}\sqrt{R_5}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{2R_4 g_m + 2R_5 g_m + 2}(C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4)}{2C_1 C_3 R_4 R_5 \sqrt{R_4 g_m + R_5 g_m + 1}}$
 K-LP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.4 LP-4 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m - 1}{2g_m + s^2(C_1 C_3 R_5 + 2C_1 C_4 R_5) + s(C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{R_5}\sqrt{g_m}\sqrt{C_3 + 2C_4}}{C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1 C_3 R_5 + 2C_1 C_4 R_5}}$
 bandwidth: $\frac{C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}{\sqrt{C_1}\sqrt{R_5}\sqrt{C_3 + 2C_4}\sqrt{C_1 C_3 R_5 + 2C_1 C_4 R_5}}$
 K-LP: $\frac{R_5 g_m - 1}{2g_m}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.5 LP-5 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s^2(C_1 C_3 R_4 R_5 + 2C_1 C_4 R_4 R_5) + s(C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3 + 2C_4}\sqrt{R_4 g_m + R_5 g_m + 1}}{C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4}$
 wo: $\frac{\sqrt{2R_4 g_m + 2R_5 g_m + 2}}{\sqrt{C_1 C_3 R_4 R_5 + 2C_1 C_4 R_4 R_5}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{2R_4 g_m + 2R_5 g_m + 2}(C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4)}{2\sqrt{C_1}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3 + 2C_4}\sqrt{C_1 C_3 R_4 R_5 + 2C_1 C_4 R_4 R_5}\sqrt{R_4 g_m + R_5 g_m + 1}}$
 K-LP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.6 LP-6 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4}{C_1 C_3 R_3 R_4 R_5 s^2 + 2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s(C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4}$
 wo: $\frac{\sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}}$

bandwidth: $\frac{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4}{C_1 C_3 R_3 R_4 R_5}$
K-LP: $\frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.7 LP-7 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + s^2 (C_1 C_3 R_3 R_5 + 2C_1 C_4 R_3 R_5) + s (C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{2R_3 g_m + R_5 g_m + 1}}{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3}$
wo: $\frac{\sqrt{2R_3 g_m + R_5 g_m + 1}}{\sqrt{C_1 C_3 R_3 R_5 + 2C_1 C_4 R_3 R_5}}$
bandwidth: $\frac{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3}{\sqrt{C_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{C_1 C_3 R_3 R_5 + 2C_1 C_4 R_3 R_5}}$
K-LP: $\frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + 1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.8 LP-8 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s^2 (C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_3 R_4 R_5) + s (C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4}$
wo: $\frac{\sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{\sqrt{C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_3 R_4 R_5}}$
bandwidth: $\frac{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4}{\sqrt{C_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_3 R_4 R_5}}$
K-LP: $\frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.9 LP-9 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3}{2C_1 C_4 R_1 R_3 R_5 s^2 + 2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s (C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5}$
wo: $\frac{\sqrt{2} \sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{2\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5}{2C_1 C_4 R_1 R_3 R_5}$
K-LP: $\frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.10 LP-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2C_1 C_4 R_1 R_3 R_4 R_5 s^2 + 2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s(C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5}$
 wo: $\frac{\sqrt{2}\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5}{2C_1 C_4 R_1 R_3 R_4 R_5}$
 K-LP: $\frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.11 LP-11 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{C_1 C_3 R_1 R_4 R_5 s^2 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5}$
 wo: $\frac{\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5}{C_1 C_3 R_1 R_4 R_5}$
 K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.12 LP-12 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + s^2(C_1 C_3 R_1 R_5 + 2C_1 C_4 R_1 R_5) + s(C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}\sqrt{R_5}\sqrt{C_3+2C_4}\sqrt{2R_1 g_m+1}}{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}$
 wo: $\frac{\sqrt{2R_1 g_m+1}}{\sqrt{C_1 C_3 R_1 R_5 + 2C_1 C_4 R_1 R_5}}$
 bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}{\sqrt{C_1}\sqrt{R_1}\sqrt{R_5}\sqrt{C_3+2C_4}\sqrt{C_1 C_3 R_1 R_5 + 2C_1 C_4 R_1 R_5}}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.13 LP-13 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2(C_1 C_3 R_1 R_4 R_5 + 2C_1 C_4 R_1 R_4 R_5) + s(C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3+2C_4}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5}$
 wo: $\frac{\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{\sqrt{C_1 C_3 R_1 R_4 R_5 + 2C_1 C_4 R_1 R_4 R_5}}$
 bandwidth: $\frac{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5}{\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3+2C_4}\sqrt{C_1 C_3 R_1 R_4 R_5 + 2C_1 C_4 R_1 R_4 R_5}}$

K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.14 LP-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{C_1 C_3 R_1 R_3 R_4 R_5 s^2 + 2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s(C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5}{C_1 C_3 R_1 R_3 R_4 R_5}$
K-LP: $\frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.15 LP-15 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2(C_1 C_3 R_1 R_3 R_5 + 2C_1 C_4 R_1 R_3 R_5) + s(C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5} + 2C_1 C_4 R_1 R_3 R_5}$
bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5} + 2C_1 C_4 R_1 R_3 R_5}$
K-LP: $\frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.16 LP-16 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s^2(C_1 C_3 R_1 R_3 R_4 R_5 + 2C_1 C_4 R_1 R_3 R_4 R_5) + s(C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} + 2C_1 C_4 R_1 R_3 R_4 R_5}$
bandwidth: $\frac{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} + 2C_1 C_4 R_1 R_3 R_4 R_5}$
K-LP: $\frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, R_3, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 s + R_1 R_3 g_m}{2C_4 C_5 R_1 R_3 s^2 + R_1 g_m + s(2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_1 g_m + 1}}{2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{R_1 g_m + 1}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}} \\ \text{bandwidth: } & \frac{2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}{2C_4 C_5 R_1 R_3} \\ \text{K-LP: } & \frac{R_1 R_3 g_m}{R_1 g_m + 1} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_1 R_3}{2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, R_3, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_5 s + R_1 R_3 R_5 g_m - R_1 R_3}{2C_4 C_5 R_1 R_3 R_5 s^2 + 2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s(2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_5}\sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}{2C_4 C_5 R_1 R_3 R_5} \\ \text{K-LP: } & \frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_1 R_3 R_5}{2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \infty, R_3, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_3 g_m + s(C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{R_1 g_m + s^2(2C_4 C_5 R_1 R_3 R_5 g_m + 2C_4 C_5 R_1 R_3 + 2C_4 C_5 R_3 R_5) + s(2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_3}R_5 g_m\sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_3}\sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}R_5\sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}}{2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5} \\ \text{wo: } & \sqrt{\frac{R_1 g_m + 1}{2C_4 C_5 R_1 R_3 R_5 g_m + 2C_4 C_5 R_1 R_3 + 2C_4 C_5 R_3 R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_1 g_m + 1}{2C_4 C_5 R_1 R_3 R_5 g_m + 2C_4 C_5 R_1 R_3 + 2C_4 C_5 R_3 R_5}}(2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5)}{\sqrt{2}\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_3}R_5 g_m\sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_3}\sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}R_5\sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}} \\ \text{K-LP: } & \frac{R_1 R_3 g_m}{R_1 g_m + 1} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3}{2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(R_1, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 s + R_1 R_3 R_4 g_m}{2C_4 C_5 R_1 R_3 R_4 s^2 + 2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s(2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$$

$$\begin{aligned} \text{wo: } & \frac{\sqrt{2}\sqrt{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{2C_4R_1R_3R_4g_m+2C_4R_3R_4+2C_5R_1R_3R_4g_m+2C_5R_1R_3+C_5R_1R_4+C_5R_3R_4}{2C_4C_5R_1R_3R_4} \\ \text{K-LP: } & \frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5R_1R_3R_4}{2C_4R_1R_3R_4g_m+2C_4R_3R_4+2C_5R_1R_3R_4g_m+2C_5R_1R_3+C_5R_1R_4+C_5R_3R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(R_1, \infty, R_3, \frac{R_4}{C_4R_4s+1}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_1R_3R_4R_5s + R_1R_3R_4R_5g_m - R_1R_3R_4}{2C_4C_5R_1R_3R_4R_5s^2 + 2R_1R_3R_4g_m + 2R_1R_3R_5g_m + 2R_1R_3 + R_1R_4R_5g_m + R_1R_4 + R_3R_4 + 2R_3R_5 + R_4R_5 + s(2C_4R_1R_3R_4R_5g_m + 2C_4R_1R_3R_4 + 2C_4R_3R_4R_5 + 2C_5R_1R_3R_4R_5g_m + 2C_5R_1R_3R_5 + C_5R_1R_4R_5 + C_5R_3R_4R_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}{2C_4R_1R_3R_4R_5g_m+2C_4R_1R_3R_4+2C_4R_3R_4R_5+2C_5R_1R_3R_4R_5g_m+2C_5R_1R_3R_5+C_5R_1R_4R_5+C_5R_3R_4R_5} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{2C_4R_1R_3R_4R_5g_m+2C_4R_1R_3R_4+2C_4R_3R_4R_5+2C_5R_1R_3R_4R_5g_m+2C_5R_1R_3R_5+C_5R_1R_4R_5+C_5R_3R_4R_5}{2C_4C_5R_1R_3R_4R_5} \\ \text{K-LP: } & \frac{R_1R_3R_4R_5g_m - R_1R_3R_4}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5R_1R_3R_4R_5}{2C_4R_1R_3R_4R_5g_m+2C_4R_1R_3R_4+2C_4R_3R_4R_5+2C_5R_1R_3R_4R_5g_m+2C_5R_1R_3R_5+C_5R_1R_4R_5+C_5R_3R_4R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.6 X-INVALID-NUMER-6 $Z(s) = \left(R_1, \infty, R_3, \frac{R_4}{C_4R_4s+1}, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{R_1R_3R_4g_m + s(C_5R_1R_3R_4R_5g_m - C_5R_1R_3R_4)}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^2(2C_4C_5R_1R_3R_4R_5g_m + 2C_4C_5R_1R_3R_4 + 2C_4C_5R_3R_4R_5) + s(2C_4R_1R_3R_4g_m + 2C_4R_3R_4 + 2C_5R_1R_3R_4g_m + 2C_5R_1R_3R_5g_m + 2C_5R_1R_3 + C_5R_1R_4R_5g_m + C_5R_1R_4 + C_5R_3R_4 + 2C_5R_3R_5 + C_5R_4R_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}}{2C_4R_1R_3R_4g_m+2C_4R_3R_4+2C_5R_1R_3R_4g_m+2C_5R_1R_3R_5g_m+2C_5R_1R_3+C_5R_1R_4R_5g_m+C_5R_1R_4+C_5R_3R_4+2C_5R_3R_5+C_5R_4R_5} \\ \text{wo: } & \sqrt{\frac{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}{2C_4C_5R_1R_3R_4R_5g_m+2C_4C_5R_1R_3R_4+2C_4C_5R_3R_4R_5}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}}{\sqrt{2C_4C_5R_1R_3R_4R_5g_m+2C_4C_5R_1R_3R_4+2C_4C_5R_3R_4R_5}(2C_4R_1R_3R_4g_m+2C_4R_3R_4+2C_5R_1R_3R_4g_m+2C_5R_1R_3R_5g_m+2C_5R_1R_3+C_5R_1R_4R_5g_m+C_5R_1R_4+C_5R_3R_4+2C_5R_3R_5+C_5R_4R_5)} \\ \text{K-LP: } & \frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_5R_1R_3R_4R_5g_m - C_5R_1R_3R_4}{2C_4R_1R_3R_4g_m+2C_4R_3R_4+2C_5R_1R_3R_4g_m+2C_5R_1R_3R_5g_m+2C_5R_1R_3+C_5R_1R_4R_5g_m+C_5R_1R_4+C_5R_3R_4+2C_5R_3R_5+C_5R_4R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.7 X-INVALID-NUMER-7 $Z(s) = \left(R_1, \infty, \frac{1}{C_3s}, R_4, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{-C_5R_1R_4s + R_1R_4g_m}{C_3C_5R_1R_4s^2 + 2R_1g_m + s(C_3R_1R_4g_m + C_3R_4 + 2C_5R_1R_4g_m + 2C_5R_1 + C_5R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_3}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1g_m+1}}{C_3R_1R_4g_m+C_3R_4+2C_5R_1R_4g_m+2C_5R_1+C_5R_4} \\ \text{wo: } & \frac{\sqrt{2R_1g_m+2}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{2R_1g_m+2}(C_3R_1R_4g_m+C_3R_4+2C_5R_1R_4g_m+2C_5R_1+C_5R_4)}{2C_3C_5R_1R_4\sqrt{R_1g_m+1}} \\ \text{K-LP: } & \frac{R_1R_4g_m}{2R_1g_m+2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5R_1R_4}{C_3R_1R_4g_m+C_3R_4+2C_5R_1R_4g_m+2C_5R_1+C_5R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.8 X-INVALID-NUMER-8 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4}{C_3 C_5 R_1 R_4 R_5 s^2 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_5} \sqrt{R_1} \sqrt{R_4} \sqrt{R_5} \sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{\sqrt{C_3} \sqrt{C_5} \sqrt{R_1} \sqrt{R_4} \sqrt{R_5}}$
bandwidth: $\frac{C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}{C_3 C_5 R_1 R_4 R_5}$
K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_4 R_5}{C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s(C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s^2(C_3 C_5 R_1 R_4 R_5 g_m + C_3 C_5 R_1 R_4 + C_3 C_5 R_4 R_5) + s(C_3 R_1 R_4 g_m + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_3} \sqrt{C_5} R_1 \sqrt{R_4} R_5 g_m \sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_3} \sqrt{C_5} R_1 \sqrt{R_4} \sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_3} \sqrt{C_5} \sqrt{R_4} R_5 \sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}}{C_3 R_1 R_4 g_m + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5}$
wo: $\sqrt{\frac{2R_1 g_m + 2}{C_3 C_5 R_1 R_4 R_5 g_m + C_3 C_5 R_1 R_4 + C_3 C_5 R_4 R_5}}$
bandwidth: $\frac{\sqrt{\frac{2R_1 g_m + 2}{C_3 C_5 R_1 R_4 R_5 g_m + C_3 C_5 R_1 R_4 + C_3 C_5 R_4 R_5}} (C_3 R_1 R_4 g_m + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5)}{\sqrt{2} \sqrt{C_3} \sqrt{C_5} R_1 \sqrt{R_4} R_5 g_m \sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_3} \sqrt{C_5} R_1 \sqrt{R_4} \sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_3} \sqrt{C_5} \sqrt{R_4} R_5 \sqrt{\frac{R_1 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}}$
K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
K-HP: 0
K-BP: $\frac{C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4}{C_3 R_1 R_4 g_m + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{2R_1 g_m + s^2(C_3 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_5) + s(C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{R_1} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{2R_1 g_m + 1}}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_3} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5} + 2C_4 C_5 R_1 R_5}$
bandwidth: $\frac{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}{\sqrt{C_5} \sqrt{R_1} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{C_3 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_5}}$
K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_5}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
Qz: None
Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{2R_1 g_m + s^2(C_3 C_5 R_1 R_4 + 2C_4 C_5 R_1 R_4) + s(C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_5} \sqrt{R_1} \sqrt{R_4} \sqrt{C_3 + 2C_4} \sqrt{R_1 g_m + 1}}{C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}$

$$\begin{aligned}
\text{wo: } & \frac{\sqrt{2R_1g_m+2}}{\sqrt{C_3C_5R_1R_4+2C_4C_5R_1R_4}} \\
\text{bandwidth: } & \frac{\sqrt{2}\sqrt{2R_1g_m+2}(C_3R_1R_4g_m+C_3R_4+2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1+C_5R_4)}{2\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{R_1g_m+1}\sqrt{C_3C_5R_1R_4+2C_4C_5R_1R_4}} \\
\text{K-LP: } & \frac{R_1R_4g_m}{2R_1g_m+2} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_1R_4}{C_3R_1R_4g_m+C_3R_4+2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1+C_5R_4} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

$$8.12 \quad \mathbf{X-INVALID-NUMER-12} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{1}{C_3s}, \quad \frac{R_4}{C_4R_4s+1}, \quad \frac{R_5}{C_5R_5s+1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5R_1R_4R_5s + R_1R_4R_5g_m - R_1R_4}{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^2(C_3C_5R_1R_4R_5 + 2C_4C_5R_1R_4R_5) + s(C_3R_1R_4R_5g_m + C_3R_1R_4 + C_3R_4R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_4R_5 + 2C_5R_1R_4R_5g_m + 2C_5R_1R_5 + C_5R_4R_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3+2C_4}\sqrt{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}{C_3R_1R_4R_5g_m+C_3R_1R_4+C_3R_4R_5+2C_4R_1R_4R_5g_m+2C_4R_1R_4+2C_4R_4R_5+2C_5R_1R_4R_5g_m+2C_5R_1R_5+C_5R_4R_5} \\
\text{wo: } & \frac{\sqrt{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}{\sqrt{C_3C_5R_1R_4R_5+2C_4C_5R_1R_4R_5}} \\
\text{bandwidth: } & \frac{C_3R_1R_4R_5g_m+C_3R_1R_4+C_3R_4R_5+2C_4R_1R_4R_5g_m+2C_4R_1R_4+2C_4R_4R_5+2C_5R_1R_4R_5g_m+2C_5R_1R_5+C_5R_4R_5}{\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3+2C_4}\sqrt{C_3C_5R_1R_4R_5+2C_4C_5R_1R_4R_5}} \\
\text{K-LP: } & \frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_1R_4R_5}{C_3R_1R_4R_5g_m+C_3R_1R_4+C_3R_4R_5+2C_4R_1R_4R_5g_m+2C_4R_1R_4+2C_4R_4R_5+2C_5R_1R_4R_5g_m+2C_5R_1R_5+C_5R_4R_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

$$8.13 \quad \mathbf{X-INVALID-NUMER-13} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{1}{C_3s}, \quad \frac{R_4}{C_4R_4s+1}, \quad R_5 + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{R_1R_4g_m + s(C_5R_1R_4R_5g_m - C_5R_1R_4)}{2R_1g_m + s^2(C_3C_5R_1R_4R_5g_m + C_3C_5R_1R_4 + C_3C_5R_4R_5 + 2C_4C_5R_1R_4R_5g_m + 2C_4C_5R_1R_4 + 2C_4C_5R_4R_5) + s(C_3R_1R_4g_m + C_3R_4 + 2C_4R_1R_4g_m + 2C_4R_4 + 2C_5R_1R_4g_m + 2C_5R_1R_5g_m + 2C_5R_1 + C_5R_4 + 2C_5R_5) + 2}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{2}C_3\sqrt{C_5}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_1g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5} + \frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}} + \sqrt{2}C_3\sqrt{C_5}R_1\sqrt{R_4}\sqrt{\frac{R_1g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5} + \frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}} + \sqrt{2}C_3\sqrt{C_5}\sqrt{R_4}R_5\sqrt{\frac{R_1g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5} + \frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}}{\sqrt{\frac{2R_1g_m+2}{C_3C_5R_1R_4R_5g_m+C_3C_5R_1R_4+C_3C_5R_4R_5+2C_4C_5R_1R_4R_5g_m+2C_4C_5R_1R_4+2C_4C_5R_4R_5}}} \\
\text{wo: } & \sqrt{\frac{2R_1g_m+2}{C_3C_5R_1R_4R_5g_m+C_3C_5R_1R_4+C_3C_5R_4R_5+2C_4C_5R_1R_4R_5g_m+2C_4C_5R_1R_4+2C_4C_5R_4R_5}} \\
\text{bandwidth: } & \frac{\sqrt{\frac{2R_1g_m+2}{C_3C_5R_1R_4R_5g_m+C_3C_5R_1R_4+C_3C_5R_4R_5+2C_4C_5R_1R_4R_5g_m+2C_4C_5R_1R_4+2C_4C_5R_4R_5}}}{\sqrt{2}C_3\sqrt{C_5}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_1g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5} + \frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}} + \sqrt{2}C_3\sqrt{C_5}R_1\sqrt{R_4}\sqrt{\frac{R_1g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5} + \frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}} + \sqrt{2}C_3\sqrt{C_5}\sqrt{R_4}R_5\sqrt{\frac{R_1g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5} + \frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}} \\
\text{K-LP: } & \frac{R_1R_4g_m}{2R_1g_m+2} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_5R_1R_4R_5g_m-C_5R_1R_4}{C_3R_1R_4g_m+C_3R_4+2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1R_5g_m+2C_5R_1+C_5R_4+2C_5R_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

$$8.14 \quad \mathbf{X-INVALID-NUMER-14} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{1}{C_3s}, \quad R_4 + \frac{1}{C_4s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_1R_5g_m - R_1 + s(C_4R_1R_4R_5g_m - C_4R_1R_4)}{2R_1g_m + s^2(C_3C_4R_1R_4R_5g_m + C_3C_4R_1R_4 + C_3C_4R_4R_5) + s(C_3R_1R_5g_m + C_3R_1 + C_3R_5 + 2C_4R_1R_4g_m + 2C_4R_1R_5g_m + 2C_4R_1 + C_4R_4 + 2C_4R_5) + 1}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_3}\sqrt{C_4}\sqrt{R_4}R_5\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}}}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5} \\
\text{wo: } & \sqrt{\frac{2R_1g_m+1}{C_3C_4R_1R_4R_5g_m+C_3C_4R_1R_4+C_3C_4R_4R_5}} \\
\text{bandwidth: } & \frac{\sqrt{\frac{2R_1g_m+1}{C_3C_4R_1R_4R_5g_m+C_3C_4R_1R_4+C_3C_4R_4R_5}}(C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5)}{\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_3}\sqrt{C_4}\sqrt{R_4}R_5\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}}} \\
\text{K-LP: } & \frac{R_1R_5g_m-R_1}{2R_1g_m+1} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_4R_1R_4R_5g_m-C_4R_1R_4}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5} \\
\text{Qz: } & \text{None}
\end{aligned}$$

Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 s + R_1 R_3 R_4 g_m}{C_3 C_5 R_1 R_3 R_4 s^2 + 2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s(C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{\sqrt{C_3} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4}}$
bandwidth: $\frac{C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}{C_3 C_5 R_1 R_3 R_4}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4}{C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
Qz: None
Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 R_5 s + R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{C_3 C_5 R_1 R_3 R_4 R_5 s^2 + 2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s(C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{\sqrt{C_3} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5}}$
bandwidth: $\frac{C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}{C_3 C_5 R_1 R_3 R_4 R_5}$
K-LP: $\frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4 R_5}{C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
Qz: None
Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m + s(C_5 R_1 R_3 R_4 R_5 g_m - C_5 R_1 R_3 R_4)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^2(C_3 C_5 R_1 R_3 R_4 R_5 g_m + C_3 C_5 R_1 R_3 R_4 + C_3 C_5 R_3 R_4 R_5) + s(C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 R_5 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 R_5 g_m + C_5 R_1 R_4 + C_5 R_3 R_4 + 2C_5 R_3 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_5} R_1 \sqrt{R_3} \sqrt{R_4} R_5 g_m \sqrt{\frac{2R_1 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_1 R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_3}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_4}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_3} \sqrt{C_5} R_1 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{2R_1 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_1 R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_3}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_4}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_3} \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} R_5 \sqrt{\frac{2R_1 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_1 R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_3}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_4}{R_1 R_5 g_m + R_1 + R_5}}}{C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 R_5 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 R_5 g_m + C_5 R_1 R_4 + C_5 R_3 R_4 + 2C_5 R_3 R_5 + C_5 R_4 R_5}$
wo: $\sqrt{\frac{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}{C_3 C_5 R_1 R_3 R_4 R_5 g_m + C_3 C_5 R_1 R_3 R_4 + C_3 C_5 R_3 R_4 R_5}}$
bandwidth: $\frac{\sqrt{\frac{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}{C_3 C_5 R_1 R_3 R_4 R_5 g_m + C_3 C_5 R_1 R_3 R_4 + C_3 C_5 R_3 R_4 R_5}} (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 R_5 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 R_5 g_m + C_5 R_1 R_4 + C_5 R_3 R_4 + 2C_5 R_3 R_5 + C_5 R_4 R_5)}{\sqrt{C_3} \sqrt{C_5} R_1 \sqrt{R_3} \sqrt{R_4} R_5 g_m \sqrt{\frac{2R_1 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_1 R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_3}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_4}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_3} \sqrt{C_5} R_1 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{2R_1 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_1 R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_3}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_4}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_3} \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} R_5 \sqrt{\frac{2R_1 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_1 R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_3}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_4}{R_1 R_5 g_m + R_1 + R_5}}}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
K-HP: 0
K-BP: $\frac{C_5 R_1 R_3 R_4 R_5 g_m - C_5 R_1 R_3 R_4}{C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 R_5 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 R_5 g_m + C_5 R_1 R_4 + C_5 R_3 R_4 + 2C_5 R_3 R_5 + C_5 R_4 R_5}$
Qz: None
Wz: None

8.18 X-INVALID-NUMER-18 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 s + R_1 R_3 g_m}{R_1 g_m + s^2 (C_3 C_5 R_1 R_3 + 2C_4 C_5 R_1 R_3) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{C_3 + 2C_4} \sqrt{R_1 g_m + 1}}{C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3} \\ \text{wo: } & \frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_3 C_5 R_1 R_3 + 2C_4 C_5 R_1 R_3}} \\ \text{bandwidth: } & \frac{C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}{\sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{C_3 + 2C_4} \sqrt{C_3 C_5 R_1 R_3 + 2C_4 C_5 R_1 R_3}} \\ \text{K-LP: } & \frac{R_1 R_3 g_m}{R_1 g_m + 1} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_1 R_3}{C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.19 X-INVALID-NUMER-19 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_5 s + R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2 (C_3 C_5 R_1 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_5) + s (C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5} \\ \text{wo: } & \frac{\sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{\sqrt{C_3 C_5 R_1 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_5}} \\ \text{bandwidth: } & \frac{C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}{\sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{C_3 C_5 R_1 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_5}} \\ \text{K-LP: } & \frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_1 R_3 R_5}{C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.20 X-INVALID-NUMER-20 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_3 g_m + s (C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{R_1 g_m + s^2 (C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_3 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_5 g_m + 2C_4 C_5 R_1 R_3 + 2C_4 C_5 R_3 R_5) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{C_3 \sqrt{C_5} R_1 \sqrt{R_3} R_5 g_m \sqrt{\frac{R_1 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}} + C_3 \sqrt{C_5} R_1 \sqrt{R_3} \sqrt{\frac{R_1 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}} + C_3 \sqrt{C_5} \sqrt{R_3} R_5 \sqrt{\frac{R_1 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}}}{\sqrt{\frac{R_1 g_m + 1}{C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_3 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_5 g_m + 2C_4 C_5 R_1 R_3 + 2C_4 C_5 R_3 R_5}}} \\ \text{wo: } & \sqrt{\frac{R_1 g_m + 1}{C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_3 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_5 g_m + 2C_4 C_5 R_1 R_3 + 2C_4 C_5 R_3 R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_1 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}} + C_3 \sqrt{C_5} R_1 \sqrt{R_3} \sqrt{\frac{R_1 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}} + C_3 \sqrt{C_5} \sqrt{R_3} R_5 \sqrt{\frac{R_1 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}}}{\sqrt{\frac{R_1 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}} + C_3 \sqrt{C_5} \sqrt{R_3} R_5 \sqrt{\frac{R_1 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}}} \\ \text{K-LP: } & \frac{R_1 R_3 g_m}{R_1 g_m + 1} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3}{C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.21 X-INVALID-NUMER-21 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 s + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^2 (C_3 C_5 R_1 R_3 R_4 + 2C_4 C_5 R_1 R_3 R_4) + s (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{C_3 + 2C_4} \sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$$

$$\begin{aligned}
\text{wo: } & \frac{\sqrt{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}{\sqrt{C_3C_5R_1R_3R_4+2C_4C_5R_1R_3R_4}} \\
\text{bandwidth: } & \frac{C_3R_1R_3R_4g_m+C_3R_3R_4+2C_4R_1R_3R_4g_m+2C_4R_3R_4+2C_5R_1R_3R_4g_m+2C_5R_1R_3+C_5R_1R_4+C_5R_3R_4}{\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{C_3C_5R_1R_3R_4+2C_4C_5R_1R_3R_4}} \\
\text{K-LP: } & \frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_1R_3R_4}{C_3R_1R_3R_4g_m+C_3R_3R_4+2C_4R_1R_3R_4g_m+2C_4R_3R_4+2C_5R_1R_3R_4g_m+2C_5R_1R_3+C_5R_1R_4+C_5R_3R_4} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.22 X-INVALID-NUMER-22 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3R_3s+1}, \frac{R_4}{C_4R_4s+1}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_1R_3R_4R_5s + R_1R_3R_4R_5g_m - R_1R_3R_4}{2R_1R_3R_4g_m + 2R_1R_3R_5g_m + 2R_1R_3 + R_1R_4R_5g_m + R_1R_4 + R_3R_4 + 2R_3R_5 + R_4R_5 + s^2(C_3C_5R_1R_3R_4R_5 + 2C_4C_5R_1R_3R_4R_5) + s(C_3R_1R_3R_4R_5g_m + C_3R_1R_3R_4 + C_3R_3R_4R_5 + 2C_4R_1R_3R_4R_5g_m + 2C_4R_1R_3R_4 + 2C_4R_3R_4R_5 + 2C_5R_1R_3R_4R_5g_m + 2C_5R_1R_3R_4 + 2C_5R_3R_4R_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3+2C_4}\sqrt{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}{C_3R_1R_3R_4R_5g_m+C_3R_1R_3R_4+C_3R_3R_4R_5+2C_4R_1R_3R_4R_5g_m+2C_4R_1R_3R_4+2C_4R_3R_4R_5+2C_5R_1R_3R_4R_5g_m+2C_5R_1R_3R_5+C_5R_1R_4R_5+C_5R_3R_4R_5} \\
\text{wo: } & \frac{\sqrt{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}{\sqrt{C_3C_5R_1R_3R_4R_5+2C_4C_5R_1R_3R_4R_5}} \\
\text{bandwidth: } & \frac{C_3R_1R_3R_4R_5g_m+C_3R_1R_3R_4+C_3R_3R_4R_5+2C_4R_1R_3R_4R_5g_m+2C_4R_1R_3R_4+2C_4R_3R_4R_5+2C_5R_1R_3R_4R_5g_m+2C_5R_1R_3R_5+C_5R_1R_4R_5+C_5R_3R_4R_5}{\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3+2C_4}\sqrt{C_3C_5R_1R_3R_4R_5+2C_4C_5R_1R_3R_4R_5}} \\
\text{K-LP: } & \frac{R_1R_3R_4R_5g_m-R_1R_3R_4}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_1R_3R_4R_5}{C_3R_1R_3R_4R_5g_m+C_3R_1R_3R_4+C_3R_3R_4R_5+2C_4R_1R_3R_4R_5g_m+2C_4R_1R_3R_4+2C_4R_3R_4R_5+2C_5R_1R_3R_4R_5g_m+2C_5R_1R_3R_5+C_5R_1R_4R_5+C_5R_3R_4R_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.23 X-INVALID-NUMER-23 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3R_3s+1}, \frac{R_4}{C_4R_4s+1}, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{R_1R_3R_4g_m + s(C_5R_1R_3R_4R_5g_m - C_5R_1R_3R_4)}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^2(C_3C_5R_1R_3R_4R_5g_m + C_3C_5R_1R_3R_4 + C_3C_5R_3R_4R_5 + 2C_4C_5R_1R_3R_4R_5g_m + 2C_4C_5R_1R_3R_4 + 2C_4C_5R_3R_4R_5) + s(C_3R_1R_3R_4g_m + C_3R_3R_4 + 2C_4R_1R_3R_4g_m + 2C_4R_3R_4 + 2C_5R_1R_3R_4g_m + 2C_5R_1R_3R_5g_m + 2C_5R_1R_3R_4)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{C_3\sqrt{C_5}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1R_3g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{R_1R_4g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{2R_3}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{R_4}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+C_3\sqrt{C_5}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_1R_3g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{R_1R_4g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{2R_3}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{R_4}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}}{\sqrt{\frac{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}{C_3C_5R_1R_3R_4R_5g_m+C_3C_5R_1R_3R_4+C_3C_5R_3R_4R_5+2C_4C_5R_1R_3R_4R_5g_m+2C_4C_5R_1R_3R_4+2C_4C_5R_3R_4R_5}}} \\
\text{wo: } & \frac{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}{C_3C_5R_1R_3R_4R_5g_m+C_3C_5R_1R_3R_4+C_3C_5R_3R_4R_5+2C_4C_5R_1R_3R_4R_5g_m+2C_4C_5R_1R_3R_4+2C_4C_5R_3R_4R_5} \\
\text{bandwidth: } & \frac{C_3\sqrt{C_5}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1R_3g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{R_1R_4g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{2R_3}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{R_4}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+C_3\sqrt{C_5}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_1R_3g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{R_1R_4g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{2R_3}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{R_4}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}}{\frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}} \\
\text{K-LP: } & \frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_1R_3R_4R_5g_m-C_5R_1R_3R_4}{C_3R_1R_3R_4g_m+C_3R_3R_4+2C_4R_1R_3R_4g_m+2C_4R_3R_4+2C_5R_1R_3R_4g_m+2C_5R_1R_3R_5g_m+2C_5R_1R_3+C_5R_1R_4R_5g_m+C_5R_1R_4+C_5R_3R_4+2C_5R_3R_5+C_5R_4R_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.24 X-INVALID-NUMER-24 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3R_3s+1}, R_4 + \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_1R_3R_5g_m - R_1R_3 + s(C_4R_1R_3R_4R_5g_m - C_4R_1R_3R_4)}{2R_1R_3g_m + R_1R_5g_m + R_1 + R_3 + R_5 + s^2(C_3C_4R_1R_3R_4R_5g_m + C_3C_4R_1R_3R_4 + C_3C_4R_3R_4R_5) + s(C_3R_1R_3R_5g_m + C_3R_1R_3 + C_3R_3R_5 + 2C_4R_1R_3R_4g_m + 2C_4R_1R_3R_5g_m + 2C_4R_1R_3 + C_4R_1R_4R_5g_m + C_4R_1R_4 + C_4R_3R_4 + 2C_4R_3R_5 + C_4R_4R_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_5}{R_1R_5g_m+R_1+R_5}}+\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_5}{R_1R_5g_m+R_1+R_5}}+\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_5}{R_1R_5g_m+R_1+R_5}}}{C_3R_1R_3R_5g_m+C_3R_1R_3+C_3R_3R_5+2C_4R_1R_3R_4g_m+2C_4R_1R_3R_5g_m+2C_4R_1R_3+C_4R_1R_4R_5g_m+C_4R_1R_4+C_4R_3R_4+2C_4R_3R_5+C_4R_4R_5} \\
\text{wo: } & \sqrt{\frac{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}{C_3C_4R_1R_3R_4R_5g_m+C_3C_4R_1R_3R_4+C_3C_4R_3R_4R_5}} \\
\text{bandwidth: } & \frac{\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_5}{R_1R_5g_m+R_1+R_5}}+\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_5}{R_1R_5g_m+R_1+R_5}}+\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_1R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_5}{R_1R_5g_m+R_1+R_5}}}{\frac{R_1R_3R_5g_m-R_1R_3}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}} \\
\text{K-LP: } & \frac{R_1R_3R_5g_m-R_1R_3}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_4R_1R_3R_4R_5g_m-C_4R_1R_3R_4}{C_3R_1R_3R_5g_m+C_3R_1R_3+C_3R_3R_5+2C_4R_1R_3R_4g_m+2C_4R_1R_3R_5g_m+2C_4R_1R_3+C_4R_1R_4R_5g_m+C_4R_1R_4+C_4R_3R_4+2C_4R_3R_5+C_4R_4R_5} \\
\text{Qz: } & \text{None}
\end{aligned}$$

Wz: None

8.25 X-INVALID-NUMER-25 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3)}{2 R_1 g_m + s^2 (2 C_3 C_4 R_1 R_3 R_5 g_m + 2 C_3 C_4 R_1 R_3 + 2 C_3 C_4 R_3 R_5) + s (2 C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5 + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5) + 1}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}R_5g_m\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}R_5\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}}{2C_3R_1R_3g_m+C_3R_1R_5g_m+C_3R_1+C_3R_3+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}$$

$$\text{wo: } \sqrt{\frac{2R_1g_m+1}{2C_3C_4R_1R_3R_5g_m+2C_3C_4R_1R_3+2C_3C_4R_3R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_1g_m+1}{2C_3C_4R_1R_3R_5g_m+2C_3C_4R_1R_3+2C_3C_4R_3R_5}}(2C_3R_1R_3g_m+C_3R_1R_5g_m+C_3R_1+C_3R_3+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5)}{\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}R_5g_m\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}R_5\sqrt{\frac{2R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}}$$

$$\text{K-LP: } \frac{R_1R_5g_m-R_1}{2R_1g_m+1}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_3R_1R_3R_5g_m-C_3R_1R_3}{2C_3R_1R_3g_m+C_3R_1R_5g_m+C_3R_1+C_3R_3+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}$$

$$\text{Qz: None}$$

Wz: None

8.26 X-INVALID-NUMER-26 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s (C_3 R_1 R_3 R_4 R_5 g_m - C_3 R_1 R_3 R_4)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^2 (2 C_3 C_4 R_1 R_3 R_4 R_5 g_m + 2 C_3 C_4 R_1 R_3 R_4 + 2 C_3 C_4 R_3 R_4 R_5) + s (2 C_3 R_1 R_3 R_4 g_m + 2 C_3 R_1 R_3 R_5 g_m + 2 C_3 R_1 R_3 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_3 R_4 + 2 C_3 R_3 R_5 + C_3 R_4 R_5 + 2 C_4 R_1 R_4 R_5 g_m + 2 C_4 R_1 R_4 + 2 C_4 R_4 R_5)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}+\frac{2R_5}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}+\frac{2R_5}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}+\frac{2R_5}{R_1R_5g_m+R_1+R_5}}}{2C_3R_1R_3R_4g_m+2C_3R_1R_3R_5g_m+2C_3R_1R_3+C_3R_1R_4R_5g_m+C_3R_1R_4+C_3R_3R_4+2C_3R_3R_5+C_3R_4R_5+2C_4R_1R_4R_5g_m+2C_4R_1R_4+2C_4R_4R_5}$$

$$\text{wo: } \sqrt{\frac{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}{2C_3C_4R_1R_3R_4R_5g_m+2C_3C_4R_1R_3R_4+2C_3C_4R_3R_4R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}{2C_3C_4R_1R_3R_4R_5g_m+2C_3C_4R_1R_3R_4+2C_3C_4R_3R_4R_5}}(2C_3R_1R_3R_4g_m+2C_3R_1R_3R_5g_m+2C_3R_1R_3+C_3R_1R_4R_5g_m+C_3R_1R_4+C_3R_3R_4+2C_3R_3R_5+C_3R_4R_5+2C_4R_1R_4R_5g_m+2C_4R_1R_4+2C_4R_4R_5)}{\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}+\frac{2R_5}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}+\frac{2R_5}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_1R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_1}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}+\frac{2R_5}{R_1R_5g_m+R_1+R_5}}}$$

$$\text{K-LP: } \frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_3R_1R_3R_4R_5g_m-C_3R_1R_3R_4}{2C_3R_1R_3R_4g_m+2C_3R_1R_3R_5g_m+2C_3R_1R_3+C_3R_1R_4R_5g_m+C_3R_1R_4+C_3R_3R_4+2C_3R_3R_5+C_3R_4R_5+2C_4R_1R_4R_5g_m+2C_4R_1R_4+2C_4R_4R_5}$$

$$\text{Qz: None}$$

Wz: None

8.27 X-INVALID-NUMER-27 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 s + R_3 R_4 g_m}{C_1 C_5 R_3 R_4 s^2 + 2 R_3 g_m + R_4 g_m + s (2 C_1 R_3 + C_1 R_4 + 2 C_5 R_3 R_4 g_m + 2 C_5 R_3 + C_5 R_4)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{C_1}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{2R_3+R_4}}{2C_1R_3+C_1R_4+2C_5R_3R_4g_m+2C_5R_3+C_5R_4}$$

$$\text{wo: } \frac{\sqrt{2R_3g_m+R_4g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{2R_3g_m+R_4g_m}(2C_1R_3+C_1R_4+2C_5R_3R_4g_m+2C_5R_3+C_5R_4)}{C_1C_5R_3R_4\sqrt{g_m}\sqrt{2R_3+R_4}}$$

$$\text{K-LP: } \frac{R_3R_4}{2R_3+R_4}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } -\frac{C_5R_3R_4}{2C_1R_3+C_1R_4+2C_5R_3R_4g_m+2C_5R_3+C_5R_4}$$

$$\text{Qz: None}$$

Wz: None

8.28 X-INVALID-NUMER-28 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 R_5 s + R_3 R_4 R_5 g_m - R_3 R_4}{C_1 C_5 R_3 R_4 R_5 s^2 + 2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s(C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}$
 wo: $\frac{\sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}{C_1 C_5 R_3 R_4 R_5}$
 K-LP: $\frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_3 R_4 R_5}{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}$
 Qz: None
 Wz: None

8.29 X-INVALID-NUMER-29 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 R_4 g_m + s(C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{2R_3 g_m + R_4 g_m + s^2(C_1 C_5 R_3 R_4 + 2C_1 C_5 R_3 R_5 + C_1 C_5 R_4 R_5) + s(2C_1 R_3 + C_1 R_4 + 2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{g_m} \sqrt{2R_3 + R_4} \sqrt{R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{2C_1 R_3 + C_1 R_4 + 2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4}$
 wo: $\frac{\sqrt{2R_3 g_m + R_4 g_m}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_3 R_4 + 2C_1 C_5 R_3 R_5 + C_1 C_5 R_4 R_5}}$
 bandwidth: $\frac{\sqrt{2R_3 g_m + R_4 g_m} (2C_1 R_3 + C_1 R_4 + 2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4)}{\sqrt{C_1} \sqrt{C_5} \sqrt{g_m} \sqrt{2R_3 + R_4} \sqrt{R_3 R_4 + 2R_3 R_5 + R_4 R_5} \sqrt{C_1 C_5 R_3 R_4 + 2C_1 C_5 R_3 R_5 + C_1 C_5 R_4 R_5}}$
 K-LP: $\frac{R_3 R_4}{2R_3 + R_4}$
 K-HP: 0
 K-BP: $\frac{C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4}{2C_1 R_3 + C_1 R_4 + 2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4}$
 Qz: None
 Wz: None

8.30 X-INVALID-NUMER-30 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 s + R_3 g_m}{g_m + s^2(2C_1 C_4 R_3 + C_1 C_5 R_3 + 2C_4 C_5 R_3) + s(C_1 + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5)}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{g_m} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}}{C_1 + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{2C_1 C_4 R_3 + C_1 C_5 R_3 + 2C_4 C_5 R_3}}$
 bandwidth: $\frac{C_1 + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5}{\sqrt{R_3} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5} \sqrt{2C_1 C_4 R_3 + C_1 C_5 R_3 + 2C_4 C_5 R_3}}$
 K-LP: R_3
 K-HP: 0
 K-BP: $-\frac{C_5 R_3}{C_1 + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5}$
 Qz: None
 Wz: None

8.31 X-INVALID-NUMER-31 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_5 s + R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + s^2(2C_1 C_4 R_3 R_5 + C_1 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_5) + s(C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{R_5} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5} \sqrt{2R_3 g_m + R_5 g_m + 1}}{C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{2R_3 g_m + R_5 g_m + 1}}{\sqrt{2C_1 C_4 R_3 R_5 + C_1 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_5}}$
 bandwidth: $\frac{C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}{\sqrt{R_3} \sqrt{R_5} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5} \sqrt{2C_1 C_4 R_3 R_5 + C_1 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_5}}$

K-LP: $\frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_3 R_5}{C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}$
Qz: None
Wz: None

8.32 X-INVALID-NUMER-32 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 s + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^2 (2C_1 C_4 R_3 R_4 + C_1 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4) + s (2C_1 R_3 + C_1 R_4 + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4)}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{2R_3 + R_4} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}}{2C_1 R_3 + C_1 R_4 + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4}$
wo: $\frac{\sqrt{2R_3 g_m + R_4 g_m}}{\sqrt{2C_1 C_4 R_3 R_4 + C_1 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4}}$
bandwidth: $\frac{\sqrt{2R_3 g_m + R_4 g_m} (2C_1 R_3 + C_1 R_4 + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4)}{\sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{2R_3 + R_4} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5} \sqrt{2C_1 C_4 R_3 R_4 + C_1 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4}}$
K-LP: $\frac{R_3 R_4}{2R_3 + R_4}$
K-HP: 0
K-BP: $-\frac{C_5 R_3 R_4}{2C_1 R_3 + C_1 R_4 + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4}$
Qz: None
Wz: None

8.33 X-INVALID-NUMER-33 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 R_5 s + R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s^2 (2C_1 C_4 R_3 R_4 R_5 + C_1 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5) + s (C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5} \sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}$
wo: $\frac{\sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{\sqrt{2C_1 C_4 R_3 R_4 R_5 + C_1 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5}}$
bandwidth: $\frac{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}{\sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5} \sqrt{2C_1 C_4 R_3 R_4 R_5 + C_1 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5}}$
K-LP: $\frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}$
K-HP: 0
K-BP: $-\frac{C_5 R_3 R_4 R_5}{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}$
Qz: None
Wz: None

8.34 X-INVALID-NUMER-34 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s (C_4 R_3 R_4 R_5 g_m - C_4 R_3 R_4)}{2R_3 g_m + R_5 g_m + s^2 (C_1 C_4 R_3 R_4 + 2C_1 C_4 R_3 R_5 + C_1 C_4 R_4 R_5) + s (C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_4} \sqrt{R_3 R_4 + 2R_3 R_5 + R_4 R_5} \sqrt{2R_3 g_m + R_5 g_m + 1}}{C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4}$
wo: $\frac{\sqrt{2R_3 g_m + R_5 g_m + 1}}{\sqrt{C_1 C_4 R_3 R_4 + 2C_1 C_4 R_3 R_5 + C_1 C_4 R_4 R_5}}$
bandwidth: $\frac{C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4}{\sqrt{C_1} \sqrt{C_4} \sqrt{R_3 R_4 + 2R_3 R_5 + R_4 R_5} \sqrt{C_1 C_4 R_3 R_4 + 2C_1 C_4 R_3 R_5 + C_1 C_4 R_4 R_5}}$
K-LP: $\frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_4 R_3 R_4 R_5 g_m - C_4 R_3 R_4}{C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4}$
Qz: None
Wz: None

8.35 X-INVALID-NUMER-35 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{2g_m + s^2 (C_1 C_3 R_4 + C_1 C_5 R_4 + C_3 C_5 R_4) + s (2C_1 + C_3 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{R_4}\sqrt{g_m}\sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5}}{2C_1 + C_3 R_4 g_m + 2C_5 R_4 g_m + 2C_5} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1 C_3 R_4 + C_1 C_5 R_4 + C_3 C_5 R_4}} \\ \text{bandwidth: } & \frac{2C_1 + C_3 R_4 g_m + 2C_5 R_4 g_m + 2C_5}{\sqrt{R_4}\sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5}\sqrt{C_1 C_3 R_4 + C_1 C_5 R_4 + C_3 C_5 R_4}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_4}{2C_1 + C_3 R_4 g_m + 2C_5 R_4 g_m + 2C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.36 X-INVALID-NUMER-36 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s^2 (C_1 C_3 R_4 R_5 + C_1 C_5 R_4 R_5 + C_3 C_5 R_4 R_5) + s (C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{R_4}\sqrt{R_5}\sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5}\sqrt{R_4 g_m + R_5 g_m + 1}}{C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5} \\ \text{wo: } & \frac{\sqrt{2R_4 g_m + 2R_5 g_m + 2}}{\sqrt{C_1 C_3 R_4 R_5 + C_1 C_5 R_4 R_5 + C_3 C_5 R_4 R_5}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{2R_4 g_m + 2R_5 g_m + 2}(C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5)}{2\sqrt{R_4}\sqrt{R_5}\sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5}\sqrt{R_4 g_m + R_5 g_m + 1}\sqrt{C_1 C_3 R_4 R_5 + C_1 C_5 R_4 R_5 + C_3 C_5 R_4 R_5}} \\ \text{K-LP: } & \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_4 R_5}{C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.37 X-INVALID-NUMER-37 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m - 1}{2g_m + s^2 (C_1 C_3 R_5 + 2C_1 C_4 R_5 + C_1 C_5 R_5 + C_3 C_5 R_5 + 2C_4 C_5 R_5) + s (C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{R_5}\sqrt{g_m}\sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}}{C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1 C_3 R_5 + 2C_1 C_4 R_5 + C_1 C_5 R_5 + C_3 C_5 R_5 + 2C_4 C_5 R_5}} \\ \text{bandwidth: } & \frac{C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m}{\sqrt{R_5}\sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}\sqrt{C_1 C_3 R_5 + 2C_1 C_4 R_5 + C_1 C_5 R_5 + C_3 C_5 R_5 + 2C_4 C_5 R_5}} \\ \text{K-LP: } & \frac{R_5 g_m - 1}{2g_m} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_5}{C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.38 X-INVALID-NUMER-38 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{2g_m + s^2 (C_1 C_3 R_4 + 2C_1 C_4 R_4 + C_1 C_5 R_4 + C_3 C_5 R_4 + 2C_4 C_5 R_4) + s (2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{R_4}\sqrt{g_m}\sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}}{2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1 C_3 R_4 + 2C_1 C_4 R_4 + C_1 C_5 R_4 + C_3 C_5 R_4 + 2C_4 C_5 R_4}} \end{aligned}$$

bandwidth: $\frac{2C_1+C_3R_4g_m+2C_4R_4g_m+2C_5R_4g_m+2C_5}{\sqrt{R_4}\sqrt{C_1C_3+2C_1C_4+C_1C_5+C_3C_5+2C_4C_5}\sqrt{C_1C_3R_4+2C_1C_4R_4+C_1C_5R_4+C_3C_5R_4+2C_4C_5R_4}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $-\frac{C_5R_4}{2C_1+C_3R_4g_m+2C_4R_4g_m+2C_5R_4g_m+2C_5}$
Qz: None
Wz: None

8.39 X-INVALID-NUMER-39 $Z(s) = \left(\frac{1}{C_1s}, \infty, \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_4R_5s + R_4R_5g_m - R_4}{2R_4g_m + 2R_5g_m + s^2(C_1C_3R_4R_5 + 2C_1C_4R_4R_5 + C_1C_5R_4R_5 + C_3C_5R_4R_5 + 2C_4C_5R_4R_5) + s(C_1R_4 + 2C_1R_5 + C_3R_4R_5g_m + C_3R_4 + 2C_4R_4R_5g_m + 2C_4R_4 + 2C_5R_4R_5g_m + 2C_5R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{R_4}\sqrt{R_5}\sqrt{R_4g_m+R_5g_m+1}\sqrt{C_1C_3+2C_1C_4+C_1C_5+C_3C_5+2C_4C_5}}{C_1R_4+2C_1R_5+C_3R_4R_5g_m+C_3R_4+2C_4R_4R_5g_m+2C_4R_4+2C_5R_4R_5g_m+2C_5R_5}$
wo: $\frac{\sqrt{2R_4g_m+2R_5g_m+2}}{\sqrt{C_1C_3R_4R_5+2C_1C_4R_4R_5+C_1C_5R_4R_5+C_3C_5R_4R_5+2C_4C_5R_4R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_4g_m+2R_5g_m+2}(C_1R_4+2C_1R_5+C_3R_4R_5g_m+C_3R_4+2C_4R_4R_5g_m+2C_4R_4+2C_5R_4R_5g_m+2C_5R_5)}{2\sqrt{R_4}\sqrt{R_5}\sqrt{R_4g_m+R_5g_m+1}\sqrt{C_1C_3+2C_1C_4+C_1C_5+C_3C_5+2C_4C_5}\sqrt{C_1C_3R_4R_5+2C_1C_4R_4R_5+C_1C_5R_4R_5+C_3C_5R_4R_5+2C_4C_5R_4R_5}}$
K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$
K-HP: 0
K-BP: $-\frac{C_5R_4R_5}{C_1R_4+2C_1R_5+C_3R_4R_5g_m+C_3R_4+2C_4R_4R_5g_m+2C_4R_4+2C_5R_4R_5g_m+2C_5R_5}$
Qz: None
Wz: None

8.40 X-INVALID-NUMER-40 $Z(s) = \left(\frac{1}{C_1s}, \infty, \frac{R_3}{C_3R_3s+1}, R_4, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{-C_5R_3R_4s + R_3R_4g_m}{2R_3g_m + R_4g_m + s^2(C_1C_3R_3R_4 + C_1C_5R_3R_4 + C_3C_5R_3R_4) + s(2C_1R_3 + C_1R_4 + C_3R_3R_4g_m + 2C_5R_3R_4g_m + 2C_5R_3 + C_5R_4)}$$

Parameters:

Q: $\frac{\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{2R_3+R_4}\sqrt{C_1C_3+C_1C_5+C_3C_5}}{2C_1R_3+C_1R_4+C_3R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3+C_5R_4}$
wo: $\frac{\sqrt{2R_3g_m+R_4g_m}}{\sqrt{C_1C_3R_3R_4+C_1C_5R_3R_4+C_3C_5R_3R_4}}$
bandwidth: $\frac{\sqrt{2R_3g_m+R_4g_m}(2C_1R_3+C_1R_4+C_3R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3+C_5R_4)}{\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{2R_3+R_4}\sqrt{C_1C_3+C_1C_5+C_3C_5}\sqrt{C_1C_3R_3R_4+C_1C_5R_3R_4+C_3C_5R_3R_4}}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $-\frac{C_5R_3R_4}{2C_1R_3+C_1R_4+C_3R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3+C_5R_4}$
Qz: None
Wz: None

8.41 X-INVALID-NUMER-41 $Z(s) = \left(\frac{1}{C_1s}, \infty, \frac{R_3}{C_3R_3s+1}, R_4, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_3R_4R_5s + R_3R_4R_5g_m - R_3R_4}{2R_3R_4g_m + 2R_3R_5g_m + 2R_3 + R_4R_5g_m + R_4 + s^2(C_1C_3R_3R_4R_5 + C_1C_5R_3R_4R_5 + C_3C_5R_3R_4R_5) + s(C_1R_3R_4 + 2C_1R_3R_5 + C_1R_4R_5 + C_3R_3R_4R_5g_m + C_3R_3R_4 + 2C_5R_3R_4R_5g_m + 2C_5R_3R_5 + C_5R_4R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{C_1C_3+C_1C_5+C_3C_5}\sqrt{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}{C_1R_3R_4+2C_1R_3R_5+C_1R_4R_5+C_3R_3R_4R_5g_m+C_3R_3R_4+2C_5R_3R_4R_5g_m+2C_5R_3R_5+C_5R_4R_5}$
wo: $\frac{\sqrt{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}{\sqrt{C_1C_3R_3R_4R_5+C_1C_5R_3R_4R_5+C_3C_5R_3R_4R_5}}$
bandwidth: $\frac{C_1R_3R_4+2C_1R_3R_5+C_1R_4R_5+C_3R_3R_4R_5g_m+C_3R_3R_4+2C_5R_3R_4R_5g_m+2C_5R_3R_5+C_5R_4R_5}{\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{C_1C_3+C_1C_5+C_3C_5}\sqrt{C_1C_3R_3R_4R_5+C_1C_5R_3R_4R_5+C_3C_5R_3R_4R_5}}$
K-LP: $\frac{R_3R_4R_5g_m-R_3R_4}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}$
K-HP: 0
K-BP: $-\frac{C_5R_3R_4R_5}{C_1R_3R_4+2C_1R_3R_5+C_1R_4R_5+C_3R_3R_4R_5g_m+C_3R_3R_4+2C_5R_3R_4R_5g_m+2C_5R_3R_5+C_5R_4R_5}$
Qz: None
Wz: None

8.42 X-INVALID-NUMER-42 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 s + R_3 g_m}{g_m + s^2 (C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_5 R_3 + C_3 C_5 R_3 + 2C_4 C_5 R_3) + s (C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5)}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{g_m} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}}{C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_5 R_3 + C_3 C_5 R_3 + 2C_4 C_5 R_3}} \frac{C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5}{\sqrt{R_3} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_5 R_3 + C_3 C_5 R_3 + 2C_4 C_5 R_3}}$
 K-LP: R_3
 K-HP: 0
 K-BP: $-\frac{C_5 R_3}{C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5}$
 Qz: None
 Wz: None

8.43 X-INVALID-NUMER-43 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_5 s + R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + s^2 (C_1 C_3 R_3 R_5 + 2C_1 C_4 R_3 R_5 + C_1 C_5 R_3 R_5 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_5) + s (C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{R_5} \sqrt{2R_3 g_m + R_5 g_m + 1} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}}{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{2R_3 g_m + R_5 g_m + 1}}{\sqrt{C_1 C_3 R_3 R_5 + 2C_1 C_4 R_3 R_5 + C_1 C_5 R_3 R_5 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_5}} \frac{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}{\sqrt{R_3} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_3 R_5 + 2C_1 C_4 R_3 R_5 + C_1 C_5 R_3 R_5 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_5}}$
 K-LP: $\frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_3 R_5}{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}$
 Qz: None
 Wz: None

8.44 X-INVALID-NUMER-44 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 s + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^2 (C_1 C_3 R_3 R_4 + 2C_1 C_4 R_3 R_4 + C_1 C_5 R_3 R_4 + C_3 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4) + s (2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4)}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{2R_3 + R_4} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}}{2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4}$
 wo: $\frac{\sqrt{2R_3 g_m + R_4 g_m}}{\sqrt{C_1 C_3 R_3 R_4 + 2C_1 C_4 R_3 R_4 + C_1 C_5 R_3 R_4 + C_3 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4}} \frac{\sqrt{2R_3 g_m + R_4 g_m} (2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4)}{\sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{2R_3 + R_4} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_3 R_4 + 2C_1 C_4 R_3 R_4 + C_1 C_5 R_3 R_4 + C_3 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4}}$
 K-LP: $\frac{R_3 R_4}{2R_3 + R_4}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_3 R_4}{2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4}$
 Qz: None
 Wz: None

8.45 X-INVALID-NUMER-45 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 R_5 s + R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s^2 (C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_3 R_4 R_5 + C_1 C_5 R_3 R_4 R_5 + C_3 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5) + s (C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_4 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}$
 wo: $\frac{\sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{\sqrt{C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_3 R_4 R_5 + C_1 C_5 R_3 R_4 R_5 + C_3 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5}} \frac{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}{\sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_3 R_4 R_5 + C_1 C_5 R_3 R_4 R_5 + C_3 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5}}$
 bandwidth: $\frac{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}{\sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_3 R_4 R_5 + C_1 C_5 R_3 R_4 R_5 + C_3 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5}}$

K-LP: $\frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}$
K-HP: 0
K-BP: $-\frac{C_5 R_3 R_4 R_5}{C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}$
Qz: None
Wz: None

8.46 X-INVALID-NUMER-46 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s (C_3 R_3 R_4 R_5 g_m - C_3 R_3 R_4)}{2R_4 g_m + 2R_5 g_m + s^2 (C_1 C_3 R_3 R_4 + 2C_1 C_3 R_3 R_5 + C_1 C_3 R_4 R_5) + s (C_1 R_4 + 2C_1 R_5 + 2C_3 R_3 R_4 g_m + 2C_3 R_3 R_5 g_m + 2C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{R_3 R_4 + 2R_3 R_5 + R_4 R_5}\sqrt{R_4 g_m + R_5 g_m + 1}}{C_1 R_4 + 2C_1 R_5 + 2C_3 R_3 R_4 g_m + 2C_3 R_3 R_5 g_m + 2C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4}$
wo: $\frac{\sqrt{2R_4 g_m + 2R_5 g_m + 2}}{\sqrt{C_1 C_3 R_3 R_4 + 2C_1 C_3 R_3 R_5 + C_1 C_3 R_4 R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_4 g_m + 2R_5 g_m + 2}(C_1 R_4 + 2C_1 R_5 + 2C_3 R_3 R_4 g_m + 2C_3 R_3 R_5 g_m + 2C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4)}{2\sqrt{C_1}\sqrt{C_3}\sqrt{R_3 R_4 + 2R_3 R_5 + R_4 R_5}\sqrt{R_4 g_m + R_5 g_m + 1}\sqrt{C_1 C_3 R_3 R_4 + 2C_1 C_3 R_3 R_5 + C_1 C_3 R_4 R_5}}$
K-LP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$
K-HP: 0
K-BP: $\frac{C_3 R_3 R_4 R_5 g_m - C_3 R_3 R_4}{C_1 R_4 + 2C_1 R_5 + 2C_3 R_3 R_4 g_m + 2C_3 R_3 R_5 g_m + 2C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4}$
Qz: None
Wz: None

8.47 X-INVALID-NUMER-47 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 s + R_1 R_3 R_4 g_m}{C_1 C_5 R_1 R_3 R_4 s^2 + 2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}{C_1 C_5 R_1 R_3 R_4}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4}{2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
Qz: None
Wz: None

8.48 X-INVALID-NUMER-48 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 R_5 s + R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{C_1 C_5 R_1 R_3 R_4 R_5 s^2 + 2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s (C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}{C_1 C_5 R_1 R_3 R_4 R_5}$
K-LP: $\frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4 R_5}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
Qz: None
Wz: None

8.49 X-INVALID-NUMER-49 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m + s (C_5 R_1 R_3 R_4 R_5 g_m - C_5 R_1 R_3 R_4)}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^2 (C_1 C_5 R_1 R_3 R_4 + 2 C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_1 R_4 R_5) + s (2 C_1 R_1 R_3 + C_1 R_1 R_4 + 2 C_5 R_1 R_3 R_4 g_m + 2 C_5 R_1 R_3 R_5 g_m + 2 C_5 R_1 R_3 + C_5 R_1 R_4 R_5 g_m + C_5 R_1 R_4 + C_5 R_3 R_4 + 2 C_5 R_3 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3 R_4 + 2 R_3 R_5 + R_4 R_5} \sqrt{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}}{2 C_1 R_1 R_3 + C_1 R_1 R_4 + 2 C_5 R_1 R_3 R_4 g_m + 2 C_5 R_1 R_3 R_5 g_m + 2 C_5 R_1 R_3 + C_5 R_1 R_4 R_5 g_m + C_5 R_1 R_4 + C_5 R_3 R_4 + 2 C_5 R_3 R_5 + C_5 R_4 R_5}$
wo: $\frac{\sqrt{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}}{\sqrt{C_1 C_5 R_1 R_3 R_4 + 2 C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_1 R_4 R_5}}$
bandwidth: $\frac{2 C_1 R_1 R_3 + C_1 R_1 R_4 + 2 C_5 R_1 R_3 R_4 g_m + 2 C_5 R_1 R_3 R_5 g_m + 2 C_5 R_1 R_3 + C_5 R_1 R_4 R_5 g_m + C_5 R_1 R_4 + C_5 R_3 R_4 + 2 C_5 R_3 R_5 + C_5 R_4 R_5}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3 R_4 + 2 R_3 R_5 + R_4 R_5} \sqrt{C_1 C_5 R_1 R_3 R_4 + 2 C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_1 R_4 R_5}}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
K-HP: 0
K-BP: $\frac{C_5 R_1 R_3 R_4 R_5 g_m - C_5 R_1 R_3 R_4}{2 C_1 R_1 R_3 + C_1 R_1 R_4 + 2 C_5 R_1 R_3 R_4 g_m + 2 C_5 R_1 R_3 R_5 g_m + 2 C_5 R_1 R_3 + C_5 R_1 R_4 R_5 g_m + C_5 R_1 R_4 + C_5 R_3 R_4 + 2 C_5 R_3 R_5 + C_5 R_4 R_5}$
Qz: None
Wz: None

8.50 X-INVALID-NUMER-50 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 s + R_1 R_3 g_m}{R_1 g_m + s^2 (2 C_1 C_4 R_1 R_3 + C_1 C_5 R_1 R_3 + 2 C_4 C_5 R_1 R_3) + s (C_1 R_1 + 2 C_4 R_1 R_3 g_m + 2 C_4 R_3 + 2 C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_1 g_m + 1} \sqrt{2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5}}{C_1 R_1 + 2 C_4 R_1 R_3 g_m + 2 C_4 R_3 + 2 C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{2 C_1 C_4 R_1 R_3 + C_1 C_5 R_1 R_3 + 2 C_4 C_5 R_1 R_3}}$
bandwidth: $\frac{C_1 R_1 + 2 C_4 R_1 R_3 g_m + 2 C_4 R_3 + 2 C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}{\sqrt{R_1} \sqrt{R_3} \sqrt{2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5} \sqrt{2 C_1 C_4 R_1 R_3 + C_1 C_5 R_1 R_3 + 2 C_4 C_5 R_1 R_3}}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3}{C_1 R_1 + 2 C_4 R_1 R_3 g_m + 2 C_4 R_3 + 2 C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}$
Qz: None
Wz: None

8.51 X-INVALID-NUMER-51 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_5 s + R_1 R_3 R_5 g_m - R_1 R_3}{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2 (2 C_1 C_4 R_1 R_3 R_5 + C_1 C_5 R_1 R_3 R_5 + 2 C_4 C_5 R_1 R_3 R_5) + s (C_1 R_1 R_3 + C_1 R_1 R_5 + 2 C_4 R_1 R_3 R_5 g_m + 2 C_4 R_1 R_3 + 2 C_4 R_3 R_5 + 2 C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5} \sqrt{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{C_1 R_1 R_3 + C_1 R_1 R_5 + 2 C_4 R_1 R_3 R_5 g_m + 2 C_4 R_1 R_3 + 2 C_4 R_3 R_5 + 2 C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}$
wo: $\frac{\sqrt{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{\sqrt{2 C_1 C_4 R_1 R_3 R_5 + C_1 C_5 R_1 R_3 R_5 + 2 C_4 C_5 R_1 R_3 R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_5 + 2 C_4 R_1 R_3 R_5 g_m + 2 C_4 R_1 R_3 + 2 C_4 R_3 R_5 + 2 C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}{\sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5} \sqrt{2 C_1 C_4 R_1 R_3 R_5 + C_1 C_5 R_1 R_3 R_5 + 2 C_4 C_5 R_1 R_3 R_5}}$
K-LP: $\frac{R_1 R_3 R_5 g_m - R_1 R_3}{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_5}{C_1 R_1 R_3 + C_1 R_1 R_5 + 2 C_4 R_1 R_3 R_5 g_m + 2 C_4 R_1 R_3 + 2 C_4 R_3 R_5 + 2 C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}$
Qz: None
Wz: None

8.52 X-INVALID-NUMER-52 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 s + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^2 (2 C_1 C_4 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + 2 C_4 C_5 R_1 R_3 R_4) + s (2 C_1 R_1 R_3 + C_1 R_1 R_4 + 2 C_4 R_1 R_3 R_4 g_m + 2 C_4 R_3 R_4 + 2 C_5 R_1 R_3 R_4 g_m + 2 C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5} \sqrt{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}}{2 C_1 R_1 R_3 + C_1 R_1 R_4 + 2 C_4 R_1 R_3 R_4 g_m + 2 C_4 R_3 R_4 + 2 C_5 R_1 R_3 R_4 g_m + 2 C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
wo: $\frac{\sqrt{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}}{\sqrt{2 C_1 C_4 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + 2 C_4 C_5 R_1 R_3 R_4}}$
bandwidth: $\frac{2 C_1 R_1 R_3 + C_1 R_1 R_4 + 2 C_4 R_1 R_3 R_4 g_m + 2 C_4 R_3 R_4 + 2 C_5 R_1 R_3 R_4 g_m + 2 C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5} \sqrt{2 C_1 C_4 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + 2 C_4 C_5 R_1 R_3 R_4}}$

K-LP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4}{2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
Qz: None
Wz: None

8.53 X-INVALID-NUMER-53 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 R_5 s + R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s^2 (2C_1 C_4 R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5) + s (C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5} \sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{\sqrt{2C_1 C_4 R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5}}$
bandwidth: $\frac{\sqrt{2C_1 C_4 R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5}}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
K-LP: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5} \sqrt{2R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5}}{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4 R_5}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
Qz: None
Wz: None

8.54 X-INVALID-NUMER-54 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3 + s (C_4 R_1 R_3 R_4 R_5 g_m - C_4 R_1 R_3 R_4)}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2 (C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_5 + C_1 C_4 R_1 R_4 R_5) + s (C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + C_4 R_1 R_4 R_5 g_m + C_4 R_1 R_4 + C_4 R_3 R_4 + 2C_4 R_3 R_5 + C_4 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_3 R_4 + 2R_3 R_5 + R_4 R_5} \sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + C_4 R_1 R_4 R_5 g_m + C_4 R_1 R_4 + C_4 R_3 R_4 + 2C_4 R_3 R_5 + C_4 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{\sqrt{C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_5 + C_1 C_4 R_1 R_4 R_5}}$
bandwidth: $\frac{\sqrt{C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_5 + C_1 C_4 R_1 R_4 R_5}}{C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + C_4 R_1 R_4 R_5 g_m + C_4 R_1 R_4 + C_4 R_3 R_4 + 2C_4 R_3 R_5 + C_4 R_4 R_5}$
K-LP: $\frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}$
K-HP: 0
K-BP: $\frac{C_4 R_1 R_3 R_4 R_5 g_m - C_4 R_1 R_3 R_4}{C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + C_4 R_1 R_4 R_5 g_m + C_4 R_1 R_4 + C_4 R_3 R_4 + 2C_4 R_3 R_5 + C_4 R_4 R_5}$
Qz: None
Wz: None

8.55 X-INVALID-NUMER-55 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{2R_1 g_m + s^2 (C_1 C_3 R_1 R_4 + C_1 C_5 R_1 R_4 + C_3 C_5 R_1 R_4) + s (2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{R_1} \sqrt{R_4} \sqrt{R_1 g_m + 1} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5}}{2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}$
wo: $\frac{\sqrt{2R_1 g_m + 2}}{\sqrt{C_1 C_3 R_1 R_4 + C_1 C_5 R_1 R_4 + C_3 C_5 R_1 R_4}}$
bandwidth: $\frac{\sqrt{2} \sqrt{2R_1 g_m + 2} (2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4)}{2\sqrt{R_1} \sqrt{R_4} \sqrt{R_1 g_m + 1} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_1 R_4 + C_1 C_5 R_1 R_4 + C_3 C_5 R_1 R_4}}$
K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_4}{2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}$
Qz: None
Wz: None

8.56 X-INVALID-NUMER-56 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2 (C_1 C_3 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + C_3 C_5 R_1 R_4 R_5) + s (C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
 wo: $\frac{\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{\sqrt{C_1 C_3 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + C_3 C_5 R_1 R_4 R_5}}$
 bandwidth: $\frac{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}{\sqrt{R_1} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + C_3 C_5 R_1 R_4 R_5}}$
 K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_4 R_5}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
 Qz: None
 Wz: None

8.57 X-INVALID-NUMER-57 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{2R_1 g_m + s^2 (C_1 C_3 R_1 R_5 + 2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_5) + s (C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_5} \sqrt{2R_1 g_m + 1} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}}{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1 C_3 R_1 R_5 + 2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_5}}$
 bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}{\sqrt{R_1} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_1 R_5 + 2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_5}}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_5}{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 Qz: None
 Wz: None

8.58 X-INVALID-NUMER-58 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{2R_1 g_m + s^2 (C_1 C_3 R_1 R_4 + 2C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + C_3 C_5 R_1 R_4 + 2C_4 C_5 R_1 R_4) + s (2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{R_1} \sqrt{R_4} \sqrt{R_1 g_m + 1} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}}{2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}$
 wo: $\frac{\sqrt{2R_1 g_m + 2}}{\sqrt{C_1 C_3 R_1 R_4 + 2C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + C_3 C_5 R_1 R_4 + 2C_4 C_5 R_1 R_4}}$
 bandwidth: $\frac{\sqrt{2} \sqrt{2R_1 g_m + 2} (2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4)}{2\sqrt{R_1} \sqrt{R_4} \sqrt{R_1 g_m + 1} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_1 R_4 + 2C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + C_3 C_5 R_1 R_4 + 2C_4 C_5 R_1 R_4}}$
 K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_4}{2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}$
 Qz: None
 Wz: None

8.59 X-INVALID-NUMER-59 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2 (C_1 C_3 R_1 R_4 R_5 + 2C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + C_3 C_5 R_1 R_4 R_5 + 2C_4 C_5 R_1 R_4 R_5) + s (C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
 wo: $\frac{\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{\sqrt{C_1 C_3 R_1 R_4 R_5 + 2C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + C_3 C_5 R_1 R_4 R_5 + 2C_4 C_5 R_1 R_4 R_5}}$

bandwidth: $\frac{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}{\sqrt{R_1} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_1 R_4 R_5 + 2C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + C_3 C_5 R_1 R_4 R_5 + 2C_4 C_5 R_1 R_4 R_5}}$
K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_4 R_5}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
Qz: None
Wz: None

8.60 X-INVALID-NUMER-60 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 s + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + C_3 C_5 R_1 R_3 R_4) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{\sqrt{C_1 C_3 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + C_3 C_5 R_1 R_3 R_4}}$
bandwidth: $\frac{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + C_3 C_5 R_1 R_3 R_4}}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4}{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
Qz: None
Wz: None

8.61 X-INVALID-NUMER-61 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 R_5 s + R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + C_3 C_5 R_1 R_3 R_4 R_5) + s (C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{\sqrt{C_1 C_3 R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + C_3 C_5 R_1 R_3 R_4 R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + C_3 C_5 R_1 R_3 R_4 R_5}}$
K-LP: $\frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4 R_5}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
Qz: None
Wz: None

8.62 X-INVALID-NUMER-62 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 s + R_1 R_3 g_m}{R_1 g_m + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_1 C_5 R_1 R_3 + C_3 C_5 R_1 R_3 + 2C_4 C_5 R_1 R_3) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_1 g_m + 1} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}}{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_1 C_5 R_1 R_3 + C_3 C_5 R_1 R_3 + 2C_4 C_5 R_1 R_3}}$
bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}{\sqrt{R_1} \sqrt{R_3} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_1 C_5 R_1 R_3 + C_3 C_5 R_1 R_3 + 2C_4 C_5 R_1 R_3}}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3}{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}$
Qz: None
Wz: None

8.63 X-INVALID-NUMER-63 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_5 s + R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2 (C_1 C_3 R_1 R_3 R_5 + 2C_1 C_4 R_1 R_3 R_5 + C_1 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_5) + s (C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_3 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{\sqrt{C_1 C_3 R_1 R_3 R_5 + 2C_1 C_4 R_1 R_3 R_5 + C_1 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}{\sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_1 R_3 R_5 + 2C_1 C_4 R_1 R_3 R_5 + C_1 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_5}}$
K-LP: $\frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_5}{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}$
Qz: None
Wz: None

8.64 X-INVALID-NUMER-64 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 s + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + C_3 C_5 R_1 R_3 R_4 + 2C_4 C_5 R_1 R_3 R_4) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5}}{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{\sqrt{C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + C_3 C_5 R_1 R_3 R_4 + 2C_4 C_5 R_1 R_3 R_4}}$
bandwidth: $\frac{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + C_3 C_5 R_1 R_3 R_4 + 2C_4 C_5 R_1 R_3 R_4}}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4}{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4}$
Qz: None
Wz: None

8.65 X-INVALID-NUMER-65 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 R_5 s + R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 + 2C_1 C_4 R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + C_3 C_5 R_1 R_3 R_4 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5) + s (C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}}{\sqrt{C_1 C_3 R_1 R_3 R_4 R_5 + 2C_1 C_4 R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + C_3 C_5 R_1 R_3 R_4 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}{\sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5} \sqrt{C_1 C_3 R_1 R_3 R_4 R_5 + 2C_1 C_4 R_1 R_3 R_4 R_5 + C_1 C_5 R_1 R_3 R_4 R_5 + C_3 C_5 R_1 R_3 R_4 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5}}$
K-LP: $\frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_4 R_5}{C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5 + C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5}$
Qz: None
Wz: None

8.66 X-INVALID-NUMER-66 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s (C_3 R_1 R_3 R_4 R_5 g_m - C_3 R_1 R_3 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2 (C_1 C_3 R_1 R_3 R_4 + 2C_1 C_3 R_1 R_3 R_5 + C_1 C_3 R_1 R_4 R_5) + s (C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_3 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_3 R_4 + 2C_3 R_3 R_5 + C_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3 R_4 + 2R_3 R_5 + R_4 R_5} \sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_3 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_3 R_4 + 2C_3 R_3 R_5 + C_3 R_4 R_5}$
wo: $\frac{\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{\sqrt{C_1 C_3 R_1 R_3 R_4 + 2C_1 C_3 R_1 R_3 R_5 + C_1 C_3 R_1 R_4 R_5}}$
bandwidth: $\frac{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_3 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_3 R_4 + 2C_3 R_3 R_5 + C_3 R_4 R_5}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3 R_4 + 2R_3 R_5 + R_4 R_5} \sqrt{C_1 C_3 R_1 R_3 R_4 + 2C_1 C_3 R_1 R_3 R_5 + C_1 C_3 R_1 R_4 R_5}}$

K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
K-HP: 0
K-BP: $\frac{C_3 R_1 R_3 R_4 R_5 g_m - C_3 R_1 R_3 R_4}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_3 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_3 R_4 + 2C_3 R_3 R_5 + C_3 R_4 R_5}$
Qz: None
Wz: None

8.67 X-INVALID-NUMER-67 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s(C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3)}{2R_3 g_m + R_5 g_m + s^2(2C_1 C_4 R_1 R_3 R_5 g_m + 2C_1 C_4 R_1 R_3 + 2C_1 C_4 R_3 R_5) + s(2C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_5 g_m + 2C_4 R_3) + 1}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}R_5g_m\sqrt{\frac{2R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{\frac{2R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_3}R_5\sqrt{\frac{2R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}}{2C_1R_1R_3g_m+C_1R_1R_5g_m+C_1R_1+C_1R_3+C_1R_5+2C_4R_3R_5g_m+2C_4R_3}$$

$$\text{wo: } \sqrt{\frac{2R_3g_m+R_5g_m+1}{2C_1C_4R_1R_3R_5g_m+2C_1C_4R_1R_3+2C_1C_4R_3R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_3g_m+R_5g_m+1}{2C_1C_4R_1R_3R_5g_m+2C_1C_4R_1R_3+2C_1C_4R_3R_5}}(2C_1R_1R_3g_m+C_1R_1R_5g_m+C_1R_1+C_1R_3+C_1R_5+2C_4R_3R_5g_m+2C_4R_3)}{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}R_5g_m\sqrt{\frac{2R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{\frac{2R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_3}R_5\sqrt{\frac{2R_3g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}}$$

$$\text{K-LP: } \frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + 1}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3}{2C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_5 g_m + 2C_4 R_3}$$

$$\text{Qz: } \text{None}$$

$$\text{Wz: } \text{None}$$

8.68 X-INVALID-NUMER-68 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4 + s(C_1 R_1 R_3 R_4 R_5 g_m - C_1 R_1 R_3 R_4)}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s^2(2C_1 C_4 R_1 R_3 R_4 R_5 g_m + 2C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_3 R_4 R_5) + s(2C_1 R_1 R_3 R_4 g_m + 2C_1 R_1 R_3 R_5 g_m + 2C_1 R_1 R_3 + C_1 R_1 R_4 R_5 g_m + C_1 R_1 R_4 + C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}}{2C_1R_1R_3R_4g_m+2C_1R_1R_3R_5g_m+2C_1R_1R_3+C_1R_1R_4R_5g_m+C_1R_1R_4+C_1R_3R_4+2C_1R_3R_5+C_1R_4R_5+2C_4R_3R_4R_5g_m+2C_4R_3R_4}$$

$$\text{wo: } \sqrt{\frac{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}{2C_1C_4R_1R_3R_4R_5g_m+2C_1C_4R_1R_3R_4+2C_1C_4R_3R_4R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}{2C_1C_4R_1R_3R_4R_5g_m+2C_1C_4R_1R_3R_4+2C_1C_4R_3R_4R_5}}(2C_1R_1R_3R_4g_m+2C_1R_1R_3R_5g_m+2C_1R_1R_3+C_1R_1R_4R_5g_m+C_1R_1R_4+C_1R_3R_4+2C_1R_3R_5+C_1R_4R_5+2C_4R_3R_4R_5g_m+2C_4R_3R_4)}{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}}$$

$$\text{K-LP: } \frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_1 R_1 R_3 R_4 R_5 g_m - C_1 R_1 R_3 R_4}{2C_1 R_1 R_3 R_4 g_m + 2C_1 R_1 R_3 R_5 g_m + 2C_1 R_1 R_3 + C_1 R_1 R_4 R_5 g_m + C_1 R_1 R_4 + C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4}$$

$$\text{Qz: } \text{None}$$

$$\text{Wz: } \text{None}$$

8.69 X-INVALID-NUMER-69 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s(C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2R_4 g_m + 2R_5 g_m + s^2(C_1 C_3 R_1 R_4 R_5 g_m + C_1 C_3 R_1 R_4 + C_1 C_3 R_4 R_5) + s(2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4) + 2}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_4}\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{R_4}R_5\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}}{2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+C_3R_4R_5g_m+C_3R_4}$$

$$\text{wo: } \sqrt{\frac{2R_4g_m+2R_5g_m+2}{C_1C_3R_1R_4R_5g_m+C_1C_3R_1R_4+C_1C_3R_4R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_4g_m+2R_5g_m+2}{C_1C_3R_1R_4R_5g_m+C_1C_3R_1R_4+C_1C_3R_4R_5}}(2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+C_3R_4R_5g_m+C_3R_4)}{\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_4}\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{R_4}R_5\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}}$$

$$\text{K-LP: } \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4}{2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4}$$

$$\text{Qz: } \text{None}$$

$$\text{Wz: } \text{None}$$

8.70 X-INVALID-NUMER-70 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1) - 1}{2g_m + s^2 (C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + 2C_1 C_4 R_5) + s (2C_1 R_1 g_m + C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}C_3R_1R_5g_m\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\sqrt{2}\sqrt{C_1}C_3R_5\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+2\sqrt{2}\sqrt{C_1}C_4R_1R_5g_m\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}}{2C_1R_1g_m+C_1+C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}$

wo: $\sqrt{2}\sqrt{\frac{g_m}{C_1C_3R_1R_5g_m+C_1C_3R_1+C_1C_3R_5+2C_1C_4R_1R_5g_m+2C_1C_4R_1+2C_1C_4R_5}}$

bandwidth: $\frac{\sqrt{2}\sqrt{C_1}C_3R_1R_5g_m\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\sqrt{2}\sqrt{C_1}C_3R_5\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+2\sqrt{2}\sqrt{C_1}C_4R_1R_5g_m\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}}{\sqrt{2}\sqrt{C_1}C_3R_1R_5g_m\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\sqrt{2}\sqrt{C_1}C_3R_5\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+2\sqrt{2}\sqrt{C_1}C_4R_1R_5g_m\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{\frac{g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}}$

K-LP: $\frac{R_5g_m-1}{2g_m}$

K-HP: 0

K-BP: $\frac{C_1R_1R_5g_m-C_1R_1}{2C_1R_1g_m+C_1+C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}$

Qz: None

Wz: None

8.71 X-INVALID-NUMER-71 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2R_4 g_m + 2R_5 g_m + s^2 (C_1 C_3 R_1 R_4 R_5 g_m + C_1 C_3 R_1 R_4 + C_1 C_3 R_4 R_5 + 2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 R_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_4g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\frac{R_5g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4}\sqrt{\frac{R_4g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\frac{R_5g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}}{\sqrt{\frac{2R_4g_m+2R_5g_m+2}{C_1C_3R_1R_4R_5g_m+C_1C_3R_1R_4+C_1C_3R_4R_5+2C_1C_4R_1R_4R_5g_m+2C_1C_4R_1R_4+2C_1C_4R_4R_5}}}$

wo: $\sqrt{\frac{2R_4g_m+2R_5g_m+2}{C_1C_3R_1R_4R_5g_m+C_1C_3R_1R_4+C_1C_3R_4R_5+2C_1C_4R_1R_4R_5g_m+2C_1C_4R_1R_4+2C_1C_4R_4R_5}}$

bandwidth: $\frac{\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_4g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\frac{R_5g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4}\sqrt{\frac{R_4g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\frac{R_5g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}}{\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_4g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}+\frac{R_5g_m}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}+\frac{1}{C_3R_1R_5g_m+C_3R_1+C_3R_5+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5}}}$

K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$

K-HP: 0

K-BP: $\frac{C_1R_1R_4R_5g_m-C_1R_1R_4}{2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+C_3R_4R_5g_m+C_3R_4+2C_4R_4R_5g_m+2C_4R_4}$

Qz: None

Wz: None

8.72 X-INVALID-NUMER-72 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4 + s (C_1 R_1 R_3 R_4 R_5 g_m - C_1 R_1 R_3 R_4)}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 g_m + C_1 C_3 R_1 R_3 R_4 + C_1 C_3 R_3 R_4 R_5) + s (2C_1 R_1 R_3 R_4 g_m + 2C_1 R_1 R_3 R_5 g_m + 2C_1 R_1 R_3 + C_1 R_1 R_4 R_5 g_m + C_1 R_1 R_4 + C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}}{2C_1R_1R_3R_4g_m+2C_1R_1R_3R_5g_m+2C_1R_1R_3+C_1R_1R_4R_5g_m+C_1R_1R_4+C_1R_3R_4+2C_1R_3R_5+C_1R_4R_5+C_3R_3R_4R_5g_m+C_3R_3R_4}$

wo: $\sqrt{\frac{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}{C_1C_3R_1R_3R_4R_5g_m+C_1C_3R_1R_3R_4+C_1C_3R_3R_4R_5}}$

bandwidth: $\frac{\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}R_5\sqrt{\frac{2R_3R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{2R_3}{R_1R_5g_m+R_1+R_5}+\frac{R_4R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_4}{R_1R_5g_m+R_1+R_5}}}{\sqrt{\frac{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}{C_1C_3R_1R_3R_4R_5g_m+C_1C_3R_1R_3R_4+C_1C_3R_3R_4R_5}}(2C_1R_1R_3R_4g_m+2C_1R_1R_3R_5g_m+2C_1R_1R_3+C_1R_1R_4R_5g_m+C_1R_1R_4+C_1R_3R_4+2C_1R_3R_5+C_1R_4R_5+C_3R_3R_4R_5g_m+C_3R_3R_4)}$

K-LP: $\frac{R_3R_4R_5g_m-R_3R_4}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}$

K-HP: 0

K-BP: $\frac{C_1R_1R_3R_4R_5g_m-C_1R_1R_3R_4}{2C_1R_1R_3R_4g_m+2C_1R_1R_3R_5g_m+2C_1R_1R_3+C_1R_1R_4R_5g_m+C_1R_1R_4+C_1R_3R_4+2C_1R_3R_5+C_1R_4R_5+C_3R_3R_4R_5g_m+C_3R_3R_4}$

Qz: None

Wz: None

8.73 X-INVALID-NUMER-73 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s (C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3)}{2R_3 g_m + R_5 g_m + s^2 (C_1 C_3 R_1 R_3 R_5 g_m + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 R_5 + 2C_1 C_4 R_1 R_3 R_5 g_m + 2C_1 C_4 R_1 R_3 + 2C_1 C_4 R_3 R_5) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} C_3 R_1 \sqrt{R_3} R_5 g_m \sqrt{\frac{2R_3 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}} + \sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{\frac{2R_3 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}}}$

wo: $\sqrt{\frac{2R_3 g_m + R_5 g_m + 1}{C_1 C_3 R_1 R_3 R_5 g_m + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 R_5 + 2C_1 C_4 R_1 R_3 R_5 g_m + 2C_1 C_4 R_1 R_3 + 2C_1 C_4 R_3 R_5}}$

bandwidth: $\frac{\sqrt{C_1} C_3 R_1 \sqrt{R_3} R_5 g_m \sqrt{\frac{2R_3 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}} + \sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{\frac{2R_3 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{1}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}}}$

K-LP: $\frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + 1}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3}{2C_1 R_1 R_3 R_5 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3}$

Qz: None

Wz: None

8.74 X-INVALID-NUMER-74 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4 + s (C_1 R_1 R_3 R_4 R_5 g_m - C_1 R_1 R_3 R_4)}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 g_m + C_1 C_3 R_1 R_3 R_4 + C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_1 R_3 R_4 R_5 g_m + 2C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_3 R_4 R_5) + s (2C_1 R_1 R_3 R_4 g_m + 2C_1 R_1 R_3 R_5 g_m + 2C_1 R_1 R_3 + C_1 R_1 R_4 R_5 g_m + C_1 R_1 R_4 + C_1 R_3 R_4 + 2C_4 R_3 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_4} R_5 g_m \sqrt{\frac{2R_3 R_4 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{2R_3 R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{2R_3}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_4 R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_4}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}} + \sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{2R_3 R_4 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{2R_3 R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{2R_3}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_4 R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_4}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}}}$

wo: $\sqrt{\frac{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}{C_1 C_3 R_1 R_3 R_4 R_5 g_m + C_1 C_3 R_1 R_3 R_4 + C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_1 R_3 R_4 R_5 g_m + 2C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_3 R_4 R_5}}$

bandwidth: $\frac{\sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_4} R_5 g_m \sqrt{\frac{2R_3 R_4 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{2R_3 R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{2R_3}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_4 R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_4}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}} + \sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{2R_3 R_4 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{2R_3 R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{2R_3}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_4 R_5 g_m}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5} + \frac{R_4}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}}}$

K-LP: $\frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_3 R_4 R_5 g_m - C_1 R_1 R_3 R_4}{2C_1 R_1 R_3 R_4 g_m + 2C_1 R_1 R_3 R_5 g_m + 2C_1 R_1 R_3 + C_1 R_1 R_4 R_5 g_m + C_1 R_1 R_4 + C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5 + C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4}$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, R_3, R_4, R_5, \infty)$

$$H(s) = \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \infty, R_3, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 s + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s (2C_5 R_1 R_3 R_4 g_m + 2C_5 R_1 R_3 + C_5 R_1 R_4 + C_5 R_3 R_4)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \infty, R_3, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_4 R_5 s + R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s (2C_5 R_1 R_3 R_4 R_5 g_m + 2C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5 + C_5 R_3 R_4 R_5)}$$

$$\mathbf{9.4 \quad X-INVALID-ORDER-4} \quad Z(s) = \left(R_1, \infty, R_3, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 g_m + s (C_5 R_1 R_3 R_4 R_5 g_m - C_5 R_1 R_3 R_4)}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s (2 C_5 R_1 R_3 R_4 g_m + 2 C_5 R_1 R_3 R_5 g_m + 2 C_5 R_1 R_3 + C_5 R_1 R_4 R_5 g_m + C_5 R_1 R_4 + C_5 R_3 R_4 + 2 C_5 R_3 R_5 + C_5 R_4 R_5)}$$

$$\mathbf{9.5 \quad X-INVALID-ORDER-5} \quad Z(s) = \left(R_1, \infty, R_3, \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3}{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s (2 C_4 R_1 R_3 R_5 g_m + 2 C_4 R_1 R_3 + 2 C_4 R_3 R_5)}$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(R_1, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2 R_1 R_3 R_4 g_m + 2 R_1 R_3 R_5 g_m + 2 R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2 R_3 R_5 + R_4 R_5 + s (2 C_4 R_1 R_3 R_4 R_5 g_m + 2 C_4 R_1 R_3 R_4 + 2 C_4 R_3 R_4 R_5)}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(R_1, \infty, R_3, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3 + s (C_4 R_1 R_3 R_4 R_5 g_m - C_4 R_1 R_3 R_4)}{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s (2 C_4 R_1 R_3 R_4 g_m + 2 C_4 R_1 R_3 R_5 g_m + 2 C_4 R_1 R_3 + C_4 R_1 R_4 R_5 g_m + C_4 R_1 R_4 + C_4 R_3 R_4 + 2 C_4 R_3 R_5 + C_4 R_4 R_5)}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s (C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5)}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + s (C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5) + 1}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{s^2 (C_3 C_5 R_1 + 2 C_4 C_5 R_1) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_5 R_1 R_5 g_m - C_5 R_1)}{s^2 (C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_1 + C_3 C_5 R_5 + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + 2 C_4 C_5 R_5) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s (C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_4 R_5 + 2 C_4 R_1 R_4 R_5 g_m + 2 C_4 R_1 R_4 + 2 C_4 R_4 R_5)}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 s^2 + R_1 g_m + s (C_4 R_1 R_4 g_m - C_5 R_1)}{C_3 C_4 C_5 R_1 R_4 s^3 + s^2 (C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4 + C_3 C_5 R_1 + 2 C_4 C_5 R_1 R_4 g_m + 2 C_4 C_5 R_1 + C_4 C_5 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 R_5 s^2 + R_1 R_5 g_m - R_1 + s(C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 - C_5 R_1 R_5)}{C_3 C_4 C_5 R_1 R_4 R_5 s^3 + 2R_1 g_m + s^2(C_3 C_4 R_1 R_4 R_5 g_m + C_3 C_4 R_1 R_4 + C_3 C_4 R_4 R_5 + C_3 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_4 R_5 g_m + 2C_4 C_5 R_1 R_5 + C_4 C_5 R_4 R_5) + s(C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2(C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4) + s(C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^3(C_3 C_4 C_5 R_1 R_4 R_5 g_m + C_3 C_4 C_5 R_1 R_4 + C_3 C_4 C_5 R_4 R_5) + s^2(C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4 + C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_1 + C_3 C_5 R_5 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4 + 2C_4 C_5 R_5) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s(C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s(C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 + 2C_4 R_3 R_5)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s(C_3 R_1 R_3 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_3 R_4 R_5 + 2C_4 R_1 R_3 R_4 R_5 g_m + 2C_4 R_1 R_3 R_4 + 2C_4 R_3 R_4 R_5)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_3 R_4 s^2 + R_1 R_3 g_m + s(C_4 R_1 R_3 R_4 g_m - C_5 R_1 R_3)}{C_3 C_4 C_5 R_1 R_3 R_4 s^3 + R_1 g_m + s^2(C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4 + C_3 C_5 R_1 R_3 + 2C_4 C_5 R_1 R_3 R_4 g_m + 2C_4 C_5 R_1 R_3 + C_4 C_5 R_1 R_4 + C_4 C_5 R_3 R_4) + s(C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_3 R_4 R_5 s^2 + R_1 R_3 R_5 g_m - R_1 R_3 + s(C_4 R_1 R_3 R_4 R_5 g_m - C_4 R_1 R_3 R_4 - C_5 R_1 R_3 R_5)}{C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^3 + 2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2(C_3 C_4 R_1 R_3 R_4 R_5 g_m + C_3 C_4 R_1 R_3 R_4 + C_3 C_4 R_3 R_4 R_5 + C_3 C_5 R_1 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_4 C_5 R_1 R_3 R_5 + C_4 C_5 R_1 R_4 R_5 + C_4 C_5 R_3 R_4 R_5) + s(C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_5 g_m + 2C_4 R_1 R_3 R_5 + C_4 C_5 R_1 R_4 R_5 g_m + C_4 C_5 R_1 R_4 + C_4 C_5 R_3 R_4 R_5)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 R_3 g_m + s^2(C_4 C_5 R_1 R_3 R_4 R_5 g_m - C_4 C_5 R_1 R_3 R_4) + s(C_4 R_1 R_3 R_4 g_m + C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{R_1 g_m + s^3(C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m + C_3 C_4 C_5 R_1 R_3 R_4 + C_3 C_4 C_5 R_3 R_4 R_5) + s^2(C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4 + C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_3 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_4 g_m + 2C_4 C_5 R_1 R_3 R_5 g_m + 2C_4 C_5 R_1 R_3 + C_4 C_5 R_1 R_4 R_5 g_m + C_4 C_5 R_1 R_4 + C_4 C_5 R_3 R_4 R_5)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s(C_3 R_1 R_3 R_4 R_5 g_m - C_3 R_1 R_3 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_3 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_3 R_4 + 2C_3 R_3 R_5 + C_3 R_4 R_5)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 s^2 + R_1 g_m + s(C_3 R_1 R_3 g_m - C_5 R_1)}{2C_3 C_4 C_5 R_1 R_3 s^3 + s^2(2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3 + 2C_3 C_5 R_1 R_3 g_m + C_3 C_5 R_1 + C_3 C_5 R_3 + 2C_4 C_5 R_1) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_5 s^2 + R_1 R_5 g_m - R_1 + s(C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3 - C_5 R_1 R_5)}{2C_3 C_4 C_5 R_1 R_3 R_5 s^3 + 2R_1 g_m + s^2(2C_3 C_4 R_1 R_3 R_5 g_m + 2C_3 C_4 R_1 R_3 + 2C_3 C_4 R_3 R_5 + 2C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_5 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_1 R_5) + s(2C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2(C_3 C_5 R_1 R_3 R_5 g_m - C_3 C_5 R_1 R_3) + s(C_3 R_1 R_3 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^3(2C_3 C_4 C_5 R_1 R_3 R_5 g_m + 2C_3 C_4 C_5 R_1 R_3 + 2C_3 C_4 C_5 R_3 R_5) + s^2(2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3 + 2C_3 C_5 R_1 R_3 g_m + C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_1 + C_3 C_5 R_3 + C_3 C_5 R_5 + 2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + 2C_4 C_5 R_5) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5)}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_4 s^2 + R_1 R_4 g_m + s(C_3 R_1 R_3 R_4 g_m - C_5 R_1 R_4)}{2C_3 C_4 C_5 R_1 R_3 R_4 s^3 + 2R_1 g_m + s^2(2C_3 C_4 R_1 R_3 R_4 g_m + 2C_3 C_4 R_3 R_4 + 2C_3 C_5 R_1 R_3 R_4 g_m + 2C_3 C_5 R_1 R_3 + C_3 C_5 R_1 R_4 + C_3 C_5 R_3 R_4 + 2C_4 C_5 R_1 R_4) + s(2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_3 + C_5 R_4) + 2C_5 R_1 R_4}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_4 R_5 s^2 + R_1 R_4 R_5 g_m - R_1 R_4 + s(C_3 R_1 R_3 R_4 R_5 g_m - C_3 R_1 R_3 R_4 - C_5 R_1 R_4 R_5)}{2C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^3 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2(2C_3 C_4 R_1 R_3 R_4 R_5 g_m + 2C_3 C_4 R_1 R_3 R_4 + 2C_3 C_4 R_3 R_4 R_5 + 2C_3 C_5 R_1 R_3 R_4 R_5 g_m + 2C_3 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_4 R_5 + C_3 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_1 R_4 R_5) + s(2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_4 R_5 g_m + 2C_3 R_1 R_3 R_4 + 2C_3 R_1 R_3 R_5 + 2C_3 R_1 R_4 R_5 + 2C_3 R_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 + 2C_5 R_1 R_3 R_4 + 2C_5 R_1 R_3 R_5 + 2C_5 R_1 R_4 R_5)}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s^2(C_3 C_5 R_1 R_3 R_4 R_5 g_m - C_3 C_5 R_1 R_3 R_4) + s(C_3 R_1 R_3 R_4 g_m + C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s^3(2C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_3 C_4 C_5 R_1 R_3 R_4 + 2C_3 C_4 C_5 R_3 R_4 R_5) + s^2(2C_3 C_4 R_1 R_3 R_4 g_m + 2C_3 C_4 R_3 R_4 + 2C_3 C_5 R_1 R_3 R_4 g_m + 2C_3 C_5 R_1 R_3 R_5 g_m + 2C_3 C_5 R_1 R_3 + C_3 C_5 R_1 R_4 R_5 g_m + C_3 C_5 R_1 R_4 + C_3 C_5 R_3 R_4 + 2C_3 C_5 R_3 R_5 + C_3 C_5 R_4 R_5 + 2C_4 C_5 R_1 R_4 R_5) + s(2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_4 R_5 g_m + 2C_3 R_1 R_3 R_4 + 2C_3 R_1 R_3 R_5 + 2C_3 R_1 R_4 R_5 + 2C_3 R_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 + 2C_5 R_1 R_3 R_4 + 2C_5 R_1 R_3 R_5 + 2C_5 R_1 R_4 R_5)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_4 C_5 R_1 R_3 R_4 s^3 + R_1 g_m + s^2(C_3 C_4 R_1 R_3 R_4 g_m - C_3 C_5 R_1 R_3 - C_4 C_5 R_1 R_4) + s(C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m - C_5 R_1)}{s^3(2C_3 C_4 C_5 R_1 R_3 R_4 g_m + 2C_3 C_4 C_5 R_1 R_3 + C_3 C_4 C_5 R_1 R_4 + C_3 C_4 C_5 R_3 R_4) + s^2(2C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4 + 2C_3 C_5 R_1 R_3 g_m + C_3 C_5 R_1 + C_3 C_5 R_3 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5 R_1 + C_5 R_3 + C_5 R_4)}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^3 + R_1 R_5 g_m - R_1 + s^2(C_3 C_4 R_1 R_3 R_4 R_5 g_m - C_3 C_4 R_1 R_3 R_4 - C_3 C_5 R_1 R_3 R_5 - C_4 C_5 R_1 R_4 R_5)}{2R_1 g_m + s^3(2C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_3 C_4 C_5 R_1 R_3 R_5 + C_3 C_4 C_5 R_1 R_4 R_5 + C_3 C_4 C_5 R_3 R_4 R_5) + s^2(2C_3 C_4 R_1 R_3 R_4 g_m + 2C_3 C_4 R_1 R_3 R_5 g_m + 2C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 R_5 g_m + C_3 C_4 R_1 R_4 + C_3 C_4 R_3 R_4 + 2C_3 C_4 R_3 R_5 + C_3 C_4 R_4 R_5 + 2C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_4 R_5) + s(2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_4 R_5 g_m + 2C_3 R_1 R_3 R_4 + 2C_3 R_1 R_3 R_5 + 2C_3 R_1 R_4 R_5 + 2C_3 R_3 R_4 R_5 + 2C_4 R_1 R_4 R_5 + 2C_4 R_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 + 2C_5 R_1 R_3 R_4 + 2C_5 R_1 R_3 R_5 + 2C_5 R_1 R_4 R_5)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^3(C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m - C_3 C_4 C_5 R_1 R_3 R_4) + s^2(C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_5 R_1 R_3 R_5 g_m - C_3 C_5 R_1 R_3 + C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4) + s(C_3 R_1 R_3 R_4 g_m + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_3 R_4 + C_3 R_1 R_3 R_5 + C_3 R_1 R_4 R_5 + C_3 R_3 R_4 R_5 + C_4 R_1 R_4 R_5 + C_4 R_3 R_4 R_5 + C_5 R_1 R_4 R_5 + C_5 R_1 R_3 R_4 + C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5)}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, R_4, R_5, \infty \right)$$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s(C_1 R_3 R_4 + 2C_1 R_3 R_5 + C_1 R_4 R_5)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_3 g_m + s(C_5 R_3 R_5 g_m - C_5 R_3)}{2C_1 C_4 C_5 R_3 R_5 s^3 + g_m + s^2(2C_1 C_4 R_3 + C_1 C_5 R_3 + C_1 C_5 R_5 + 2C_4 C_5 R_3 R_5 g_m + 2C_4 C_5 R_3) + s(C_1 + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4 g_m + s (C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{2C_1 C_4 C_5 R_3 R_4 R_5 s^3 + 2R_3 g_m + R_4 g_m + s^2 (2C_1 C_4 R_3 R_4 + C_1 C_5 R_3 R_4 + 2C_1 C_5 R_3 R_5 + C_1 C_5 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5 g_m + 2C_4 C_5 R_3 R_4) + s (2C_1 R_3 + C_1 R_4 + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4)}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_3 R_4 s^2 + R_3 g_m + s (C_4 R_3 R_4 g_m - C_5 R_3)}{C_1 C_4 C_5 R_3 R_4 s^3 + g_m + s^2 (2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 C_5 R_3 + 2C_4 C_5 R_3 R_4 g_m + 2C_4 C_5 R_3 + C_4 C_5 R_4) + s (C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m + 2C_5 R_3 g_m + C_5)}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_3 R_4 R_5 s^2 + R_3 R_5 g_m - R_3 + s (C_4 R_3 R_4 R_5 g_m - C_4 R_3 R_4 - C_5 R_3 R_5)}{C_1 C_4 C_5 R_3 R_4 R_5 s^3 + 2R_3 g_m + R_5 g_m + s^2 (C_1 C_4 R_3 R_4 + 2C_1 C_4 R_3 R_5 + C_1 C_4 R_4 R_5 + C_1 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_4 R_5 g_m + 2C_4 C_5 R_3 R_5 + C_4 C_5 R_4 R_5) + s (C_1 R_3 + C_1 R_5 + 2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4 + 2C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 g_m + s^2 (C_4 C_5 R_3 R_4 R_5 g_m - C_4 C_5 R_3 R_4) + s (C_4 R_3 R_4 g_m + C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^3 (C_1 C_4 C_5 R_3 R_4 + 2C_1 C_4 C_5 R_3 R_5 + C_1 C_4 C_5 R_4 R_5) + s^2 (2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 C_5 R_3 + C_1 C_5 R_5 + 2C_4 C_5 R_3 R_4 g_m + 2C_4 C_5 R_3 R_5 g_m + 2C_4 C_5 R_3 + C_4 C_5 R_4 R_5 g_m + C_4 C_5 R_4) + s (C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{C_1 C_3 C_5 R_4 R_5 s^3 + 2g_m + s^2 (C_1 C_3 R_4 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + C_3 C_5 R_4 R_5 g_m + C_3 C_5 R_4) + s (2C_1 + C_3 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 s + g_m}{s^2 (C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s (C_5 R_5 g_m - C_5)}{s^3 (C_1 C_3 C_5 R_5 + 2C_1 C_4 C_5 R_5) + s^2 (C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_5 R_5 g_m + C_3 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.41 \quad X-INVALID-ORDER-41} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^3 (C_1 C_3 C_5 R_4 R_5 + 2C_1 C_4 C_5 R_4 R_5) + s^2 (C_1 C_3 R_4 + 2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + C_3 C_5 R_4 R_5 g_m + C_3 C_5 R_4 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4) + s (2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.42 \quad X-INVALID-ORDER-42} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{C_1 C_3 C_4 R_4 R_5 s^3 + 2g_m + s^2 (C_1 C_3 R_5 + C_1 C_4 R_4 + 2C_1 C_4 R_5 + C_3 C_4 R_4 R_5 g_m + C_3 C_4 R_4) + s (C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.43 \quad X-INVALID-ORDER-43} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 s^2 + g_m + s (C_4 R_4 g_m - C_5)}{s^3 (C_1 C_3 C_4 R_4 + C_1 C_4 C_5 R_4 + C_3 C_4 C_5 R_4) + s^2 (C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_4 R_4 g_m + C_3 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.44 \quad X-INVALID-ORDER-44} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 R_5 s^2 + R_5 g_m + s(C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2g_m + s^3(C_1 C_3 C_4 R_4 R_5 + C_1 C_4 C_5 R_4 R_5 + C_3 C_4 C_5 R_4 R_5) + s^2(C_1 C_3 R_5 + C_1 C_4 R_4 + 2C_1 C_4 R_5 + C_1 C_5 R_5 + C_3 C_4 R_4 R_5 g_m + C_3 C_4 R_4 + C_3 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5) + s(C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.45 \quad X-INVALID-ORDER-45} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^2(C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s(C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{C_1 C_3 C_4 C_5 R_4 R_5 s^4 + s^3(C_1 C_3 C_4 R_4 + C_1 C_3 C_5 R_5 + C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5 + C_3 C_4 C_5 R_4 R_5 g_m + C_3 C_4 C_5 R_4) + s^2(C_1 C_3 + 2C_1 C_4 + C_1 C_5 + C_3 C_4 R_4 g_m + C_3 C_5 R_5 g_m + C_3 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s(C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.46 \quad X-INVALID-ORDER-46} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4 g_m + s(C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{C_1 C_3 C_5 R_3 R_4 R_5 s^3 + 2R_3 g_m + R_4 g_m + s^2(C_1 C_3 R_3 R_4 + C_1 C_5 R_3 R_4 + 2C_1 C_5 R_3 R_5 + C_1 C_5 R_4 R_5 + C_3 C_5 R_3 R_4 R_5 g_m + C_3 C_5 R_3 R_4) + s(2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4)}$$

$$\mathbf{9.47 \quad X-INVALID-ORDER-47} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 g_m + s(C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^3(C_1 C_3 C_5 R_3 R_5 + 2C_1 C_4 C_5 R_3 R_5) + s^2(C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_5 R_3 + C_1 C_5 R_5 + C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_3 + 2C_4 C_5 R_3 R_5 g_m + 2C_4 C_5 R_3) + s(C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.48 \quad X-INVALID-ORDER-48} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4 g_m + s(C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{2R_3 g_m + R_4 g_m + s^3(C_1 C_3 C_5 R_3 R_4 R_5 + 2C_1 C_4 C_5 R_3 R_4 R_5) + s^2(C_1 C_3 R_3 R_4 + 2C_1 C_4 R_3 R_4 + C_1 C_5 R_3 R_4 + 2C_1 C_5 R_3 R_5 + C_1 C_5 R_4 R_5 + C_3 C_5 R_3 R_4 R_5 g_m + C_3 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4 R_5 g_m + 2C_4 C_5 R_3 R_4) + s(2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5)}$$

$$\mathbf{9.49 \quad X-INVALID-ORDER-49} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s(C_4 R_3 R_4 R_5 g_m - C_4 R_3 R_4)}{C_1 C_3 C_4 R_3 R_4 R_5 s^3 + 2R_3 g_m + R_5 g_m + s^2(C_1 C_3 R_3 R_5 + C_1 C_4 R_3 R_4 + 2C_1 C_4 R_3 R_5 + C_1 C_4 R_4 R_5 + C_3 C_4 R_3 R_4 R_5 g_m + C_3 C_4 R_3 R_4) + s(C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4) + 1}$$

$$\mathbf{9.50 \quad X-INVALID-ORDER-50} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_3 R_4 s^2 + R_3 g_m + s(C_4 R_3 R_4 g_m - C_5 R_3)}{g_m + s^3(C_1 C_3 C_4 R_3 R_4 + C_1 C_4 C_5 R_3 R_4 + C_3 C_4 C_5 R_3 R_4) + s^2(C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 C_5 R_3 + C_3 C_4 R_3 R_4 g_m + C_3 C_5 R_3 + 2C_4 C_5 R_3 R_4 g_m + 2C_4 C_5 R_3 + C_4 C_5 R_4) + s(C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + C_4 R_4 g_m + 2C_5 R_3 g_m + C_5)}$$

$$\mathbf{9.51 \quad X-INVALID-ORDER-51} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_3 R_4 R_5 s^2 + R_3 R_5 g_m - R_3 + s(C_4 R_3 R_4 R_5 g_m - C_4 R_3 R_4 - C_5 R_3 R_5)}{2R_3 g_m + R_5 g_m + s^3(C_1 C_3 C_4 R_3 R_4 R_5 + C_1 C_4 C_5 R_3 R_4 R_5 + C_3 C_4 C_5 R_3 R_4 R_5) + s^2(C_1 C_3 R_3 R_5 + C_1 C_4 R_3 R_4 + 2C_1 C_4 R_3 R_5 + C_1 C_4 R_4 R_5 + C_1 C_5 R_3 R_5 + C_3 C_4 R_3 R_4 R_5 g_m + C_3 C_4 R_3 R_4 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_4 R_5 g_m + 2C_4 C_5 R_3 R_5 + C_4 C_5 R_4 R_5) + s(C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4)}$$

$$\mathbf{9.52 \quad X-INVALID-ORDER-52} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 g_m + s^2(C_4 C_5 R_3 R_4 R_5 g_m - C_4 C_5 R_3 R_4) + s(C_4 R_3 R_4 g_m + C_5 R_3 R_5 g_m - C_5 R_3)}{C_1 C_3 C_4 C_5 R_3 R_4 R_5 s^4 + g_m + s^3(C_1 C_3 C_4 R_3 R_4 + C_1 C_3 C_5 R_3 R_5 + C_1 C_4 C_5 R_3 R_4 + 2C_1 C_4 C_5 R_3 R_5 + C_1 C_4 C_5 R_4 R_5 + C_3 C_4 C_5 R_3 R_4 R_5 g_m + C_3 C_4 C_5 R_3 R_4) + s^2(C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 C_5 R_3 + C_1 C_5 R_5 + C_3 C_4 R_3 R_4 g_m + C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_3) + s(C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.53 \quad X-INVALID-ORDER-53} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 R_4 s^2 + R_4 g_m + s(C_3 R_3 R_4 g_m - C_5 R_4)}{C_1 C_3 C_5 R_3 R_4 s^3 + 2g_m + s^2(2C_1 C_3 R_3 + C_1 C_3 R_4 + C_1 C_5 R_4 + 2C_3 C_5 R_3 R_4 g_m + 2C_3 C_5 R_3 + C_3 C_5 R_4) + s(2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.54 \quad X-INVALID-ORDER-54} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 R_4 R_5 s^2 + R_4 R_5 g_m - R_4 + s (C_3 R_3 R_4 R_5 g_m - C_3 R_3 R_4 - C_5 R_4 R_5)}{C_1 C_3 C_5 R_3 R_4 R_5 s^3 + 2 R_4 g_m + 2 R_5 g_m + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 C_3 R_3 R_5 + C_1 C_3 R_4 R_5 + C_1 C_5 R_4 R_5 + 2 C_3 C_5 R_3 R_4 R_5 g_m + 2 C_3 C_5 R_3 R_5 + C_3 C_5 R_4 R_5) + s (C_1 R_4 + 2 C_1 R_5 + 2 C_3 R_3 R_4 g_m + 2 C_3 R_3 R_5 g_m + 2 C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_5) + 2}$$

$$\mathbf{9.55 \quad X-INVALID-ORDER-55} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^2 (C_3 C_5 R_3 R_4 R_5 g_m - C_3 C_5 R_3 R_4) + s (C_3 R_3 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2 g_m + s^3 (C_1 C_3 C_5 R_3 R_4 + 2 C_1 C_3 C_5 R_3 R_5 + C_1 C_3 C_5 R_4 R_5) + s^2 (2 C_1 C_3 R_3 + C_1 C_3 R_4 + C_1 C_5 R_4 + 2 C_1 C_5 R_5 + 2 C_3 C_5 R_3 R_4 g_m + 2 C_3 C_5 R_3 R_5 g_m + 2 C_3 C_5 R_3 + C_3 C_5 R_4 R_5 g_m + C_3 C_5 R_4) + s (2 C_1 + 2 C_3 R_3 g_m + C_3 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.56 \quad X-INVALID-ORDER-56} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s (C_3 R_3 R_5 g_m - C_3 R_3) - 1}{2 C_1 C_3 C_4 R_3 R_5 s^3 + 2 g_m + s^2 (C_1 C_3 R_3 + C_1 C_3 R_5 + 2 C_1 C_4 R_5 + 2 C_3 C_4 R_3 R_5 g_m + 2 C_3 C_4 R_3) + s (C_1 + 2 C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 s^2 + g_m + s (C_3 R_3 g_m - C_5)}{s^3 (2 C_1 C_3 C_4 R_3 + C_1 C_3 C_5 R_3 + 2 C_3 C_4 C_5 R_3) + s^2 (C_1 C_3 + 2 C_1 C_4 + C_1 C_5 + 2 C_3 C_4 R_3 g_m + 2 C_3 C_5 R_3 g_m + C_3 C_5 + 2 C_4 C_5) + s (C_3 g_m + 2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 R_5 s^2 + R_5 g_m + s (C_3 R_3 R_5 g_m - C_3 R_3 - C_5 R_5) - 1}{2 g_m + s^3 (2 C_1 C_3 C_4 R_3 R_5 + C_1 C_3 C_5 R_3 R_5 + 2 C_3 C_4 C_5 R_3 R_5) + s^2 (C_1 C_3 R_3 + C_1 C_3 R_5 + 2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_3 C_4 R_3 R_5 g_m + 2 C_3 C_4 R_3 + 2 C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_5 + 2 C_4 C_5 R_5) + s (C_1 + 2 C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_3 C_5 R_3 R_5 g_m - C_3 C_5 R_3) + s (C_3 R_3 g_m + C_5 R_5 g_m - C_5)}{2 C_1 C_3 C_4 C_5 R_3 R_5 s^4 + s^3 (2 C_1 C_3 C_4 R_3 + C_1 C_3 C_5 R_3 + C_1 C_3 C_5 R_5 + 2 C_1 C_4 C_5 R_5 + 2 C_3 C_4 C_5 R_3 R_5 g_m + 2 C_3 C_4 C_5 R_3) + s^2 (C_1 C_3 + 2 C_1 C_4 + C_1 C_5 + 2 C_3 C_4 R_3 g_m + 2 C_3 C_5 R_3 g_m + C_3 C_5 R_5 g_m + C_3 C_5 + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (C_3 g_m + 2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s (C_3 R_3 R_4 R_5 g_m - C_3 R_3 R_4)}{2 C_1 C_3 C_4 R_3 R_4 R_5 s^3 + 2 R_4 g_m + 2 R_5 g_m + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 C_3 R_3 R_5 + C_1 C_3 R_4 R_5 + 2 C_1 C_4 R_4 R_5 + 2 C_3 C_4 R_3 R_4 R_5 g_m + 2 C_3 C_4 R_3 R_4) + s (C_1 R_4 + 2 C_1 R_5 + 2 C_3 R_3 R_4 g_m + 2 C_3 R_3 R_5 g_m + 2 C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4 + 2 C_4 R_4 R_5 g_m + 2 C_4 R_4) + 2}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 R_4 s^2 + R_4 g_m + s (C_3 R_3 R_4 g_m - C_5 R_4)}{2 g_m + s^3 (2 C_1 C_3 C_4 R_3 R_4 + C_1 C_3 C_5 R_3 R_4 + 2 C_3 C_4 C_5 R_3 R_4) + s^2 (2 C_1 C_3 R_3 + C_1 C_3 R_4 + 2 C_1 C_4 R_4 + C_1 C_5 R_4 + 2 C_3 C_4 R_3 R_4 g_m + 2 C_3 C_5 R_3 R_4 g_m + 2 C_3 C_5 R_3 + C_3 C_5 R_4 + 2 C_4 C_5 R_4) + s (2 C_1 + 2 C_3 R_3 g_m + C_3 R_4 g_m + 2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 R_4 R_5 s^2 + R_4 R_5 g_m - R_4 + s (C_3 R_3 R_4 R_5 g_m - C_3 R_3 R_4 - C_5 R_4 R_5)}{2 R_4 g_m + 2 R_5 g_m + s^3 (2 C_1 C_3 C_4 R_3 R_4 R_5 + C_1 C_3 C_5 R_3 R_4 R_5 + 2 C_3 C_4 C_5 R_3 R_4 R_5) + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 C_3 R_3 R_5 + C_1 C_3 R_4 R_5 + 2 C_1 C_4 R_4 R_5 + C_1 C_5 R_4 R_5 + 2 C_3 C_4 R_3 R_4 R_5 g_m + 2 C_3 C_4 R_3 R_4 + 2 C_3 C_5 R_3 R_4 R_5 g_m + 2 C_3 C_5 R_3 R_5 + C_3 C_5 R_4 R_5 + 2 C_4 C_5 R_4 R_5) + s (C_1 R_4 + 2 C_1 R_5 + 2 C_3 R_3 R_4 g_m + 2 C_3 R_3 R_5 g_m + 2 C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4 + 2 C_4 R_4 R_5 g_m + 2 C_4 R_4) + 2}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^2 (C_3 C_5 R_3 R_4 R_5 g_m - C_3 C_5 R_3 R_4) + s (C_3 R_3 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2 C_1 C_3 C_4 C_5 R_3 R_4 R_5 s^4 + 2 g_m + s^3 (2 C_1 C_3 C_4 R_3 R_4 + C_1 C_3 C_5 R_3 R_4 + 2 C_1 C_3 C_5 R_3 R_5 + C_1 C_3 C_5 R_4 R_5 + 2 C_1 C_4 C_5 R_4 R_5 + 2 C_3 C_4 C_5 R_3 R_4 R_5 g_m + 2 C_3 C_4 C_5 R_3 R_4) + s^2 (2 C_1 C_3 R_3 + C_1 C_3 R_4 + 2 C_1 C_4 R_4 + C_1 C_5 R_4 + 2 C_1 C_5 R_5 + 2 C_3 C_4 R_3 R_4 g_m + 2 C_3 C_5 R_3 R_4 g_m + 2 C_3 C_5 R_3 + C_3 C_5 R_4 R_5 g_m + C_3 C_5 R_4 R_5) + s (C_3 g_m + 2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^2 (C_3 C_4 R_3 R_4 R_5 g_m - C_3 C_4 R_3 R_4) + s (C_3 R_3 R_5 g_m - C_3 R_3 + C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2g_m + s^3 (C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 C_4 R_3 R_5 + C_1 C_3 C_4 R_4 R_5) + s^2 (C_1 C_3 R_3 + C_1 C_3 R_5 + C_1 C_4 R_4 + 2C_1 C_4 R_5 + 2C_3 C_4 R_3 R_4 g_m + 2C_3 C_4 R_3 R_5 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4 R_5 g_m + C_3 C_4 R_4) + s (C_1 + 2C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_3 C_4 C_5 R_3 R_4 s^3 + g_m + s^2 (C_3 C_4 R_3 R_4 g_m - C_3 C_5 R_3 - C_4 C_5 R_4) + s (C_3 R_3 g_m + C_4 R_4 g_m - C_5)}{C_1 C_3 C_4 C_5 R_3 R_4 s^4 + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_1 C_3 C_5 R_3 + C_1 C_4 C_5 R_4 + 2C_3 C_4 C_5 R_3 R_4 g_m + 2C_3 C_4 C_5 R_3 + C_3 C_4 C_5 R_4) + s^2 (C_1 C_3 + 2C_1 C_4 + C_1 C_5 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m + 2C_3 C_5 R_3 g_m + C_3 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_4)}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_3 C_4 C_5 R_3 R_4 R_5 s^3 + R_5 g_m + s^2 (C_3 C_4 R_3 R_4 R_5 g_m - C_3 C_4 R_3 R_4 - C_3 C_5 R_3 R_5 - C_4 C_5 R_4 R_5) + s (C_3 R_3 R_5 g_m - C_3 R_3 R_5 + C_4 R_4 R_5 g_m - C_4 R_4 R_5) - 1}{C_1 C_3 C_4 C_5 R_3 R_4 R_5 s^4 + 2g_m + s^3 (C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 C_4 R_3 R_5 + C_1 C_3 C_4 R_4 R_5 + C_1 C_3 C_5 R_3 R_5 + C_1 C_4 C_5 R_4 R_5 + 2C_3 C_4 C_5 R_3 R_4 R_5 g_m + 2C_3 C_4 C_5 R_3 R_5 + C_3 C_4 C_5 R_4 R_5) + s^2 (C_1 C_3 R_3 + C_1 C_3 R_5 + C_1 C_4 R_4 + 2C_1 C_4 R_5 + C_1 C_5 R_5 + 2C_3 C_4 R_3 R_4 g_m + 2C_3 C_4 R_3 R_5 + C_3 C_4 R_3 + C_3 C_4 R_4 R_5 g_m + C_3 C_4 R_4) + s (C_3 R_3 R_5 g_m - C_3 R_3 R_5 + C_4 R_4 R_5 g_m - C_4 R_4 R_5)}$$

$$\mathbf{9.67 \quad X-INVALID-ORDER-67} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^3 (C_3 C_4 C_5 R_3 R_4 R_5 g_m - C_3 C_4 C_5 R_3 R_4) + s^2 (C_3 C_4 R_3 R_4 g_m + C_3 C_5 R_3 R_5 g_m - C_3 C_5 R_3 + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s (C_3 R_3 g_m + C_3 R_3 + C_4 R_4 g_m - C_4 R_4) - 1}{s^4 (C_1 C_3 C_4 C_5 R_3 R_4 + 2C_1 C_3 C_4 C_5 R_3 R_5 + C_1 C_3 C_4 C_5 R_4 R_5) + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_1 C_3 C_5 R_3 + C_1 C_3 C_5 R_5 + C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5 + 2C_3 C_4 C_5 R_3 R_4 g_m + 2C_3 C_4 C_5 R_3 R_5 g_m + 2C_3 C_4 C_5 R_3 + C_3 C_4 C_5 R_4 R_5 g_m + C_3 C_4 C_5 R_4) + s^2 (C_1 C_3 + 2C_1 C_4 + C_1 C_5 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m + 2C_3 C_5 R_3 g_m + C_3 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_4)}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad R_4, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5 + s (C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + C_1 R_1 R_4 R_5)}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_3 g_m + s (C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{2C_1 C_4 C_5 R_1 R_3 R_5 s^3 + R_1 g_m + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_3 R_5 g_m + 2C_4 C_5 R_1 R_3 + 2C_4 C_5 R_3 R_5) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + 2C_4 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5) + 1}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 g_m + s (C_5 R_1 R_3 R_4 R_5 g_m - C_5 R_1 R_3 R_4)}{2C_1 C_4 C_5 R_1 R_3 R_4 R_5 s^3 + 2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^2 (2C_1 C_4 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + 2C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_1 R_4 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_4 C_5 R_1 R_3 R_4 + 2C_4 C_5 R_3 R_4 R_5) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2C_5 R_1 R_3 R_4 g_m + C_5 R_1 R_3 R_4 + C_5 R_1 R_3 R_5 + C_5 R_1 R_4 R_5)}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_3 R_4 s^2 + R_1 R_3 g_m + s (C_4 R_1 R_3 R_4 g_m - C_5 R_1 R_3)}{C_1 C_4 C_5 R_1 R_3 R_4 s^3 + R_1 g_m + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_3 + 2C_4 C_5 R_1 R_3 R_4 g_m + 2C_4 C_5 R_1 R_3 + C_4 C_5 R_1 R_4 + C_4 C_5 R_3 R_4) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_3 R_4 R_5 s^2 + R_1 R_3 R_5 g_m - R_1 R_3 + s (C_4 R_1 R_3 R_4 R_5 g_m - C_4 R_1 R_3 R_4 - C_5 R_1 R_3 R_5)}{C_1 C_4 C_5 R_1 R_3 R_4 R_5 s^3 + 2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2 (C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_5 + C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_3 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_4 C_5 R_1 R_3 R_5 + C_4 C_5 R_1 R_4 R_5 + C_4 C_5 R_3 R_4 R_5) + s (C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_1 R_3 R_5 + C_4 R_1 R_4 R_5)}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_3 g_m + s^2 (C_4 C_5 R_1 R_3 R_4 R_5 g_m - C_4 C_5 R_1 R_3 R_4) + s (C_4 R_1 R_3 R_4 g_m + C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{R_1 g_m + s^3 (C_1 C_4 C_5 R_1 R_3 R_4 + 2C_1 C_4 C_5 R_1 R_3 R_5 + C_1 C_4 C_5 R_1 R_4 R_5) + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_3 R_4 g_m + 2C_4 C_5 R_1 R_3 R_5 g_m + 2C_4 C_5 R_1 R_3 + C_4 C_5 R_1 R_4 R_5 g_m + C_4 C_5 R_1 R_4 + C_4 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_5 + C_4 C_5 R_4 R_5) + s (C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_1 R_3 R_5 + C_4 R_1 R_4 R_5)}$$

$$\mathbf{9.74 \quad X-INVALID-ORDER-74} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{C_1 C_3 C_5 R_1 R_4 R_5 s^3 + 2 R_1 g_m + s^2 (C_1 C_3 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_4 R_5 g_m + C_3 C_5 R_1 R_4 + C_3 C_5 R_4 R_5) + s (2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 R_5 g_m + 2 C_5 R_1 + C_5 R_4 + 2 C_5 R_5) + 2}$$

$$\mathbf{9.75 \quad X-INVALID-ORDER-75} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_1 C_5 R_1 + C_3 C_5 R_1 + 2 C_4 C_5 R_1) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.76 \quad X-INVALID-ORDER-76} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_5 R_1 R_5 g_m - C_5 R_1)}{s^3 (C_1 C_3 C_5 R_1 R_5 + 2 C_1 C_4 C_5 R_1 R_5) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_1 C_5 R_1 + C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_1 + C_3 C_5 R_5 + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + 2 C_4 C_5 R_5) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.77 \quad X-INVALID-ORDER-77} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2 R_1 g_m + s^3 (C_1 C_3 C_5 R_1 R_4 R_5 + 2 C_1 C_4 C_5 R_1 R_4 R_5) + s^2 (C_1 C_3 R_1 R_4 + 2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_4 R_5 g_m + C_3 C_5 R_1 R_4 + C_3 C_5 R_4 R_5 + 2 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 R_1 R_4 + 2 C_4 C_5 R_4 R_5) + s (2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2 C_4 R_1 R_4 g_m)}$$

$$\mathbf{9.78 \quad X-INVALID-ORDER-78} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{C_1 C_3 C_4 R_1 R_4 R_5 s^3 + 2 R_1 g_m + s^2 (C_1 C_3 R_1 R_5 + C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5 + C_3 C_4 R_1 R_4 R_5 g_m + C_3 C_4 R_1 R_4 + C_3 C_4 R_4 R_5) + s (C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + C_4 R_4 + 2 C_4 R_5) + 1}$$

$$\mathbf{9.79 \quad X-INVALID-ORDER-79} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 s^2 + R_1 g_m + s (C_4 R_1 R_4 g_m - C_5 R_1)}{s^3 (C_1 C_3 C_4 R_1 R_4 + C_1 C_4 C_5 R_1 R_4 + C_3 C_4 C_5 R_1 R_4) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_1 C_5 R_1 + C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4 + C_3 C_5 R_1 + 2 C_4 C_5 R_1 R_4 g_m + 2 C_4 C_5 R_1 + C_4 C_5 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.80 \quad X-INVALID-ORDER-80} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 R_5 s^2 + R_1 R_5 g_m - R_1 + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 - C_5 R_1 R_5)}{2 R_1 g_m + s^3 (C_1 C_3 C_4 R_1 R_4 R_5 + C_1 C_4 C_5 R_1 R_4 R_5 + C_3 C_4 C_5 R_1 R_4 R_5) + s^2 (C_1 C_3 R_1 R_5 + C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + C_3 C_4 R_1 R_4 R_5 g_m + C_3 C_4 R_1 R_4 + C_3 C_4 R_4 R_5 + C_3 C_5 R_1 R_5 + 2 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 R_1 R_5 + C_4 C_5 R_4 R_5) + s (C_1 R_1 + C_3 R_1 R_4 g_m)}$$

$$\mathbf{9.81 \quad X-INVALID-ORDER-81} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4) + s (C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_3 C_4 C_5 R_1 R_4 R_5 s^4 + s^3 (C_1 C_3 C_4 R_1 R_4 + C_1 C_3 C_5 R_1 R_5 + C_1 C_4 C_5 R_1 R_4 + 2 C_1 C_4 C_5 R_1 R_5 + C_3 C_4 C_5 R_1 R_4 R_5 g_m + C_3 C_4 C_5 R_1 R_4 + C_3 C_4 C_5 R_4 R_5) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_1 C_5 R_1 + C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4 + C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_1 + C_3 C_5 R_5 + 2 C_4 C_5 R_1 R_4 R_5 g_m)}$$

$$\mathbf{9.82 \quad X-INVALID-ORDER-82} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 g_m + s (C_5 R_1 R_3 R_4 R_5 g_m - C_5 R_1 R_3 R_4)}{C_1 C_3 C_5 R_1 R_3 R_4 R_5 s^3 + 2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + 2 C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_1 R_4 R_5 + C_3 C_5 R_1 R_3 R_4 R_5 g_m + C_3 C_5 R_1 R_3 R_4 + C_3 C_5 R_3 R_4 R_5) + s (2 C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2 C_5 R_1 R_3 R_4 g_m + 2 C_5 R_1 R_3 R_5 g_m)}$$

$$\mathbf{9.83 \quad X-INVALID-ORDER-83} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_3 g_m + s (C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{R_1 g_m + s^3 (C_1 C_3 C_5 R_1 R_3 R_5 + 2 C_1 C_4 C_5 R_1 R_3 R_5) + s^2 (C_1 C_3 R_1 R_3 + 2 C_1 C_4 R_1 R_3 + C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_3 + C_3 C_5 R_3 R_5 + 2 C_4 C_5 R_1 R_3 R_5 g_m + 2 C_4 C_5 R_1 R_3 + 2 C_4 C_5 R_3 R_5) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2 C_4 R_1 R_3 g_m + 2 C_4 R_1 R_3 R_5 g_m)}$$

9.84 X-INVALID-ORDER-84 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m + s(C_5 R_1 R_3 R_4 R_5 g_m - C_5 R_1 R_3 R_4 R_5)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3(C_1 C_3 C_5 R_1 R_3 R_4 R_5 + 2C_1 C_4 C_5 R_1 R_3 R_4 R_5) + s^2(C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 + C_1 C_5 R_1 R_3 R_4 + 2C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_1 R_4 R_5 + C_3 C_5 R_1 R_3 R_4 R_5 g_m + C_3 C_5 R_1 R_3 R_4 + C_3 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_4 C_5 R_1 R_3 R_4 R_5)}$$

9.85 X-INVALID-ORDER-85 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3 + s(C_4 R_1 R_3 R_4 R_5 g_m - C_4 R_1 R_3 R_4)}{C_1 C_3 C_4 R_1 R_3 R_4 R_5 s^3 + 2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2(C_1 C_3 R_1 R_3 R_5 + C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_5 + C_1 C_4 R_1 R_4 R_5 + C_3 C_4 R_1 R_3 R_4 R_5 g_m + C_3 C_4 R_1 R_3 R_4 + C_3 C_4 R_3 R_4 R_5) + s(C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_4 R_1 R_3 R_4)}$$

9.86 X-INVALID-ORDER-86 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_1 R_3 R_4 s^2 + R_1 R_3 g_m + s(C_4 R_1 R_3 R_4 g_m - C_5 R_1 R_3)}{R_1 g_m + s^3(C_1 C_3 C_4 R_1 R_3 R_4 + C_1 C_4 C_5 R_1 R_3 R_4 + C_3 C_4 C_5 R_1 R_3 R_4) + s^2(C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_3 + C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4 + C_3 C_5 R_1 R_3 + 2C_4 C_5 R_1 R_3 R_4 g_m + 2C_4 C_5 R_1 R_3 + C_4 C_5 R_1 R_4 + C_4 C_5 R_3 R_4) + s(C_1 R_1 + C_3 R_1 R_4)}$$

9.87 X-INVALID-ORDER-87 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_1 R_3 R_4 R_5 s^2 + R_1 H}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 R_5 + C_1 C_4 C_5 R_1 R_3 R_4 R_5 + C_3 C_4 C_5 R_1 R_3 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 R_5 + C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_5 + C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_3 R_5 + C_3 C_4 R_1 R_3 R_4 R_5 g_m + C_3 C_4 R_1 R_3 R_4 + C_3 C_4 R_3 R_4 R_5 + C_3 C_5}$$

9.88 X-INVALID-ORDER-88 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^4 + R_1 g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 + C_1 C_3 C_5 R_1 R_3 R_5 + C_1 C_4 C_5 R_1 R_3 R_4 + 2 C_1 C_4 C_5 R_1 R_3 R_5 + C_1 C_4 C_5 R_1 R_4 R_5 + C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m + C_3 C_4 C_5 R_1 R_3 R_4 + C_3 C_4 C_5 R_3 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 + 2 C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_3 +$$

9.89 X-INVALID-ORDER-89 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_4 s^2 + R_1 R_4 g_m + s(C_3 R_1 R_3 R_4 g_m - C_5 R_1 R_4)}{C_1 C_3 C_5 R_1 R_3 R_4 s^3 + 2R_1 g_m + s^2(2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + C_1 C_5 R_1 R_4 + 2C_3 C_5 R_1 R_3 R_4 g_m + 2C_3 C_5 R_1 R_3 + C_3 C_5 R_1 R_4 + C_3 C_5 R_3 R_4) + s(2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

9.90 X-INVALID-ORDER-90 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_4 R_5 s^2 + R_1 R_4 R_5 g_m - R_1 R_4 + s(C_3 R_1 R_3 R_4 R_5 g_m - C_3 R_1 R_3 R_4 - C_5 R_1 R_4 R_5)}{C_1 C_3 C_5 R_1 R_3 R_4 R_5 s^3 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2(C_1 C_3 R_1 R_3 R_4 + 2C_1 C_3 R_1 R_3 R_5 + C_1 C_3 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + 2C_3 C_5 R_1 R_3 R_4 R_5 g_m + 2C_3 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_4 R_5 + C_3 C_5 R_3 R_4 R_5) + s(C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 + 2C_3 R_1 R_4 g_m + 2C_3 R_1 R_4 R_5 + 2C_5 R_1 R_4 R_5)}$$

9.91 X-INVALID-ORDER-91 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s^2 (C_3 C_5 R_1 R_3 R_4 R_5 g_m - C_3 C_5 R_1 R_3 R_4) + s (C_3 R_1 R_3 R_4 g_m + C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2 R_1 g_m + s^3 (C_1 C_3 C_5 R_1 R_3 R_4 + 2 C_1 C_3 C_5 R_1 R_3 R_5 + C_1 C_3 C_5 R_1 R_4 R_5) + s^2 (2 C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_1 C_5 R_1 R_5 + 2 C_3 C_5 R_1 R_3 R_4 g_m + 2 C_3 C_5 R_1 R_3 R_5 g_m + 2 C_3 C_5 R_1 R_3 + C_3 C_5 R_1 R_4 R_5 g_m + C_3 C_5 R_1 R_4 + C_3 C_5 R_3 R_4 + 2 C_3 C_5 R_3 R_5 + C_3 C_5 R_4 R_5)}.$$

9.92 X-INVALID-ORDER-92 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s(C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3)}{2C_1 C_3 C_4 R_1 R_3 R_5 s^3 + 2R_1 g_m + s^2(C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_5 + 2C_1 C_4 R_1 R_5 + 2C_3 C_4 R_1 R_3 R_5 g_m + 2C_3 C_4 R_1 R_3 + 2C_3 C_4 R_3 R_5) + s(C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5) + 1}$$

9.93 X-INVALID-ORDER-93 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 s^2 + R_1 g_m + s(C_3 R_1 R_3 g_m - C_5 R_1)}{s^3(2C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_5 R_1 R_3 + 2C_3 C_4 C_5 R_1 R_3) + s^2(C_1 C_3 R_1 + 2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3 + 2C_3 C_5 R_1 R_3 g_m + C_3 C_5 R_1 + C_3 C_5 R_3 + 2C_4 C_5 R_1) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.94 \quad X-INVALID-ORDER-94} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_5 s^2 + R_1 R_5 g_m - R_1 + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3 - C_5 R_1 R_5)}{2 R_1 g_m + s^3 (2 C_1 C_3 C_4 R_1 R_3 R_5 + C_1 C_3 C_5 R_1 R_3 R_5 + 2 C_3 C_4 C_5 R_1 R_3 R_5) + s^2 (C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_5 + 2 C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2 C_3 C_4 R_1 R_3 R_5 g_m + 2 C_3 C_4 R_1 R_3 + 2 C_3 C_4 R_3 R_5 + 2 C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_5 + C_3 C_5 R_3 R_5 + 2 C_4 C_5 R_1 R_5) + s (C_1 R_1 + 2 C_1 R_1 R_3 + 2 C_1 R_1 R_5 + 2 C_3 R_1 R_3 R_5 g_m + 2 C_3 R_1 R_3 + 2 C_3 R_1 R_5 + 2 C_3 R_3 R_5 + 2 C_4 R_1 R_5 + 2 C_4 R_3 R_5 + 2 C_5 R_1 R_5)}$$

$$\mathbf{9.95 \quad X-INVALID-ORDER-95} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_3 C_5 R_1 R_3 R_5 g_m - C_3 C_5 R_1 R_3) + s (C_3 R_1 R_3 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{2 C_1 C_3 C_4 C_5 R_1 R_3 R_5 s^4 + s^3 (2 C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_1 R_5 + 2 C_1 C_4 C_5 R_1 R_5 + 2 C_3 C_4 C_5 R_1 R_3 R_5 g_m + 2 C_3 C_4 C_5 R_1 R_3 + 2 C_3 C_4 C_5 R_3 R_5) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_3 C_4 R_1 R_3 g_m + 2 C_3 C_4 R_3 + 2 C_3 C_5 R_1 R_3 g_m + C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_3 R_5) + s (C_1 R_1 + 2 C_1 R_1 R_3 + 2 C_1 R_1 R_5 + 2 C_3 R_1 R_3 R_5 g_m + 2 C_3 R_1 R_3 + 2 C_3 R_1 R_5 + 2 C_3 R_3 R_5 + 2 C_4 R_1 R_5 + 2 C_4 R_3 R_5 + 2 C_5 R_1 R_5)}$$

$$\mathbf{9.96 \quad X-INVALID-ORDER-96} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s (C_3 R_1 R_3 R_4 R_5 g_m - C_3 R_1 R_3 R_4)}{2 C_1 C_3 C_4 R_1 R_3 R_4 R_5 s^3 + 2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^2 (C_1 C_3 R_1 R_3 R_4 + 2 C_1 C_3 R_1 R_3 R_5 + C_1 C_3 R_1 R_4 R_5 + 2 C_1 C_4 R_1 R_4 R_5 + 2 C_3 C_4 R_1 R_3 R_4 R_5 g_m + 2 C_3 C_4 R_1 R_3 R_4 + 2 C_3 C_4 R_3 R_4 R_5) + s (C_1 R_1 R_4 + 2 C_1 R_1 R_5 + 2 C_3 R_1 R_3 R_4 g_m + 2 C_3 R_1 R_3 R_5 g_m + 2 C_3 R_1 R_4 + 2 C_3 R_1 R_5 + 2 C_3 R_3 R_4 + 2 C_4 R_1 R_5 + 2 C_4 R_3 R_5 + 2 C_5 R_1 R_5)}$$

$$\mathbf{9.97 \quad X-INVALID-ORDER-97} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_4 s^2 + R_1 R_4 g_m + s (C_3 R_1 R_3 R_4 g_m - C_5 R_1 R_4)}{2 R_1 g_m + s^3 (2 C_1 C_3 C_4 R_1 R_3 R_4 + C_1 C_3 C_5 R_1 R_3 R_4 + 2 C_3 C_4 C_5 R_1 R_3 R_4) + s^2 (2 C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_3 C_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 R_3 R_4 + 2 C_3 C_5 R_1 R_3 R_4 g_m + 2 C_3 C_5 R_1 R_3 + C_3 C_5 R_1 R_4 + C_3 C_5 R_3 R_4 + 2 C_4 C_5 R_1 R_4) + s (2 C_1 R_1 + 2 C_1 R_1 R_3 + 2 C_1 R_1 R_4 + 2 C_3 R_1 R_3 R_4 g_m + 2 C_3 R_1 R_3 + 2 C_3 R_1 R_4 + 2 C_3 R_3 R_4 + 2 C_4 R_1 R_4 + 2 C_4 R_3 R_4 + 2 C_5 R_1 R_4)}$$

$$\mathbf{9.98 \quad X-INVALID-ORDER-98} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_4 R_5 s^2 + R_1 R_5 g_m - R_1 + s^2 (C_3 C_4 R_1 R_3 R_4 R_5 g_m - C_3 C_4 R_1 R_3 R_4) + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3 + C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^3 (2 C_1 C_3 C_4 R_1 R_3 R_4 R_5 + C_1 C_3 C_5 R_1 R_3 R_4 R_5 + 2 C_3 C_4 C_5 R_1 R_3 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 R_4 + 2 C_1 C_3 R_1 R_3 R_5 + C_1 C_3 R_1 R_4 R_5 + 2 C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + 2 C_3 C_4 R_1 R_3 R_4 R_5 g_m + 2 C_3 C_4 R_1 R_3 R_4 + 2 C_3 C_4 R_3 R_4 R_5) + s (C_1 R_1 + 2 C_1 R_1 R_3 + 2 C_1 R_1 R_4 + 2 C_3 R_1 R_3 R_4 g_m + 2 C_3 R_1 R_3 + 2 C_3 R_1 R_4 + 2 C_3 R_3 R_4 + 2 C_4 R_1 R_4 + 2 C_4 R_3 R_4 + 2 C_5 R_1 R_4)}$$

$$\mathbf{9.99 \quad X-INVALID-ORDER-99} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^3 + R_1 g_m + s^2 (C_3 C_4 R_1 R_3 R_4 g_m - C_3 C_5 R_1 R_3 - C_4 C_5 R_1 R_4) + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m - C_5 R_1)}{2 C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^4 + 2 R_1 g_m + s^3 (2 C_1 C_3 C_4 R_1 R_3 R_4 + C_1 C_3 C_5 R_1 R_3 R_4 + 2 C_1 C_3 C_5 R_1 R_3 R_5 + C_1 C_3 C_5 R_1 R_4 R_5 + 2 C_1 C_4 C_5 R_1 R_4 R_5 + 2 C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2 C_3 C_4 C_5 R_1 R_3 R_4 + 2 C_3 C_4 C_5 R_3 R_4 R_5) + s^2 (2 C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_3 C_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 R_1 R_3 + 2 C_3 C_4 R_3 R_4 + 2 C_3 C_5 R_1 R_3 R_4 g_m + 2 C_3 C_5 R_1 R_3 + C_3 C_5 R_1 R_4 + C_3 C_5 R_3 R_4 + 2 C_4 C_5 R_1 R_4 + 2 C_4 C_5 R_3 R_4 + 2 C_5 R_1 R_4)}$$

$$\mathbf{9.100 \quad X-INVALID-ORDER-100} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^2 (C_3 C_4 R_1 R_3 R_4 R_5 g_m - C_3 C_4 R_1 R_3 R_4) + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3 + C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{2 R_1 g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 + 2 C_1 C_3 C_4 R_1 R_3 R_5 + C_1 C_3 C_4 R_1 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_5 + C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5 + 2 C_3 C_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 R_1 R_3 R_5 g_m + 2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 R_5 g_m + C_3 C_4 R_1 R_4 + C_3 C_4 R_3 R_4 + 2 C_3 C_4 R_3 R_5 + C_3 C_4 R_4 R_5) + s (C_1 R_1 + 2 C_1 R_1 R_3 + 2 C_1 R_1 R_4 + 2 C_3 R_1 R_3 R_4 g_m + 2 C_3 R_1 R_3 + 2 C_3 R_1 R_4 + 2 C_3 R_3 R_4 + 2 C_4 R_1 R_4 + 2 C_4 R_3 R_4 + 2 C_5 R_1 R_4)}$$

$$\mathbf{9.101 \quad X-INVALID-ORDER-101} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_4 C_5 R_1 R_3 R_4 s^3 + R_1 g_m + s^2 (C_3 C_4 R_1 R_3 R_4 g_m - C_3 C_5 R_1 R_3 - C_4 C_5 R_1 R_4) + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m - C_5 R_1)}{C_1 C_3 C_4 C_5 R_1 R_3 R_4 s^4 + s^3 (2 C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_4 R_1 R_4 + C_1 C_3 C_5 R_1 R_3 + C_1 C_4 C_5 R_1 R_4 + 2 C_3 C_4 C_5 R_1 R_3 R_4 g_m + 2 C_3 C_4 C_5 R_1 R_3 + C_3 C_4 C_5 R_1 R_4 + C_3 C_4 C_5 R_3 R_4) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2 C_3 C_4 R_3 + C_3 C_4 R_4 + 2 C_3 C_5 R_1 R_3 R_4 g_m + 2 C_3 C_5 R_1 R_3 + C_3 C_5 R_1 R_4 + C_3 C_5 R_3 R_4 + 2 C_4 C_5 R_1 R_4 + 2 C_4 C_5 R_3 R_4 + 2 C_5 R_1 R_4)}$$

$$\mathbf{9.102 \quad X-INVALID-ORDER-102} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_4 C_5 R_1 R_3 R_5 s^2 + R_1 R_5 g_m - R_1 + s^2 (C_3 C_4 R_1 R_3 R_4 R_5 g_m - C_3 C_4 R_1 R_3 R_4) + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3 + C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^4 + 2 R_1 g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 + 2 C_1 C_3 C_4 R_1 R_3 R_5 + C_1 C_3 C_4 R_1 R_4 R_5 + C_1 C_3 C_5 R_1 R_3 R_5 + C_1 C_4 C_5 R_1 R_4 R_5 + 2 C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2 C_3 C_4 C_5 R_1 R_3 R_5 + C_3 C_4 C_5 R_1 R_4 R_5 + C_3 C_4 C_5 R_3 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_5 + C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5 + 2 C_3 C_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 R_1 R_3 R_5 g_m + 2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 R_5 g_m + C_3 C_4 R_1 R_4 + C_3 C_4 R_3 R_4 + 2 C_3 C_4 R_3 R_5 + C_3 C_4 R_4 R_5) + s (C_1 R_1 + 2 C_1 R_1 R_3 + 2 C_1 R_1 R_4 + 2 C_3 R_1 R_3 R_4 g_m + 2 C_3 R_1 R_3 + 2 C_3 R_1 R_4 + 2 C_3 R_3 R_4 + 2 C_4 R_1 R_4 + 2 C_4 R_3 R_4 + 2 C_5 R_1 R_4)}$$

$$\mathbf{9.103 \quad X-INVALID-ORDER-103} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^3 (C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m - C_3 C_4 C_5 R_1 R_3 R_4) + s^2 (C_3 C_4 R_1 R_3 R_4 R_5 g_m - C_3 C_4 R_1 R_3 R_4) + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3 + C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{s^4 (C_1 C_3 C_4 C_5 R_1 R_3 R_4 + 2 C_1 C_3 C_4 C_5 R_1 R_3 R_5 + C_1 C_3 C_4 C_5 R_1 R_4 R_5) + s^3 (2 C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_4 R_1 R_4 + C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_1 R_5 + C_1 C_4 C_5 R_1 R_4 + 2 C_1 C_4 C_5 R_1 R_5 + 2 C_3 C_4 C_5 R_1 R_3 R_4 g_m + 2 C_3 C_4 C_5 R_1 R_3 R_5 g_m + 2 C_3 C_4 C_5 R_1 R_3 + C_3 C_4 C_5 R_1 R_4 R_5 g_m + C_3 C_4 C_5 R_1 R_4 + C_3 C_4 C_5 R_3 R_4 + 2 C_4 C_5 R_1 R_4 + 2 C_4 C_5 R_3 R_4 + 2 C_5 R_1 R_4)}$$

$$\mathbf{9.104 \quad X-INVALID-ORDER-104} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4 + s (C_1 R_1 R_3 R_4 R_5 g_m - C_1 R_1 R_3 R_4)}{2 R_3 R_4 g_m + 2 R_3 R_5 g_m + 2 R_3 + R_4 R_5 g_m + R_4 + s (2 C_1 R_1 R_3 R_4 g_m + 2 C_1 R_1 R_3 R_5 g_m + 2 C_1 R_1 R_3 + C_1 R_1 R_4 R_5 g_m + C_1 R_1 R_4 + C_1 R_3 R_4 + 2 C_1 R_3 R_5 + C_1 R_4 R_5)}$$

$$\mathbf{9.105 \quad X-INVALID-ORDER-105} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 s^2 + R_3 g_m + s (C_1 R_1 R_3 g_m - C_5 R_3)}{2 C_1 C_4 C_5 R_1 R_3 s^3 + g_m + s^2 (2 C_1 C_4 R_1 R_3 g_m + 2 C_1 C_4 R_3 + 2 C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 + C_1 C_5 R_3 + 2 C_4 C_5 R_3) + s (C_1 R_1 g_m + C_1 + 2 C_4 R_3 g_m + 2 C_5 R_3 g_m + C_5)}$$

$$\mathbf{9.106 \quad X-INVALID-ORDER-106} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_5 s^2 + R_3 R_5 g_m - R_3 + s (C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3 - C_5 R_3 R_5)}{2 C_1 C_4 C_5 R_1 R_3 R_5 s^3 + 2 R_3 g_m + R_5 g_m + s^2 (2 C_1 C_4 R_1 R_3 R_5 g_m + 2 C_1 C_4 R_1 R_3 + 2 C_1 C_4 R_3 R_5 + 2 C_1 C_5 R_1 R_3 R_5 g_m + C_1 C_5 R_1 R_5 + C_1 C_5 R_3 R_5 + 2 C_4 C_5 R_3 R_5) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + 2 C_4 R_3 R_5 g_m + 2 C_4 R_3 + 2 C_5 R_3 R_5 g_m + C_5 R_5)}$$

$$\mathbf{9.107 \quad X-INVALID-ORDER-107} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 g_m + s^2 (C_1 C_5 R_1 R_3 R_5 g_m - C_1 C_5 R_1 R_3) + s (C_1 R_1 R_3 g_m + C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^3 (2 C_1 C_4 C_5 R_1 R_3 R_5 g_m + 2 C_1 C_4 C_5 R_1 R_3 + 2 C_1 C_4 C_5 R_3 R_5) + s^2 (2 C_1 C_4 R_1 R_3 g_m + 2 C_1 C_4 R_3 + 2 C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_1 + C_1 C_5 R_3 + C_1 C_5 R_5 + 2 C_4 C_5 R_3 R_5 g_m + 2 C_4 C_5 R_3) + s (C_1 R_1 g_m + C_1 + 2 C_4 R_3 g_m + 2 C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.108 \quad X-INVALID-ORDER-108} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_4 s^2 + R_3 R_4 g_m + s (C_1 R_1 R_3 R_4 g_m - C_5 R_3 R_4)}{2 C_1 C_4 C_5 R_1 R_3 R_4 s^3 + 2 R_3 g_m + R_4 g_m + s^2 (2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4 + 2 C_1 C_5 R_1 R_3 R_4 g_m + 2 C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_4 + C_1 C_5 R_3 R_4 + 2 C_4 C_5 R_3 R_4) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m + 2 C_5 R_3 R_4 g_m + 2 C_5 R_3 + C_5 R_4)}$$

$$\mathbf{9.109 \quad X-INVALID-ORDER-109} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_4 R_5 s^2 + R_3 R_4 R_5 g_m - R_3 R_4 + s (C_1 R_1 R_3 R_4 R_5 g_m - C_1 R_1 R_3 R_4 - C_5 R_3 R_4 R_5)}{2 C_1 C_4 C_5 R_1 R_3 R_4 R_5 s^3 + 2 R_3 R_4 g_m + 2 R_3 R_5 g_m + 2 R_3 + R_4 R_5 g_m + R_4 + s^2 (2 C_1 C_4 R_1 R_3 R_4 R_5 g_m + 2 C_1 C_4 R_1 R_3 R_4 + 2 C_1 C_4 R_3 R_4 R_5 + 2 C_1 C_5 R_1 R_3 R_4 R_5 g_m + 2 C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_1 R_4 R_5 + C_1 C_5 R_3 R_4 R_5 + 2 C_4 C_5 R_3 R_4 R_5) + s (2 C_1 R_1 R_3 R_4 g_m + 2 C_1 R_1 R_3 R_5 g_m + 2 C_1 R_1 R_3 R_4 + 2 C_1 R_1 R_4 R_5 g_m + 2 C_1 R_1 R_4 R_5 + 2 C_1 R_3 R_4 R_5 g_m + 2 C_1 R_3 R_4 R_5 + 2 C_1 R_4 R_5 g_m + 2 C_1 R_4 R_5 + 2 C_4 R_3 R_4 R_5 g_m + 2 C_4 R_3 R_4 R_5 + 2 C_4 R_4 R_5 g_m + 2 C_4 R_4 R_5 + 2 C_5 R_3 R_4 R_5 g_m + 2 C_5 R_3 R_4 R_5 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_4 R_5 + 2 C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.110 \quad X-INVALID-ORDER-110} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4 g_m + s^2 (C_1 C_5 R_1 R_3 R_4 R_5 g_m - C_1 C_5 R_1 R_3 R_4) + s (C_1 R_1 R_3 R_4 g_m + C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{2 R_3 g_m + R_4 g_m + s^3 (2 C_1 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2 C_1 C_4 C_5 R_1 R_3 R_4 + 2 C_1 C_4 C_5 R_3 R_4 R_5) + s^2 (2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4 + 2 C_1 C_5 R_1 R_3 R_4 g_m + 2 C_1 C_5 R_1 R_3 R_5 g_m + 2 C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_4 R_5 g_m + C_1 C_5 R_1 R_4 + C_1 C_5 R_3 R_4 + 2 C_1 C_5 R_3 R_5 + C_1 C_5 R_4 R_5 + 2 C_4 C_5 R_3 R_4 R_5) + s (2 C_1 R_1 R_3 R_4 g_m + 2 C_1 R_1 R_3 R_5 g_m + 2 C_1 R_1 R_3 R_4 + 2 C_1 R_1 R_4 R_5 g_m + 2 C_1 R_1 R_4 R_5 + 2 C_1 R_3 R_4 R_5 g_m + 2 C_1 R_3 R_4 R_5 + 2 C_1 R_4 R_5 g_m + 2 C_1 R_4 R_5 + 2 C_4 R_3 R_4 R_5 g_m + 2 C_4 R_3 R_4 R_5 + 2 C_4 R_4 R_5 g_m + 2 C_4 R_4 R_5 + 2 C_5 R_3 R_4 R_5 g_m + 2 C_5 R_3 R_4 R_5 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_4 R_5 + 2 C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.111 \quad X-INVALID-ORDER-111} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 R_1 R_3 R_4 s^3 + R_3 g_m + s^2 (C_1 C_4 R_1 R_3 R_4 g_m - C_1 C_5 R_1 R_3 - C_4 C_5 R_3 R_4) + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m - C_5 R_3)}{g_m + s^3 (2 C_1 C_4 C_5 R_1 R_3 R_4 g_m + 2 C_1 C_4 C_5 R_1 R_3 + C_1 C_4 C_5 R_1 R_4 + C_1 C_4 C_5 R_3 R_4) + s^2 (2 C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_3 + C_1 C_4 R_4 + 2 C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 + C_1 C_5 R_3 + 2 C_4 C_5 R_3 R_4 g_m + 2 C_4 C_5 R_3 + C_4 C_5 R_4) + s (C_1 R_1 g_m + C_1 + 2 C_4 R_3 g_m + 2 C_5 R_3 g_m + C_5)}$$

$$\mathbf{9.112 \quad X-INVALID-ORDER-112} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 R_1 R_3 R_4 R_5 s^3 + R_3 R_5 g_m - R_3 + s^2 (C_1 C_4 R_1 R_3 R_4 R_5 g_m - C_1 C_4 R_1 R_3 R_4 - C_1 C_5 R_1 R_3 R_5 - C_4 C_5 R_3 R_4 R_5) + s (C_1 R_1 R_3 R_4 R_5 g_m + C_4 R_3 R_4 R_5 g_m - C_5 R_3 R_4 R_5)}{2 R_3 g_m + R_5 g_m + s^3 (2 C_1 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2 C_1 C_4 C_5 R_1 R_3 R_5 + C_1 C_4 C_5 R_1 R_4 R_5 + C_1 C_4 C_5 R_3 R_4 R_5) + s^2 (2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_1 R_3 R_5 g_m + 2 C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 R_5 g_m + C_1 C_4 R_1 R_4 + C_1 C_4 R_3 R_4 + 2 C_1 C_4 R_3 R_5 + C_1 C_4 R_4 R_5 + 2 C_1 C_5 R_1 R_3 R_5 g_m + 2 C_1 C_5 R_1 R_3 R_5 + 2 C_1 C_5 R_4 R_5 g_m + 2 C_1 C_5 R_4 R_5 + 2 C_4 C_5 R_3 R_4 R_5) + s (2 C_1 R_1 R_3 R_4 R_5 g_m + 2 C_1 R_1 R_3 R_4 R_5 + 2 C_1 R_1 R_4 R_5 g_m + 2 C_1 R_1 R_4 R_5 + 2 C_1 R_3 R_4 R_5 g_m + 2 C_1 R_3 R_4 R_5 + 2 C_1 R_4 R_5 g_m + 2 C_1 R_4 R_5 + 2 C_4 R_3 R_4 R_5 g_m + 2 C_4 R_3 R_4 R_5 + 2 C_4 R_4 R_5 g_m + 2 C_4 R_4 R_5 + 2 C_5 R_3 R_4 R_5 g_m + 2 C_5 R_3 R_4 R_5 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_4 R_5 + 2 C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.113 \quad X-INVALID-ORDER-113} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 g_m + s^3 (C_1 C_4 C_5 R_1 R_3 R_4 R_5 g_m - C_1 C_4 C_5 R_1 R_3 R_4) + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 C_5 R_1 R_3 R_5 g_m - C_1 C_5 R_1 R_3 + C_4 C_5 R_3 R_4 R_5 g_m - C_4 C_5 R_3 R_4)}{g_m + s^3 (2 C_1 C_4 C_5 R_1 R_3 R_4 g_m + 2 C_1 C_4 C_5 R_1 R_3 R_5 g_m + 2 C_1 C_4 C_5 R_1 R_3 + C_1 C_4 C_5 R_1 R_4 R_5 g_m + C_1 C_4 C_5 R_1 R_4 + C_1 C_4 C_5 R_3 R_4 + 2 C_1 C_4 C_5 R_3 R_5 + C_1 C_4 C_5 R_4 R_5) + s^2 (2 C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_3 + C_1 C_4 R_4 + 2 C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_4 + C_1 C_5 R_3 R_4 + 2 C_1 C_5 R_3 R_5 + C_1 C_5 R_4 R_5) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_1 R_3 + C_1 R_1 R_4 + 2 C_1 R_3 R_4 + 2 C_1 R_3 R_5 + C_1 R_4 R_5 g_m + C_1 R_4 R_5 + 2 C_4 R_3 R_4 + 2 C_4 R_3 R_5 + C_4 R_4 R_5 g_m + C_4 R_4 R_5 + 2 C_5 R_3 R_4 + 2 C_5 R_3 R_5 + C_5 R_4 R_5 g_m + C_5 R_4 R_5 + 2 C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.114 \quad X-INVALID-ORDER-114} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m - C_5 R_4)}{C_1 C_3 C_5 R_1 R_4 s^3 + 2g_m + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + C_3 C_5 R_4) + s (2C_1 R_1 g_m + 2C_1 + C_3 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.115 \quad X-INVALID-ORDER-115} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 R_5 s^2 + R_4 R_5 g_m - R_4 + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 - C_5 R_4 R_5)}{C_1 C_3 C_5 R_1 R_4 R_5 s^3 + 2R_4 g_m + 2R_5 g_m + s^2 (C_1 C_3 R_1 R_4 R_5 g_m + C_1 C_3 R_1 R_4 + C_1 C_3 R_4 R_5 + 2C_1 C_5 R_1 R_4 R_5 g_m + 2C_1 C_5 R_1 R_5 + C_1 C_5 R_4 R_5 + C_3 C_5 R_4 R_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5)}$$

$$\mathbf{9.116 \quad X-INVALID-ORDER-116} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^2 (C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4) + s (C_1 R_1 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^3 (C_1 C_3 C_5 R_1 R_4 R_5 g_m + C_1 C_3 C_5 R_1 R_4 + C_1 C_3 C_5 R_4 R_5) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + C_3 C_5 R_4 R_5 g_m + C_3 C_5 R_4) + s (2C_1 R_1 g_m + 2C_1 + C_3 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.117 \quad X-INVALID-ORDER-117} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 s^2 + g_m + s (C_1 R_1 g_m - C_5)}{s^3 (C_1 C_3 C_5 R_1 + 2C_1 C_4 C_5 R_1) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + C_3 C_5 + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.118 \quad X-INVALID-ORDER-118} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 s^2 + R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{2g_m + s^3 (C_1 C_3 C_5 R_1 R_5 + 2C_1 C_4 C_5 R_1 R_5) + s^2 (C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + 2C_1 C_4 R_5 + 2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + C_3 C_5 R_5 + 2C_4 C_5 R_5) + s (2C_1 R_1 g_m + C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.119 \quad X-INVALID-ORDER-119} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{s^3 (C_1 C_3 C_5 R_1 R_5 g_m + C_1 C_3 C_5 R_1 + C_1 C_3 C_5 R_5 + 2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + 2C_1 C_4 C_5 R_5) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + C_3 C_5 R_5 g_m + C_3 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.120 \quad X-INVALID-ORDER-120} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m - C_5 R_4)}{2g_m + s^3 (C_1 C_3 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_1 R_4) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + C_3 C_5 R_4 + 2C_4 C_5 R_4) + s (2C_1 R_1 g_m + 2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.121 \quad X-INVALID-ORDER-121} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 R_5 s^2 + R_4 R_5 g_m - R_4 + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 - C_5 R_4 R_5)}{2R_4 g_m + 2R_5 g_m + s^3 (C_1 C_3 C_5 R_1 R_4 R_5 + 2C_1 C_4 C_5 R_1 R_4 R_5) + s^2 (C_1 C_3 R_1 R_4 R_5 g_m + C_1 C_3 R_1 R_4 + C_1 C_3 R_4 R_5 + 2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 R_5 + 2C_1 C_5 R_1 R_4 R_5 g_m + 2C_1 C_5 R_1 R_5 + C_1 C_5 R_4 R_5 + C_3 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_3 R_4 R_5 g_m + C_3 R_4 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_4 R_5)}$$

$$\mathbf{9.122 \quad X-INVALID-ORDER-122} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^2 (C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4) + s (C_1 R_1 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^3 (C_1 C_3 C_5 R_1 R_4 R_5 g_m + C_1 C_3 C_5 R_1 R_4 + C_1 C_3 C_5 R_4 R_5 + 2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_4 R_5) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + C_3 C_5 R_4 R_5 g_m + C_3 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4 R_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_3 R_4 R_5 g_m + C_3 R_4 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_4 R_5)}$$

$$\mathbf{9.123 \quad X-INVALID-ORDER-123} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2g_m + s^3 (C_1 C_3 C_4 R_1 R_4 R_5 g_m + C_1 C_3 C_4 R_1 R_4 + C_1 C_3 C_4 R_4 R_5) + s^2 (C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 R_5 + C_3 C_4 R_4 R_5 g_m + C_3 C_4 R_4) + s (2C_1 R_1 g_m + C_1 + C_3 R_5 g_m + C_3 + 2C_4 R_4 g_m + 2C_4 R_4 R_5 g_m + 2C_4 R_4 R_5)}$$

$$\mathbf{9.124 \quad X-INVALID-ORDER-124} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 R_1 R_4 s^3 + g_m + s^2 (C_1 C_4 R_1 R_4 g_m - C_1 C_5 R_1 - C_4 C_5 R_4) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{C_1 C_3 C_4 C_5 R_1 R_4 s^4 + s^3 (C_1 C_3 C_4 R_1 R_4 g_m + C_1 C_3 C_4 R_4 + C_1 C_3 C_5 R_1 + 2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + C_3 C_4 C_5 R_4) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + C_3 C_4 R_4 g_m + C_3 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (C_1 R_1 + C_4 R_4 + C_5)}$$

$$\mathbf{9.125 \quad X-INVALID-ORDER-125} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 R_1 R_4 R_5 s^3 + R_5 g_m + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4 - C_1 C_5 R_1 R_5 - C_4 C_5 R_4 R_5) + s (C_1 R_1 g_m + C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{C_1 C_3 C_4 C_5 R_1 R_4 R_5 s^4 + 2g_m + s^3 (C_1 C_3 C_4 R_1 R_4 R_5 g_m + C_1 C_3 C_4 R_1 R_4 + C_1 C_3 C_4 R_4 R_5 + C_1 C_3 C_5 R_1 R_5 + 2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_5 + C_1 C_4 C_5 R_4 R_5 + C_3 C_4 C_5 R_4 R_5) + s^2 (C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 R_5 + C_1 C_4 R_4 R_5 + C_3 C_4 R_4 R_5) + s (C_1 R_1 + C_4 R_4 + C_5 R_5)}$$

$$\mathbf{9.126 \quad X-INVALID-ORDER-126} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 R_1 R_4) + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4)}{s^4 (C_1 C_3 C_4 C_5 R_1 R_4 R_5 g_m + C_1 C_3 C_4 C_5 R_1 R_4 + C_1 C_3 C_4 C_5 R_4 R_5) + s^3 (C_1 C_3 C_4 R_1 R_4 g_m + C_1 C_3 C_4 R_4 + C_1 C_3 C_5 R_1 R_5 g_m + C_1 C_3 C_5 R_1 + C_1 C_3 C_5 R_5 + 2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5 + C_3 C_4 C_5 R_4 R_5 g_m + C_3 C_4 C_5 R_4 R_5) + s^2 (C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 R_5 + C_1 C_4 R_4 R_5 + C_3 C_4 R_4 R_5) + s (C_1 R_1 + C_4 R_4 + C_5 R_5)}$$

$$\mathbf{9.127 \quad X-INVALID-ORDER-127} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_4 s^2 + R_3 R_4 g_m + s (C_1 R_1 R_3 R_4 g_m - C_5 R_3 R_4)}{C_1 C_3 C_5 R_1 R_3 R_4 s^3 + 2R_3 g_m + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2C_1 C_5 R_1 R_3 R_4 g_m + 2C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_4 + C_1 C_5 R_3 R_4 + C_3 C_5 R_3 R_4) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4)}$$

$$\mathbf{9.128 \quad X-INVALID-ORDER-128} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_4 R_5 s^2 + R_3 R_4 R_5 g_m - R_3 R_4 + s (C_1 R_1 R_3 R_4 R_5 g_m - C_1 R_1 R_3 R_4 - C_5 R_3 R_4 R_5)}{C_1 C_3 C_5 R_1 R_3 R_4 R_5 s^3 + 2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 g_m + C_1 C_3 R_1 R_3 R_4 + C_1 C_3 R_3 R_4 R_5 + 2C_1 C_5 R_1 R_3 R_4 R_5 g_m + 2C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_1 R_4 R_5 + C_1 C_5 R_3 R_4 R_5 + C_3 C_5 R_3 R_4 R_5) + s (2C_1 R_1 R_3 R_4 g_m + 2C_1 R_1 R_3 R_5 g_m + 2C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + 2C_1 R_1 R_4 R_5 + 2C_1 R_3 R_4 R_5 + C_3 R_3 R_4 R_5) + s (2C_1 R_1 R_3 R_4 g_m + 2C_1 R_1 R_3 R_5 g_m + 2C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + 2C_1 R_1 R_4 R_5 + 2C_1 R_3 R_4 R_5 + C_3 R_3 R_4 R_5)}$$

$$\mathbf{9.129 \quad X-INVALID-ORDER-129} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4 g_m + s^2 (C_1 C_5 R_1 R_3 R_4 R_5 g_m - C_1 C_5 R_1 R_3 R_4) + s (C_1 R_1 R_3 R_4 g_m + C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{2R_3 g_m + R_4 g_m + s^3 (C_1 C_3 C_5 R_1 R_3 R_4 R_5 g_m + C_1 C_3 C_5 R_1 R_3 R_4 + C_1 C_3 C_5 R_3 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2C_1 C_5 R_1 R_3 R_4 g_m + 2C_1 C_5 R_1 R_3 R_5 g_m + 2C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_4 R_5 g_m + C_1 C_5 R_1 R_4 + C_1 C_5 R_3 R_4 + 2C_1 C_5 R_3 R_5 + C_1 C_5 R_4 R_5 + C_3 C_5 R_3 R_4 R_5) + s (2C_1 R_1 R_3 R_4 g_m + 2C_1 R_1 R_3 R_5 g_m + 2C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 R_5 + 2C_1 R_1 R_4 R_5 + 2C_1 R_3 R_4 R_5 + C_3 R_3 R_4 R_5)}$$

$$\mathbf{9.130 \quad X-INVALID-ORDER-130} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 s^2 + R_3 g_m + s (C_1 R_1 R_3 g_m - C_5 R_3)}{g_m + s^3 (C_1 C_3 C_5 R_1 R_3 + 2C_1 C_4 C_5 R_1 R_3) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + 2C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 + C_1 C_5 R_3 + C_3 C_5 R_3 + 2C_4 C_5 R_3) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5)}$$

$$\mathbf{9.131 \quad X-INVALID-ORDER-131} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_5 s^2 + R_3 R_5 g_m - R_3 + s (C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3 - C_5 R_3 R_5)}{2R_3 g_m + R_5 g_m + s^3 (C_1 C_3 C_5 R_1 R_3 R_5 + 2C_1 C_4 C_5 R_1 R_3 R_5) + s^2 (C_1 C_3 R_1 R_3 R_5 g_m + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 R_5 + 2C_1 C_4 R_1 R_3 R_5 g_m + 2C_1 C_4 R_1 R_3 + 2C_1 C_4 R_3 R_5 + 2C_1 C_5 R_1 R_3 R_5 g_m + C_1 C_5 R_1 R_5 + C_1 C_5 R_3 R_5 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_5) + s (2C_1 R_1 R_3 g_m + 2C_1 R_1 R_3 R_5 + 2C_1 R_1 R_3 + 2C_1 R_1 R_5 + 2C_1 R_3 R_5 + C_3 R_3 R_5)}$$

$$\mathbf{9.132 \quad X-INVALID-ORDER-132} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 g_m + s^2 (C_1 C_5 R_1 R_3 R_5 g_m - C_1 C_5 R_1 R_3) + s (C_1 R_1 R_3 g_m + C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^3 (C_1 C_3 C_5 R_1 R_3 R_5 g_m + C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_3 R_5 + 2C_1 C_4 C_5 R_1 R_3 R_5 g_m + 2C_1 C_4 C_5 R_1 R_3 + 2C_1 C_4 C_5 R_3 R_5) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + 2C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_1 + C_1 C_5 R_3 + C_1 C_5 R_5 + C_3 C_5 R_3 R_5) + s (2C_1 R_1 R_3 g_m + 2C_1 R_1 R_3 R_5 + 2C_1 R_1 R_3 + 2C_1 R_1 R_5 + 2C_1 R_3 R_5 + C_3 R_3 R_5)}$$

$$\mathbf{9.133 \quad X-INVALID-ORDER-133} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_4 s^2 + R_3 R_4 g_m + s (C_1 R_1 R_3 R_4 g_m - C_5 R_3 R_4)}{2R_3 g_m + R_4 g_m + s^3 (C_1 C_3 C_5 R_1 R_3 R_4 + 2C_1 C_4 C_5 R_1 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 g_m + 2C_1 C_4 R_3 R_4 + 2C_1 C_5 R_1 R_3 R_4 g_m + 2C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_4 + C_1 C_5 R_3 R_4 + C_3 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_1 R_3 R_4 + 2C_1 R_1 R_3 + 2C_1 R_1 R_4 + 2C_1 R_3 R_4 + C_3 R_3 R_4)}$$

$$\mathbf{9.134 \quad X-INVALID-ORDER-134} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_4 R_5 s^2 + R_3 R_4 R_5 g_m - R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s^3 (C_1 C_3 C_5 R_1 R_3 R_4 R_5 + 2C_1 C_4 C_5 R_1 R_3 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 g_m + C_1 C_3 R_1 R_3 R_4 + C_1 C_3 R_3 R_4 R_5 + 2C_1 C_4 R_1 R_3 R_4 R_5 g_m + 2C_1 C_4 R_1 R_3 R_4 + 2C_1 C_4 R_3 R_4 R_5 + 2C_1 C_5 R_1 R_3 R_4 R_5 g_m + 2C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_4 R_5 g_m)}$$

$$\mathbf{9.135 \quad X-INVALID-ORDER-135} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4 g_m + s^2 (C_1 C_5 R_1 R_3 R_4 R_5 g_m - C_1 C_5 R_1 R_3 R_4 - C_1 C_5 R_4 R_5 g_m)}{2R_3 g_m + R_4 g_m + s^3 (C_1 C_3 C_5 R_1 R_3 R_4 R_5 g_m + C_1 C_3 C_5 R_1 R_3 R_4 + C_1 C_3 C_5 R_3 R_4 R_5 + 2C_1 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_3 R_4 + 2C_1 C_4 C_5 R_3 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 g_m + 2C_1 C_4 R_3 R_4 + 2C_1 C_5 R_1 R_3 R_4 g_m + 2C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_4 R_5 g_m)}$$

$$\mathbf{9.136 \quad X-INVALID-ORDER-136} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s^2 (C_1 C_4 R_1 R_3 R_4 R_5 g_m - C_1 C_4 R_1 R_3 R_4) + s (C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3 + C_4 R_5 g_m)}{2R_3 g_m + R_5 g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 R_5 g_m + C_1 C_3 C_4 R_1 R_3 R_4 + C_1 C_3 C_4 R_3 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 R_5 g_m + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 R_5 + 2C_1 C_4 R_1 R_3 R_4 g_m + 2C_1 C_4 R_1 R_3 R_5 g_m + 2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 R_5 g_m + C_1 C_4 R_1 R_4 + C_1 C_4 R_3 R_4 + 2C_1 C_4 R_3 R_5 + C_1 C_4 R_5 g_m)}$$

$$\mathbf{9.137 \quad X-INVALID-ORDER-137} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 R_1 R_3 R_4 s^3 + R_3 g_m + s^2 (C_1 C_4 R_1 R_3 R_4 g_m - C_1 C_5 R_1 R_3 - C_4 C_5 R_3 R_4) + s (C_1 R_1 R_3 g_m - C_1 R_1 R_3 + C_4 R_5 g_m)}{C_1 C_3 C_4 C_5 R_1 R_3 R_4 s^4 + g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_3 R_4 + C_1 C_3 C_5 R_1 R_3 + 2C_1 C_4 C_5 R_1 R_3 R_4 g_m + 2C_1 C_4 C_5 R_1 R_3 + C_1 C_4 C_5 R_1 R_4 + C_1 C_4 C_5 R_3 R_4 + C_3 C_4 C_5 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_5 g_m)}$$

$$\mathbf{9.138 \quad X-INVALID-ORDER-138} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s^2 (C_1 C_4 R_1 R_3 R_4 R_5 g_m - C_1 C_4 R_1 R_3 R_4 - C_1 C_5 R_1 R_3 - C_4 C_5 R_3 R_4) + s (C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3 + C_4 R_5 g_m)}{C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^4 + 2R_3 g_m + R_5 g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 R_5 g_m + C_1 C_3 C_4 R_1 R_3 R_4 + C_1 C_3 C_4 R_3 R_4 R_5 + C_1 C_3 C_5 R_1 R_3 R_5 + 2C_1 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_3 R_5 + C_1 C_4 C_5 R_1 R_4 R_5 + C_1 C_4 C_5 R_3 R_4 R_5 + C_3 C_4 C_5 R_3 R_4 R_5) + s^2 (C_1 C_3 R_1 R_3 R_5 g_m + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 R_5 + 2C_1 C_4 R_1 R_3 R_4 g_m + 2C_1 C_4 R_1 R_3 R_5 g_m + 2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 R_5 g_m + C_1 C_4 R_1 R_4 + C_1 C_4 R_3 R_4 + 2C_1 C_4 R_3 R_5 + C_1 C_4 R_5 g_m)}$$

$$\mathbf{9.139 \quad X-INVALID-ORDER-139} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s^2 (C_1 C_4 R_1 R_3 R_4 R_5 g_m - C_1 C_4 R_1 R_3 R_4 - C_1 C_5 R_1 R_3 - C_4 C_5 R_3 R_4) + s (C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3 + C_4 R_5 g_m)}{g_m + s^4 (C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m + C_1 C_3 C_4 C_5 R_1 R_3 R_4 + C_1 C_3 C_4 C_5 R_3 R_4 R_5) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_3 R_4 + C_1 C_3 C_5 R_1 R_3 R_5 g_m + C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_3 R_5 + 2C_1 C_4 C_5 R_1 R_3 R_4 g_m + 2C_1 C_4 C_5 R_1 R_3 R_5 g_m + 2C_1 C_4 C_5 R_1 R_3 + C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_4 + C_1 C_4 C_5 R_3 R_4 R_5)}$$

$$\mathbf{9.140 \quad X-INVALID-ORDER-140} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_3 C_5 R_1 R_3 R_4 s^3 + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m - C_1 C_5 R_1 R_4 - C_3 C_5 R_3 R_4) + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m - C_5 R_4)}{2g_m + s^3 (2C_1 C_3 C_5 R_1 R_3 R_4 g_m + 2C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_1 R_4 + C_1 C_3 C_5 R_3 R_4) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_3 C_5 R_3 R_4 g_m + 2C_3 C_5 R_3 + C_3 C_5 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_3 R_3 g_m)}$$

$$\mathbf{9.141 \quad X-INVALID-ORDER-141} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_3 C_5 R_1 R_3 R_4 R_5 s^3 + R_4 R_5 g_m - R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 g_m - C_1 C_3 R_1 R_3 R_4 - C_1 C_5 R_1 R_4 R_5 - C_3 C_5 R_3 R_4 R_5) + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 + C_3 R_3 R_4 R_5 g_m - C_3 R_3 R_4 + C_5 R_5 g_m)}{2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_3 C_5 R_1 R_3 R_4 R_5 g_m + 2C_1 C_3 C_5 R_1 R_3 R_5 + C_1 C_3 C_5 R_1 R_4 R_5 + C_1 C_3 C_5 R_3 R_4 R_5) + s^2 (2C_1 C_3 R_1 R_3 R_4 g_m + 2C_1 C_3 R_1 R_3 R_5 g_m + 2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 R_5 g_m + C_1 C_3 R_1 R_4 + C_1 C_3 R_3 R_4 + 2C_1 C_3 R_3 R_5 + C_1 C_3 R_4 R_5 + 2C_1 C_5 R_1 R_4 R_5 g_m + 2C_1 C_5 R_1 R_4 R_5 + C_1 C_5 R_3 R_4 R_5)}$$

$$\mathbf{9.142 \quad X-INVALID-ORDER-142} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^3 (C_1 C_3 C_5 R_1 R_3 R_4 R_5 g_m - C_1 C_3 C_5 R_1 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4 + C_3 C_5 R_3 R_4 R_5 g_m - C_3 C_5 R_3 R_4 + C_5 R_5 g_m)}{2g_m + s^3 (2C_1 C_3 C_5 R_1 R_3 R_4 g_m + 2C_1 C_3 C_5 R_1 R_3 R_5 g_m + 2C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_1 R_4 R_5 g_m + C_1 C_3 C_5 R_1 R_4 + C_1 C_3 C_5 R_3 R_4 + 2C_1 C_3 C_5 R_3 R_5 + C_1 C_3 C_5 R_4 R_5) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 R_3 + C_1 C_5 R_4 + 2C_3 C_5 R_3 R_4 g_m + 2C_3 C_5 R_3 + C_3 C_5 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_3 R_3 g_m)}$$

$$\mathbf{9.143 \quad X-INVALID-ORDER-143} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 C_3 R_1 R_3 R_5 g_m - C_1 C_3 R_1 R_3) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_3 R_3 R_5 g_m - C_3 R_3) - 1}{2g_m + s^3 (2C_1 C_3 C_4 R_1 R_3 R_5 g_m + 2C_1 C_3 C_4 R_1 R_3 + 2C_1 C_3 C_4 R_3 R_5) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + 2C_1 C_4 R_5 + 2C_3 C_4 R_3 R_5 g_m + 2C_3 C_4 R_3) + s (2C_1 R_1 g_m + C_1 + 2C_3 R_3 g_m + C_3 R_5 g_m + C_3 R_5)}$$

$$\mathbf{9.144 \quad X-INVALID-ORDER-144} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_3 C_5 R_1 R_3 s^3 + g_m + s^2 (C_1 C_3 R_1 R_3 g_m - C_1 C_5 R_1 - C_3 C_5 R_3) + s (C_1 R_1 g_m + C_3 R_3 g_m - C_5)}{2C_1 C_3 C_4 C_5 R_1 R_3 s^4 + s^3 (2C_1 C_3 C_4 R_1 R_3 g_m + 2C_1 C_3 C_4 R_3 + 2C_1 C_3 C_5 R_1 R_3 g_m + C_1 C_3 C_5 R_1 + C_1 C_3 C_5 R_3 + 2C_1 C_4 C_5 R_1 + 2C_3 C_4 C_5 R_3) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_3 C_4 R_3 g_m + 2C_3 C_5 R_3 g_m + C_3 C_5 + 2C_4 C_5) + s (C_1 R_1 + C_3 R_3 - C_5) + C_5}$$

$$\mathbf{9.145 \quad X-INVALID-ORDER-145} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_3 C_5 R_1 R_3 R_5 s^3 + R_5 g_m + s^2 (C_1 C_3 R_1 R_3 R_5 g_m - C_1 C_3 R_1 R_3 - C_1 C_5 R_1 R_5 - C_3 C_5 R_3 R_5) + s (C_1 R_1 R_5 g_m + C_3 R_3 R_5 g_m - C_5)}{2C_1 C_3 C_4 C_5 R_1 R_3 R_5 s^4 + 2g_m + s^3 (2C_1 C_3 C_4 R_1 R_3 R_5 g_m + 2C_1 C_3 C_4 R_1 R_3 + 2C_1 C_3 C_4 R_3 R_5 + 2C_1 C_3 C_5 R_1 R_3 R_5 g_m + C_1 C_3 C_5 R_1 R_5 + C_1 C_3 C_5 R_3 R_5 + 2C_1 C_4 C_5 R_1 R_5 + 2C_3 C_4 C_5 R_3 R_5) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_5 g_m + C_1 C_4 R_1 R_3 + C_1 C_4 R_3 R_5) + s (C_1 R_1 R_5 + C_3 R_3 R_5 - C_5) + C_5}$$

$$\mathbf{9.146 \quad X-INVALID-ORDER-146} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^3 (C_1 C_3 C_5 R_1 R_3 R_5 g_m - C_1 C_3 C_5 R_1 R_3) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_3 C_5 R_3 R_5 g_m - C_3 C_5 R_3 R_5) + s (C_1 R_1 R_5 g_m + C_3 R_3 R_5 g_m - C_5)}{s^4 (2C_1 C_3 C_4 C_5 R_1 R_3 R_5 g_m + 2C_1 C_3 C_4 C_5 R_1 R_3 + 2C_1 C_3 C_4 C_5 R_3 R_5) + s^3 (2C_1 C_3 C_4 R_1 R_3 g_m + 2C_1 C_3 C_4 R_3 + 2C_1 C_3 C_5 R_1 R_3 g_m + C_1 C_3 C_5 R_1 R_5 g_m + C_1 C_3 C_5 R_1 + C_1 C_3 C_5 R_3 + C_1 C_3 C_5 R_5 + 2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + 2C_1 C_4 C_5 R_5 + 2C_3 C_4 C_5 R_3 R_5 g_m + 2C_3 C_4 C_5 R_3 R_5) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_5 g_m + C_1 C_4 R_1 R_3 + C_1 C_4 R_3 R_5) + s (C_1 R_1 R_5 + C_3 R_3 R_5 - C_5) + C_5}$$

$$\mathbf{9.147 \quad X-INVALID-ORDER-147} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 g_m - C_1 C_3 R_1 R_3 R_4) + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 R_5 g_m + 2C_1 C_3 C_4 R_1 R_3 R_4 + 2C_1 C_3 C_4 R_3 R_4 R_5) + s^2 (2C_1 C_3 R_1 R_3 R_4 g_m + 2C_1 C_3 R_1 R_3 R_5 g_m + 2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 R_5 g_m + C_1 C_3 R_1 R_4 + C_1 C_3 R_3 R_4 + 2C_1 C_3 R_3 R_5 + C_1 C_3 R_4 R_5 + 2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_3 R_4 R_5) + s (C_1 R_1 R_4 + C_3 R_3 R_4 - C_5) + C_5}$$

$$\mathbf{9.148 \quad X-INVALID-ORDER-148} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_3 C_5 R_1 R_3 R_4 s^3 + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m - C_1 C_5 R_1 R_4 - C_3 C_5 R_3 R_4) + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m - C_5)}{2C_1 C_3 C_4 C_5 R_1 R_3 R_4 s^4 + 2g_m + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 C_5 R_1 R_3 R_4 g_m + 2C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_1 R_4 + C_1 C_3 C_5 R_3 R_4 + 2C_1 C_4 C_5 R_1 R_4 + 2C_3 C_4 C_5 R_3 R_4) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_1 R_3 + C_1 C_4 R_3 R_4) + s (C_1 R_1 R_4 + C_3 R_3 R_4 - C_5) + C_5}$$

$$\mathbf{9.149 \quad X-INVALID-ORDER-149} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 g_m - C_1 C_3 R_1 R_3 R_4) + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^4 + 2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 R_5 g_m + 2C_1 C_3 C_4 R_1 R_3 R_4 + 2C_1 C_3 C_4 R_3 R_4 R_5 + 2C_1 C_3 C_5 R_1 R_3 R_4 R_5 g_m + 2C_1 C_3 C_5 R_1 R_3 R_5 + C_1 C_3 C_5 R_1 R_4 R_5 + C_1 C_3 C_5 R_3 R_4 R_5 + 2C_1 C_4 C_5 R_1 R_4 R_5 + 2C_3 C_4 C_5 R_3 R_4 R_5) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_1 R_3 + C_1 C_4 R_3 R_4) + s (C_1 R_1 R_4 + C_3 R_3 R_4 - C_5) + C_5}$$

$$\mathbf{9.150 \quad X-INVALID-ORDER-150} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 R_5 g_m - C_1 C_3 R_1 R_3 R_4) + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2g_m + s^4 (2C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_1 C_3 C_4 C_5 R_1 R_3 R_4 + 2C_1 C_3 C_4 C_5 R_3 R_4 R_5) + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 C_5 R_1 R_3 R_4 g_m + 2C_1 C_3 C_5 R_1 R_3 R_5 g_m + 2C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_1 R_4 R_5 g_m + C_1 C_3 C_5 R_1 R_4 + C_1 C_3 C_5 R_3 R_4 + 2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_3 R_4 R_5) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_1 R_3 + C_1 C_4 R_3 R_4) + s (C_1 R_1 R_4 + C_3 R_3 R_4 - C_5) + C_5}$$

$$\mathbf{9.151 \quad X-INVALID-ORDER-151} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 R_5 g_m - C_1 C_3 C_4 R_1 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 R_5 g_m - C_1 C_3 R_1 R_3 + C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4 + C_3 C_4 R_3 R_4 R_5 g_m - C_3 C_4 R_3 R_4 R_5) + s (C_1 R_1 R_5 g_m + C_3 R_3 R_5 g_m - C_5)}{2g_m + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 R_1 R_3 R_5 g_m + 2C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_4 R_1 R_4 R_5 g_m + C_1 C_3 C_4 R_1 R_4 + C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 C_4 R_3 R_5 + C_1 C_3 C_4 R_4 R_5) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_3 + C_1 C_4 R_3 R_4) + s (C_1 R_1 R_5 + C_3 R_3 R_5 - C_5) + C_5}$$

$$\mathbf{9.152 \quad X-INVALID-ORDER-152} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_3 C_4 C_5 R_1 R_3 R_4 s^4 + g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m - C_1 C_3 C_5 R_1 R_3 - C_1 C_4 C_5 R_1 R_4 - C_3 C_4 C_5 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m - C_1 C_5 R_1 R_5 - C_3 C_5 R_3 R_5) + s (C_1 R_1 R_4 g_m + C_3 R_3 R_5 g_m - C_5)}{s^4 (2C_1 C_3 C_4 C_5 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 C_5 R_1 R_3 + C_1 C_3 C_4 C_5 R_1 R_4 + C_1 C_3 C_4 C_5 R_3 R_4) + s^3 (2C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + 2C_1 C_3 C_5 R_1 R_3 g_m + C_1 C_3 C_5 R_1 + C_1 C_3 C_5 R_3 + 2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2C_3 C_4 C_5 R_3 R_5 g_m + 2C_3 C_4 C_5 R_3 R_5) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_1 R_3 + C_1 C_4 R_3 R_4) + s (C_1 R_1 R_4 + C_3 R_3 R_5 - C_5) + C_5}$$

$$\mathbf{9.153 \quad X-INVALID-ORDER-153} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 s^4 + R_5 g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 R_5 g_m - C_1 C_3 C_4 R_1 R_3 R_4 - C_1 C_5 R_1 R_5 - C_3 C_5 R_3 R_5) + s (C_1 R_1 R_5 g_m + C_3 R_3 R_5 g_m - C_5)}{2g_m + s^4 (2C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2C_1 C_3 C_4 C_5 R_1 R_3 R_5 + C_1 C_3 C_4 C_5 R_1 R_4 R_5 + C_1 C_3 C_4 C_5 R_3 R_4 R_5) + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 R_1 R_3 R_5 g_m + 2C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_4 R_1 R_4 R_5 g_m + C_1 C_3 C_4 R_1 R_4 + C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 C_4 R_3 R_5 + C_1 C_3 C_4 R_4 R_5) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 R_5 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_3 + C_1 C_4 R_3 R_4) + s (C_1 R_1 R_5 + C_3 R_3 R_5 - C_5) + C_5}$$

9.154 X-INVALID-ORDER-154 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{g_m + s^4 (C_1 C_3 C_4 C_5 R_1 R_3 R_4 R_5 g_m - C_1 C_3 C_4 C_5 R_1 R_3 R_4) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_5 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_1 R_3 R_5 g_m + C_1 C_3 C_4 R_1 R_4 R_5 g_m + C_1 C_3 C_4 C_5 R_1 R_4 + C_1 C_3 C_4 C_5 R_3 R_4 + 2 C_1 C_3 C_4 C_5 R_3 R_5 + C_1 C_3 C_4 C_5 R_4 R_5) + s^2 (2 C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + 2 C_1 C_3 C_5 R_3 R_4) + s (C_4 R_1 R_3 R_4 g_m - C_5 R_1 R_3 R_4) + C_4 R_1 R_3 R_4}{s^4 (2 C_1 C_3 C_4 C_5 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 C_5 R_1 R_3 R_5 g_m + 2 C_1 C_3 C_4 C_5 R_1 R_3 + C_1 C_3 C_4 C_5 R_1 R_4 R_5 g_m + C_1 C_3 C_4 C_5 R_1 R_4 + C_1 C_3 C_4 C_5 R_3 R_4 + 2 C_1 C_3 C_4 C_5 R_3 R_5 + C_1 C_3 C_4 C_5 R_4 R_5) + s^3 (2 C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + 2 C_1 C_3 C_5 R_3 R_4) + s^2 (2 C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + 2 C_1 C_3 C_5 R_3 R_4) + s (C_4 R_1 R_3 R_4 g_m - C_5 R_1 R_3 R_4) + C_4 R_1 R_3 R_4}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \infty, R_3, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_1 R_3 R_4 s^2 + R_1 R_3 g_m + s (C_4 R_1 R_3 R_4 g_m - C_5 R_1 R_3)}{R_1 g_m + s^2 (2 C_4 C_5 R_1 R_3 R_4 g_m + 2 C_4 C_5 R_1 R_3 + C_4 C_5 R_1 R_4 + C_4 C_5 R_3 R_4) + s (2 C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2 C_4 R_3 + C_4 R_4 + 2 C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

Parameters:

Q: $\frac{2\sqrt{C_4}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+2\sqrt{C_4}\sqrt{C_5}R_1R_3\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\sqrt{C_4}\sqrt{C_5}R_1R_4\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\sqrt{C_4}\sqrt{C_5}R_3R_4\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}{2C_4R_1R_3g_m+C_4R_1R_4g_m+2C_4R_3+C_4R_4+2C_5R_1R_3g_m+C_5R_1+C_5R_3}$

wo: $\sqrt{\frac{R_1g_m+1}{2C_4C_5R_1R_3R_4g_m+2C_4C_5R_1R_3+C_4C_5R_1R_4+C_4C_5R_3R_4}}$

bandwidth: $\frac{\sqrt{\frac{R_1g_m+1}{2C_4C_5R_1R_3R_4g_m+2C_4C_5R_1R_3+C_4C_5R_1R_4+C_4C_5R_3R_4}}(2C_4R_1R_3g_m+C_4R_1R_4g_m+2C_4R_3+C_4R_4+2C_5R_1R_3g_m+C_5R_1+C_5R_3)}{2\sqrt{C_4}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+2\sqrt{C_4}\sqrt{C_5}R_1R_3\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\sqrt{C_4}\sqrt{C_5}R_1R_4\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\sqrt{C_4}\sqrt{C_5}R_3R_4\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}$

K-LP: $\frac{R_1R_3g_m}{R_1g_m+1}$

K-HP: $-\frac{R_1R_3R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}$

K-BP: $\frac{C_4R_1R_3R_4g_m-C_5R_1R_3}{2C_4R_1R_3g_m+C_4R_1R_4g_m+2C_4R_3+C_4R_4+2C_5R_1R_3g_m+C_5R_1+C_5R_3}$

Qz: None

Wz: $\frac{\sqrt{-g_m}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1, \infty, R_3, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_1 R_3 R_4 R_5 s^2 + R_1 R_3 R_5 g_m - R_1 R_3 + s (C_4 R_1 R_3 R_4 R_5 g_m - C_4 R_1 R_3 R_4 - C_5 R_1 R_3 R_5)}{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2 (2 C_4 C_5 R_1 R_3 R_4 R_5 g_m + 2 C_4 C_5 R_1 R_3 R_5 + C_4 C_5 R_1 R_4 R_5 + C_4 C_5 R_3 R_4 R_5) + s (2 C_4 R_1 R_3 R_4 g_m + 2 C_4 R_1 R_3 R_5 g_m + 2 C_4 R_1 R_3 + C_4 R_1 R_4 R_5 g_m + C_4 R_1 R_4 + C_4 R_3 R_4 + 2 C_4 R_3 R_5 + C_4 R_4 R_5 + 2 C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_1 + C_5 R_3 + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{2\sqrt{C_4}\sqrt{C_5}R_1R_3R_4\sqrt{R_5g_m}\sqrt{\frac{2R_1R_3g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_5}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+2\sqrt{C_4}\sqrt{C_5}R_1R_3\sqrt{R_5}\sqrt{\frac{2R_1R_3g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_5}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}{2\sqrt{C_4}\sqrt{C_5}R_1R_3R_4\sqrt{R_5g_m}\sqrt{\frac{2R_1R_3g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_5}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+2\sqrt{C_4}\sqrt{C_5}R_1R_3\sqrt{R_5}\sqrt{\frac{2R_1R_3g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_5}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}$

wo: $\sqrt{\frac{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}{2C_4C_5R_1R_3R_4R_5g_m+2C_4C_5R_1R_3R_5+C_4C_5R_1R_4R_5+C_4C_5R_3R_4R_5}}$

bandwidth: $\frac{\sqrt{\frac{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}{2C_4C_5R_1R_3R_4R_5g_m+2C_4C_5R_1R_3R_5+C_4C_5R_1R_4R_5+C_4C_5R_3R_4R_5}}}{2\sqrt{C_4}\sqrt{C_5}R_1R_3R_4\sqrt{R_5g_m}\sqrt{\frac{2R_1R_3g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_5}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+2\sqrt{C_4}\sqrt{C_5}R_1R_3\sqrt{R_5}\sqrt{\frac{2R_1R_3g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_5}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}$

K-LP: $\frac{R_1R_3R_5g_m-R_1R_3}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}$

K-HP: $-\frac{R_1R_3R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}$

K-BP: $\frac{C_4R_1R_3R_4R_5g_m-C_4R_1R_3R_4-C_5R_1R_3R_5}{2C_4R_1R_3R_4g_m+2C_4R_1R_3R_5g_m+2C_4R_1R_3+C_4R_1R_4R_5g_m+C_4R_1R_4+C_4R_3R_4+2C_4R_3R_5+C_4R_4R_5+2C_5R_1R_3R_5g_m+C_5R_1R_5+C_5R_3R_5}$

Qz: None

Wz: $\frac{\sqrt{-R_5g_m+1}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1, \infty, R_3, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_3 g_m + s^2 (C_4 C_5 R_1 R_3 R_4 R_5 g_m - C_4 C_5 R_1 R_3 R_4) + s (C_4 R_1 R_3 R_4 g_m + C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{R_1 g_m + s^2 (2 C_4 C_5 R_1 R_3 R_4 g_m + 2 C_4 C_5 R_1 R_3 R_5 g_m + 2 C_4 C_5 R_1 R_3 + C_4 C_5 R_1 R_4 R_5 g_m + C_4 C_5 R_1 R_4 + C_4 C_5 R_3 R_4 + 2 C_4 C_5 R_3 R_5 + C_4 C_5 R_4 R_5) + s (2 C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2 C_4 R_3 + C_4 R_4 + 2 C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{2\sqrt{C_4}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+2\sqrt{C_4}\sqrt{C_5}R_1R_3R_5g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}{2\sqrt{C_4}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+2\sqrt{C_4}\sqrt{C_5}R_1R_3R_5g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}$

wo: $\sqrt{\frac{R_1g_m+1}{2C_4C_5R_1R_3R_4g_m+2C_4C_5R_1R_3R_5g_m+2C_4C_5R_1R_3+C_4C_5R_1R_4R_5g_m+C_4C_5R_1R_4+C_4C_5R_3R_4+2C_4C_5R_3R_5+C_4C_5R_4R_5}}$

bandwidth: $\frac{\sqrt{\frac{R_1g_m+1}{2C_4C_5R_1R_3R_4g_m+2C_4C_5R_1R_3R_5g_m+2C_4C_5R_1R_3+C_4C_5R_1R_4R_5g_m+C_4C_5R_1R_4+C_4C_5R_3R_4+2C_4C_5R_3R_5+C_4C_5R_4R_5}}}{2\sqrt{C_4}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+2\sqrt{C_4}\sqrt{C_5}R_1R_3R_5g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}$

K-LP: $\frac{R_1R_3g_m}{R_1g_m+1}$

$$\begin{aligned} \text{K-HP: } & \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5} \\ \text{K-BP: } & \frac{C_4 R_1 R_3 R_4 g_m + C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3}{2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4}} \end{aligned}$$

10.4 X-INVALID-WZ-4 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_4 s^2 + R_1 R_4 g_m + s (C_3 R_1 R_3 R_4 g_m - C_5 R_1 R_4)}{2R_1 g_m + s^2 (2C_3 C_5 R_1 R_3 R_4 g_m + 2C_3 C_5 R_1 R_3 + C_3 C_5 R_1 R_4 + C_3 C_5 R_3 R_4) + s (2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_3\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_4\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_4\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}}{2C_3R_1R_3g_m+C_3R_1R_4g_m+2C_3R_3+C_3R_4+2C_5R_1R_4g_m+2C_5R_1+C_5R_4}$$

$$\text{wo: } \sqrt{\frac{2R_1g_m+2}{2C_3C_5R_1R_3R_4g_m+2C_3C_5R_1R_3+C_3C_5R_1R_4+C_3C_5R_3R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_1g_m+2}{2C_3C_5R_1R_3R_4g_m+2C_3C_5R_1R_3+C_3C_5R_1R_4+C_3C_5R_3R_4}}(2C_3R_1R_3g_m+C_3R_1R_4g_m+2C_3R_3+C_3R_4+2C_5R_1R_4g_m+2C_5R_1+C_5R_4)}{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_3\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_4\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_4\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}}$$

$$\text{K-LP: } \frac{R_1 R_4 g_m}{2R_1 g_m + 2}$$

$$\text{K-HP: } -\frac{R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 + R_1 R_4 + R_3 R_4}$$

$$\text{K-BP: } \frac{C_3 R_1 R_3 R_4 g_m - C_5 R_1 R_4}{2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}$$

Qz: None

$$\text{Wz: } \frac{\sqrt{-g_m}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}}$$

10.5 X-INVALID-WZ-5 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_4 R_5 s^2 + R_1 R_4 R_5 g_m - R_1 R_4 + s (C_3 R_1 R_3 R_4 R_5 g_m - C_3 R_1 R_3 R_4 - C_5 R_1 R_4 R_5)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2 (2C_3 C_5 R_1 R_3 R_4 R_5 g_m + 2C_3 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_4 R_5 + C_3 C_5 R_3 R_4 R_5) + s (2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_3 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_3 R_4 + 2C_3 R_3 R_5 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_3}\sqrt{C_5}R_1R_3R_4\sqrt{R_5g_m}\sqrt{\frac{2R_1R_4g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{2R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_5}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+2\sqrt{C_3}\sqrt{C_5}R_1R_3\sqrt{R_5}\sqrt{\frac{2R_1R_4g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{2R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}{\sqrt{\frac{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}{2C_3C_5R_1R_3R_4R_5g_m+2C_3C_5R_1R_3R_5+C_3C_5R_1R_4R_5+C_3C_5R_3R_4R_5}}}$$

$$\text{wo: } \sqrt{\frac{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}{2C_3C_5R_1R_3R_4R_5g_m+2C_3C_5R_1R_3R_5+C_3C_5R_1R_4R_5+C_3C_5R_3R_4R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}{2C_3C_5R_1R_3R_4R_5g_m+2C_3C_5R_1R_3R_5+C_3C_5R_1R_4R_5+C_3C_5R_3R_4R_5}}}{2\sqrt{C_3}\sqrt{C_5}R_1R_3R_4\sqrt{R_5g_m}\sqrt{\frac{2R_1R_4g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{2R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_5}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+2\sqrt{C_3}\sqrt{C_5}R_1R_3\sqrt{R_5}\sqrt{\frac{2R_1R_4g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{2R_1R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_1}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}}$$

$$\text{K-LP: } \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$$

$$\text{K-HP: } -\frac{R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 + R_1 R_4 + R_3 R_4}$$

$$\text{K-BP: } \frac{C_3 R_1 R_3 R_4 R_5 g_m - C_3 R_1 R_3 R_4 - C_5 R_1 R_4 R_5}{2C_3 R_1 R_3 R_4 g_m + 2C_3 R_1 R_3 R_5 g_m + 2C_3 R_1 R_3 + C_3 R_1 R_4 R_5 g_m + C_3 R_1 R_4 + C_3 R_3 R_4 + 2C_3 R_3 R_5 + C_3 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$$

Qz: None

$$\text{Wz: } \frac{\sqrt{-R_5 g_m + 1}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_5}}$$

10.6 X-INVALID-WZ-6 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s^2 (C_3 C_5 R_1 R_3 R_4 R_5 g_m - C_3 C_5 R_1 R_3 R_4) + s (C_3 R_1 R_3 R_4 g_m + C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s^2 (2C_3 C_5 R_1 R_3 R_4 g_m + 2C_3 C_5 R_1 R_3 R_5 g_m + 2C_3 C_5 R_1 R_3 + C_3 C_5 R_1 R_4 R_5 g_m + C_3 C_5 R_1 R_4 + C_3 C_5 R_3 R_4 + 2C_3 C_5 R_3 R_5 + C_3 C_5 R_4 R_5) + s (2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5) + 2}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_3R_5g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}{\sqrt{\frac{2R_1g_m+2}{2C_3C_5R_1R_3R_4g_m+2C_3C_5R_1R_3R_5g_m+2C_3C_5R_1R_3+C_3C_5R_1R_4R_5g_m+C_3C_5R_1R_4+C_3C_5R_3R_4+2C_3C_5R_3R_5+C_3C_5R_4R_5}}}$$

$$\text{wo: } \sqrt{\frac{2R_1g_m+2}{2C_3C_5R_1R_3R_4g_m+2C_3C_5R_1R_3R_5g_m+2C_3C_5R_1R_3+C_3C_5R_1R_4R_5g_m+C_3C_5R_1R_4+C_3C_5R_3R_4+2C_3C_5R_3R_5+C_3C_5R_4R_5}}$$

$$\text{bandwidth: } \frac{1}{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_1R_3R_5g_m\sqrt{\frac{R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}}$$

$$\text{K-LP: } \frac{R_1 R_4 g_m}{2R_1 g_m + 2}$$

$$\text{K-HP: } \frac{R_1 R_3 R_4 R_5 g_m - R_1 R_3 R_4}{2R_1 R_3 R_4 g_m + 2R_1 R_3 R_5 g_m + 2R_1 R_3 + R_1 R_4 R_5 g_m + R_1 R_4 + R_3 R_4 + 2R_3 R_5 + R_4 R_5}$$

$$\text{K-BP: } \frac{C_3 R_1 R_3 R_4 g_m + C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4}{2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5}$$

Qz: None

$$\text{Wz: } \sqrt{\frac{g_m}{C_3 C_5 R_3 R_5 g_m - C_3 C_5 R_3}}$$

10.7 X-INVALID-WZ-7 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^2 (C_3 C_4 R_1 R_3 R_4 R_5 g_m - C_3 C_4 R_1 R_3 R_4) + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3 + C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{2 R_1 g_m + s^2 (2 C_3 C_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 R_1 R_3 R_5 g_m + 2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 R_5 g_m + C_3 C_4 R_1 R_4 + C_3 C_4 R_3 R_4 + 2 C_3 C_4 R_3 R_5 + C_3 C_4 R_4 R_5) + s (2 C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + C_4 R_4 + 2 C_4 R_5) + 2 C_3 C_4 R_1 R_3 R_4 R_5 g_m - C_3 C_4 R_1 R_3 R_4}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_3}\sqrt{C_4}R_1R_3R_4g_m\sqrt{\frac{2R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+2\sqrt{C_3}\sqrt{C_4}R_1R_3R_5g_m\sqrt{\frac{2R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}$$

$$\text{wo: } \sqrt{\frac{2R_1g_m+1}{2C_3C_4R_1R_3R_4g_m+2C_3C_4R_1R_3R_5g_m+2C_3C_4R_1R_3+C_3C_4R_1R_4R_5g_m+C_3C_4R_1R_4+C_3C_4R_3R_4+2C_3C_4R_3R_5+C_3C_4R_4R_5}}$$

$$\text{bandwidth: } \frac{2\sqrt{C_3}\sqrt{C_4}R_1R_3R_4g_m\sqrt{\frac{2R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+2\sqrt{C_3}\sqrt{C_4}R_1R_3R_5g_m\sqrt{\frac{2R_1g_m}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}+\frac{1}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}}$$

$$\text{K-LP: } \frac{R_1R_5g_m-R_1}{2R_1g_m+1}$$

$$\text{K-HP: } \frac{R_1R_3R_4R_5g_m-R_1R_3R_4}{2R_1R_3R_4g_m+2R_1R_3R_5g_m+2R_1R_3+R_1R_4R_5g_m+R_1R_4+R_3R_4+2R_3R_5+R_4R_5}$$

$$\text{K-BP: } \frac{C_3R_1R_3R_5g_m-C_3R_1R_3+C_4R_1R_4R_5g_m-C_4R_1R_4}{2C_3R_1R_3g_m+C_3R_1R_5g_m+C_3R_1+C_3R_3+C_3R_5+2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5}$$

Qz: None

$$\text{Wz: } \frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}$$

10.8 X-INVALID-WZ-8 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_4 s^2 + R_3 R_4 g_m + s (C_1 R_1 R_3 R_4 g_m - C_5 R_3 R_4)}{2 R_3 g_m + R_4 g_m + s^2 (2 C_1 C_5 R_1 R_3 R_4 g_m + 2 C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_4 + C_1 C_5 R_3 R_4) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + 2 C_5 R_3 R_4 g_m + 2 C_5 R_3 + C_5 R_4)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_1}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}\sqrt{2R_3+R_4}+2\sqrt{C_1}\sqrt{C_5}R_1R_3\sqrt{\frac{g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}\sqrt{2R_3+R_4}+\sqrt{C_1}\sqrt{C_5}R_1R_4\sqrt{\frac{g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}\sqrt{2R_3+R_4}+\sqrt{C_1}\sqrt{C_5}R_3R_4\sqrt{\frac{g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}\sqrt{2R_3+R_4}}{2C_1R_1R_3g_m+C_1R_1R_4g_m+2C_1R_3+C_1R_4+2C_5R_3R_4g_m+2C_5R_3+C_5R_4}$$

$$\text{wo: } \sqrt{\frac{2R_3g_m+R_4g_m}{2C_1C_5R_1R_3R_4g_m+2C_1C_5R_1R_3+C_1C_5R_1R_4+C_1C_5R_3R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_3g_m+R_4g_m}{2C_1C_5R_1R_3R_4g_m+2C_1C_5R_1R_3+C_1C_5R_1R_4+C_1C_5R_3R_4}}(2C_1R_1R_3g_m+C_1R_1R_4g_m+2C_1R_3+C_1R_4+2C_5R_3R_4g_m+2C_5R_3+C_5R_4)}{2\sqrt{C_1}\sqrt{C_5}R_1R_3R_4g_m\sqrt{\frac{g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}\sqrt{2R_3+R_4}+2\sqrt{C_1}\sqrt{C_5}R_1R_3\sqrt{\frac{g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}\sqrt{2R_3+R_4}+\sqrt{C_1}\sqrt{C_5}R_1R_4\sqrt{\frac{g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}\sqrt{2R_3+R_4}+\sqrt{C_1}\sqrt{C_5}R_3R_4\sqrt{\frac{g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}\sqrt{2R_3+R_4}}$$

$$\text{K-LP: } \frac{R_3R_4}{2R_3+R_4}$$

$$\text{K-HP: } -\frac{R_1R_3R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}$$

$$\text{K-BP: } \frac{C_1R_1R_3R_4g_m-C_5R_3R_4}{2C_1R_1R_3g_m+C_1R_1R_4g_m+2C_1R_3+C_1R_4+2C_5R_3R_4g_m+2C_5R_3+C_5R_4}$$

Qz: None

$$\text{Wz: } \frac{\sqrt{-g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}}$$

10.9 X-INVALID-WZ-9 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_4 R_5 s^2 + R_3 R_4 R_5 g_m - R_3 R_4 + s (C_1 R_1 R_3 R_4 R_5 g_m - C_1 R_1 R_3 R_4 - C_5 R_3 R_4 R_5)}{2 R_3 R_4 g_m + 2 R_3 R_5 g_m + 2 R_3 + R_4 R_5 g_m + R_4 + s^2 (2 C_1 C_5 R_1 R_3 R_4 R_5 g_m + 2 C_1 C_5 R_1 R_3 R_5 + C_1 C_5 R_1 R_4 R_5 + C_1 C_5 R_3 R_4 R_5) + s (2 C_1 R_1 R_3 R_4 g_m + 2 C_1 R_1 R_3 R_5 g_m + 2 C_1 R_1 R_3 + C_1 R_1 R_4 R_5 g_m + C_1 R_1 R_4 + C_1 R_3 R_4 + 2 C_1 R_3 R_5 + C_1 R_4 R_5 + 2 C_5 R_3 R_4 R_5 g_m + 2 C_5 R_3 R_4 + C_5 R_4 R_5) + 2 C_1 C_5 R_1 R_3 R_4 R_5 g_m - C_1 C_5 R_1 R_3 R_4 - C_5 R_3 R_4 R_5}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_1}\sqrt{C_5}R_1R_3R_4\sqrt{R_5}g_m\sqrt{\frac{2R_3R_4g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{2R_3R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_4R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+2\sqrt{C_1}\sqrt{C_5}R_1R_3\sqrt{R_5}\sqrt{\frac{2R_3R_4g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{2R_3R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}$$

$$\text{wo: } \sqrt{\frac{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}{2C_1C_5R_1R_3R_4R_5g_m+2C_1C_5R_1R_3R_5+C_1C_5R_1R_4R_5+C_1C_5R_3R_4R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2R_3R_4g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{2R_3R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_4R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+2\sqrt{C_1}\sqrt{C_5}R_1R_3\sqrt{R_5}\sqrt{\frac{2R_3R_4g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}+\frac{2R_3R_5g_m}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{2R_3}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}+\frac{R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}}$$

$$\text{K-LP: } \frac{R_3R_4R_5g_m-R_3R_4}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}$$

$$\text{K-HP: } -\frac{R_1R_3R_4}{2R_1R_3R_4g_m+2R_1R_3+R_1R_4+R_3R_4}$$

$$\text{K-BP: } \frac{C_1R_1R_3R_4R_5g_m-C_1R_1R_3R_4-C_5R_3R_4R_5}{2C_1R_1R_3R_4g_m+2C_1R_1R_3R_5g_m+2C_1R_1R_3+C_1R_1R_4R_5g_m+C_1R_1R_4+C_1R_3R_4+2C_1R_3R_5+C_1R_4R_5+2C_5R_3R_4R_5g_m+2C_5R_3R_5+C_5R_4R_5}$$

Qz: None

$$\text{Wz: } \frac{\sqrt{-R_5g_m+1}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}}$$

